

**BUS 123 | SOLVING BUSINESS PROBLEMS WITH
TECHNOLOGY**

BUS 123 Pre-Reading Collection

Fall 2026 | Official course sequence

18 readings | Intro, Business Math M01-M12, and Excel M01-M04
Business Math M09 includes separate L01 and L02 readings.

Contents

Select a reading title to jump to that section.

01 INTRO-M01-L02 INTRO
BUS 123 — INTRO-M01-L02 —
Technology Orientation &
Technology in Business

02 MATH-M01-L01 BUSINESS
MATH
BUS 123 — MATH-M01— Whole
Numbers, Fractions & Percents

03 MATH-M02-L01 BUSINESS
MATH
BUS 123 — MATH-M02-L01 —
Solving Equations

04 MATH-M03-L01 BUSINESS
MATH
BUS 123 — MATH-M03-L01 —
Discounts: Trade and Cash

05 MATH-M04-L01 BUSINESS
MATH
BUS 123 — MATH-M04-L01 —
Markup, Markdown & Break-Even

06 MATH-M05-L01 BUSINESS
MATH
BUS 123 — MATH-M05-L01 —
Payroll and Depreciation

07 MATH-M06-L01 BUSINESS
MATH
BUS 123 — MATH-M06-L01 —
Inventory, Overhead & Cost of
Goods Sold

08 MATH-M07-L01 BUSINESS
MATH
BUS 123 — MATH-M07-L01 —
Sales, Excise & Property Taxes

09 MATH-M08-L01 BUSINESS
MATH
BUS 123 — MATH-M08-L01 —
Simple Interest, Discounting &
Present Value

10 MATH-M09-L01 BUSINESS
MATH
BUS 123 — MATH-M09-L01 —
Compound Interest & Future Value

11 MATH-M09-L02 BUSINESS
MATH
BUS 123 — MATH-M09-L02 — PMT
Function & Annuities

12 MATH-M10-L01 BUSINESS
MATH
BUS 123 — MATH-M10-L01 —
Foundations of Investing: Stocks,
Bonds, Mutual Funds & ETFs

13 MATH-M11-L01 BUSINESS
MATH
BUS 123 — MATH-M11-L01 —
Reading and Analyzing Financial
Reports

14 MATH-M12-L01 BUSINESS
MATH
BUS 123 — MATH-M12-L01 —
Business Statistics: Center, Spread,
and Correlation

15 EXCEL-M01-L01 EXCEL
Navigating Excel, Formatting & Cell
References

16 EXCEL-M02-L01 EXCEL
BUS 123 - EXCEL-M02-L01 -
Formatting and Organizing
Worksheets

17 EXCEL-M03-L01 EXCEL
Excel Functions for Business
Analysis

18 EXCEL-M04-L01 EXCEL
Excel Charts and PivotTables

BUS 123 PRE-READINGS

Intro

1 official reading

BUS 123 — INTRO-M01-L02 — Technology Orientation & Technology in Business

BUS 123 · INTRO-M01-L02 · TECHNOLOGY ORIENTATION & TECHNOLOGY IN BUSINESS

Course: Solving Business Problems with Technology · Fall 2026 **Track:** INTRO · **Module:** M01 · **Lessons:** L02-L03

CONNECT TO PRIOR KNOWLEDGE

This is the first reading lesson of BUS123 — Solving Business Problems with Technology. You are beginning with a clean slate, which means the prior knowledge here is the everyday digital experience you already bring to the room: using a smartphone, saving files to a computer, logging into a school portal.

This combined reading builds directly on that familiarity. First, you will set up the Microsoft 365 and OneDrive habits needed for the course. Then you will step back and ask a sharper question: **are you using these tools casually, or professionally?** By the end of this module, you will have the foundation to answer confidently.

WHAT IS MICROSOFT 365?

Microsoft 365 is a subscription-based suite of productivity software that includes Excel, Word, PowerPoint, OneDrive, Outlook, and Teams. As an Endicott student, you have **full access at no cost** through your college account — which includes installing the desktop applications on up to five personal devices.

Application	Primary Purpose	Used In BUS123
Excel	Numerical data, calculations, financial analysis	Primary tool — every module
Word	Professional documents, reports, memos	Selected written deliverables
PowerPoint	Presentations, pitch decks, visual communication	Final project and selected activities
OneDrive	Cloud storage and file sync across all devices	Required for all file storage and submission

WHAT IS ONEDRIVE?

OneDrive is Microsoft's cloud storage and synchronization service. When you save a file to your OneDrive folder on your laptop, it is automatically copied to Microsoft's servers and made available on every other device connected to the same account.

This means:

- You can never truly "lose" the file to a dead laptop — it lives in the cloud.

- Every device stays current — edit on your phone and the changes appear on your laptop.
- Sharing is a link, not an attachment — colleagues can see the live version, never a stale copy.
- Version history is automatic — OneDrive saves prior versions so you can roll back mistakes.

Endicott-Specific Detail

Your Endicott College Microsoft account includes **1 TB of OneDrive storage**. This is separate from any personal Microsoft account you may already have. Always confirm you are signed in to your **@mail.endicott.edu** account — not a personal Hotmail or Outlook.com account — before saving course files.

INSTALLING AND ACTIVATING MICROSOFT 365

If you have not already installed the desktop applications, follow these steps using your Endicott student credentials:

1. Go to **portal.office.com** and sign in with your @endicott.edu email address.
2. Click **Install apps** in the top right corner and select **Microsoft 365 apps**.
3. Run the downloaded installer and follow the on-screen prompts.
4. Open any Office app (Excel, Word, or PowerPoint), go to **File** → **Account**, and confirm your Endicott email appears under **User Information**.
5. If you see a different account, click **Sign Out**, then **Sign In** with your @mail.endicott.edu address.

⚠ Common Mistake

Students who already have a personal Microsoft account often find that OneDrive syncs to the wrong account. Before saving any course file, verify that the account shown in **File** → **Account** is your **@mail.endicott.edu** address — not a personal account. Files saved to a personal OneDrive are not accessible to your instructor.

SETTING UP YOUR BUS123 FOLDER STRUCTURE IN ONEDRIVE

Consistent folder organization is a professional habit, not a preference. In BUS123, all files should be saved inside a specific folder structure in your Endicott OneDrive to allow you to find and upload assignments easily. A recommended folder structure, that aligns with the class modules in Canvas is listed below:

Folder Level	Name	What Goes Here
1 — Top level	BUS123	Everything for this course
2 — Subfolders	INTRO / EXCEL / MATH	One folder per track
3 — Module folders	M01 , M02 , M03 ...	One folder per module within each track

FILE NAMING CONVENTION

Every file you save/upload in BUS123 must follow the course naming convention. This makes it easy to identify, sort, and share your work without confusion.

Format: LastName_FirstName_BUS123_ModuleN.xlsx

Example: Evitts_Bethany_BUS123_Module01.xlsx

✅ Why This Matters

I receive dozens of files with names like `homework.xlsx` or `final_FINAL_v3.xlsx`. A properly named file tells me exactly who submitted it and what it contains — before it is even opened. This is a professional habit that carries into every workplace. Get in that habit now.

SAVING, SYNCING, AND VERIFYING A FILE

Saving a file to your OneDrive folder is not the same as the file being available in the cloud. There is a brief delay between saving locally and the file completing its upload to Microsoft's servers. **Always verify sync status before submitting.**

Sync Icon	What It Means	What to Do
 Blue circle/arrows	File is currently uploading	Wait — do not submit yet
 Green checkmark	File is fully synced to the cloud	Safe to copy share link and submit
 Red X	Sync error — file did not upload	Check your internet connection and re-save

KEY VOCABULARY: MICROSOFT 365 & ONEDRIVE

Term	Definition
Microsoft 365	A subscription-based suite of productivity applications including Excel, Word, PowerPoint, OneDrive, Outlook, and Teams. Endicott students have full access at no cost through their college credentials.
OneDrive	Microsoft's cloud storage and file synchronization service. Files saved to OneDrive are automatically backed up to remote servers and accessible across all signed-in devices.
Cloud Storage	Files stored on remote servers accessed over the internet rather than on a physical hard drive. Enables multi-device access and protects against local hardware failure.
Sync	The process of ensuring the local copy of a file and the cloud copy are identical and up to date. Indicated by a green checkmark on the file icon in your OneDrive folder.
Desktop Application	A software program installed and running directly on your computer, as opposed to a browser-based version. Desktop apps have full features and work offline.
File Naming Convention	A standardized format for naming files so they are identifiable, sortable, and consistent. BUS123 format: LastName_FirstName_BUS123_ModuleN.xlsx.
Activation	The process of signing in to Microsoft 365 with your Endicott credentials to unlock full access to the desktop applications. Verified under File → Account in any Office app.
Version History	OneDrive's automatic record of prior saved states of a file. Allows you to restore an earlier version if content is accidentally deleted or overwritten.

WHY TECHNOLOGY MATTERS IN BUSINESS

In the first lessons of BUS123 you installed Microsoft 365, set up your OneDrive folder structure, and learned what this course is and how it works. You now have the tools and the expectations.

This lesson asks a sharper question: **why does any of it matter?** Before you learn how to build a payroll model or analyze inventory data, it helps to understand the larger context — how technology actually functions inside a business, and what separates someone who uses it casually from someone who uses it professionally.

PASSIVE USER VS. PROFESSIONAL OPERATOR

Most people interact with technology every day without thinking about it deliberately. A **passive user** opens an app, does a task, and closes it. A **professional operator** thinks about what output the tool needs to produce, how to structure the work so others can verify it, and how to catch errors before they matter.

Dimension	Passive User	Professional Operator
Goal	Complete the task	Produce reliable output someone else can use
File management	Saves wherever is convenient	Follows a naming and folder convention every time
Formula use	Types answers directly into cells	References input cells so results update automatically
Error response	Deletes the error and retypes a number	Traces the formula to find and fix the root cause
Sharing	Emails an attachment	Shares a live OneDrive link to the current version

The Shift This Course Is Asking You to Make

You do not need to be an expert to be a professional operator. You need to be **deliberate**. The habits you build in BUS123 — naming files correctly, using formulas instead of typed values, checking your work before submitting — are the same habits that make a junior analyst trustworthy to a manager.

TECHNOLOGY ACROSS BUSINESS SECTORS

Every industry uses technology differently, but the underlying skills are transferable. Understanding how technology shows up in the work you are heading toward helps you know which tools to prioritize.

Sector	Core Technology Problem	BUS123 Connection
Retail	Tracking inventory costs, pricing products, managing markdowns	COGS, markup/markdown, inventory math
Professional Services	Calculating employee compensation, modeling client loan scenarios	Payroll, TVM, interest calculations
Event Management	Projecting breakeven, managing seasonal cash gaps	Breakeven analysis, cash flow modeling
Healthcare	Analyzing payer mix, managing complex multi-rate payroll	Data analysis, payroll with variable rates
Finance	Modeling loan payments, evaluating investment returns	TVM, PMT, FV/PV functions

THE DIGITAL FLUENCY SPECTRUM

Employers think about digital skills in layers. Knowing where you currently sit on this spectrum — and where you want to be — helps you set a goal for what BUS123 can do for you.

Level	Description	What It Looks Like
1 — Basic User	Can open, navigate, and save files in standard applications	Types text in Word, opens spreadsheets, attaches files to email
2 — Functional User	Uses built-in tools to complete defined tasks reliably	Writes SUM formulas, formats a table, builds a basic chart
3 — Professional Operator	Structures work so it is accurate, auditable, and maintainable	Builds referenced formulas, names files correctly, checks for errors
4 — Analytical Producer	Uses technology to answer business questions and support decisions	Builds a payroll model, runs a breakeven analysis, interprets results

✓ Where BUS123 Targets You

Most students enter at Level 1 or 2. BUS123 is designed to move you to **Level 3** by the end of the EXCEL track and to **Level 4** by the end of the MATH track. Level 4 is where early-career business roles are won.

HOW TECHNOLOGY SHAPES DECISIONS

A spreadsheet does not just execute a calculation you already know how to do. It changes what questions you can even ask.

Consider a retail buyer deciding how many units to order. Without a model, the decision is a guess. With a properly built spreadsheet, the buyer can run ten scenarios in two minutes and identify the order quantity that minimizes risk under each one.

This is the core reason BUS123 exists: **technology does not replace business judgment — it amplifies it.** The analyst who can build the model gets to participate in the decision. The one who cannot has to wait for someone else's output.

WHERE AI FITS IN

AI tools, including tools like Microsoft Copilot, Claude, and ChatGPT in Excel and Word, are increasingly present in business workflows. It is worth being honest about what they do and do not change.

AI Is Good At	AI Is Not Good At
Drafting a first version of something	Knowing whether the answer is correct for your specific situation
Summarizing large amounts of text	Catching errors in a formula it did not write
Suggesting functions you did not know existed	Understanding the business context behind the numbers
Automating repetitive formatting tasks	Taking responsibility for a wrong result

The Skill AI Cannot Replace

AI tools are most valuable to people who **already know how to do the work**. If you cannot read a formula and judge whether it is correct, you cannot evaluate what AI produces. BUS123 builds the foundation that makes AI useful rather than dangerous.

TECHNOLOGY HABITS THAT PAY OFF

The research on early-career outcomes consistently shows that digital skill gaps are one of the most common reasons recent graduates underperform expectations. The habits below are simple — but most students do not practice them until they learn the hard way.

Name every file correctly, every time. File naming is not a preference. It is a professional standard. An improperly named file signals carelessness before the content is even opened.

Save to the cloud, not the device. Local saves are one hardware failure away from lost work. Cloud saves are versioned, backed up, and accessible from anywhere.

Use formulas, not typed values. A typed number in a result cell is invisible to the next person who opens the file. A formula is a documented, auditable calculation.

Check your output before you share it. Sending a file with a formula error is worse than sending it late. Spend two minutes reviewing before submitting anything.

Learn the keyboard shortcut for what you do most. Speed compounds. Ten seconds saved per task adds up to hours across a semester — and across a career.

KEY VOCABULARY: TECHNOLOGY IN BUSINESS

Term	Definition
Passive User	Someone who uses technology reactively to complete a task without thinking deliberately about structure, accuracy, or how others will use the output.
Professional Operator	Someone who uses technology deliberately to produce reliable, auditable, and maintainable output that others can depend on.
Digital Fluency	The ability to use digital tools effectively and appropriately across a range of business contexts. Fluency implies understanding, not just task completion.
Digital Fluency Spectrum	A four-level framework describing how technology competence develops: from basic user through functional user, professional operator, to analytical producer.
Analytical Producer	The highest level of the digital fluency spectrum. An analytical producer uses technology to answer business questions, model scenarios, and support decisions rather than execute predefined tasks.
Formula Integrity	The property of a spreadsheet in which all calculated values are produced by formulas referencing input cells, making the model auditable, updateable, and free of hard-coded values in result cells.
Microsoft Copilot	An AI assistant built into Microsoft 365 applications including Excel and Word. Useful for drafting, summarizing, and suggesting functions — but requires the user to evaluate and verify all output.
Payer Mix	In healthcare finance, the proportion of revenue coming from different payment sources such as private insurance, Medicare, Medicaid, and self-pay patients. Used in Harborside Medical Center case studies.

BUS 123 PRE-READINGS

Business Math

13 official readings

BUS 123 — MATH-M01— Whole Numbers, Fractions & Percents

BUS 123 · MATH-M01-L01 · WHOLE NUMBERS, FRACTIONS & PERCENTS

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M01 ·

1 · CONNECT TO PRIOR KNOWLEDGE

This module is the mathematical foundation for everything that follows in BUS 123. Every formula you will build — markup percentages, payroll calculations, breakeven analysis, and loan math — starts with the skills you'll practice here: whole number operations, fraction-to-decimal conversion, and percent calculations.

This should feel very familiar; for this class pay close attention to the **business context** and the **Excel approach**, which add layers that go beyond a standard algebra class.

Our case study company for this module is **Meridian Advisory Group**, a wealth management firm whose day-to-day operations generate exactly the kinds of arithmetic problems we will solve.

2 · CORE CONCEPTS

Part A — Whole Number Operations

The four arithmetic operations — addition, subtraction, multiplication, and division — are the building blocks of business math. In a business context, each operation maps to a specific type of question:

Operation	Business Meaning	Excel Formula
Addition	Summing revenue across accounts	=SUM(B2:B10) or =B2+B3
Subtraction	Net income = Revenue – Expenses	=B5 – C5
Multiplication	Total fees = Accounts × Fee per account	=B3 * C3
Division	Per-client fee = Total ÷ Client count	=B7 / C7

Worked Example — Meridian Advisory Group:

Meridian begins Q3 with 48 client accounts, adds 12 new clients, and loses 3. The management fee is \$4,200 per account per quarter.

- Quarter-end accounts: $48 + 12 - 3 = 57$
- Total quarterly fees: $57 \times \$4,200 = \$239,400$
- In Excel: =B2+B3-B4 for the account count, then =D5*C2 for total fees

Always reference cells rather than typing numbers directly into the formula.

⚠ Unit Mismatch Warning

Before multiplying, check that both numbers use the same unit and time period. Multiplying a quarterly account count by an annual fee rate gives a meaningless result. Label every column header with its unit (e.g., `Fee/Account/Quarter`) to catch this error early.

Part B — Fractions in Context

A fraction expresses a part of a whole: the **numerator** is the piece being measured, and the **denominator** is the total universe. In business, fractions appear as staff allocations, market share figures, and service-line proportions.

Excel converts fractions to decimals automatically when you divide the numerator by the denominator; format the result cell as a percentage or fraction if needed.

Worked Example — Meridian Advisory Group:

Meridian's 6-person advisory team is split evenly: 2 advisors each in Tax, Estate Planning, and Investment Management.

- Each service line represents $2/6 = 1/3 \approx 0.3333$ of total capacity.
- If Meridian hires 3 more Tax advisors, the Tax team grows to 5 of 9 total advisors: $5/9 \approx 0.5556 = 55.6\%$.
- Note that the denominator changed from 6 to 9 — **you must use the new total in the denominator**.

✅ Tip: The Denominator Is the Whole Universe

When the total changes (e.g., new hires), update the denominator before recalculating. Using an outdated denominator is one of the most common fraction errors in business analysis.

Part C — Percent Calculations

A percent is a fraction expressed out of 100. Business math involves **three distinct percent calculations**, each requiring a different formula structure. Confusing them is a frequent source of errors:

Operation	Excel Formula	What It Calculates
Case 1 — % of a Whole	<code>=whole * rate</code>	What is 15% of \$239,400? → <code>=239400*0.15</code>
Case 2 — % Change	<code>=(new - old) / old</code>	Fees: \$1.2M → \$1.255M → <code>=(1255000-1200000)/1200000</code>
Case 3 — Ratio as %	<code>=part / whole</code>	4 of 25 clients renewed → <code>=4/25</code> (format as %)

Worked Example — Forward and Backward:

Meridian earned \$1,200,000 in AUM fees last year and projects a 7.5% increase.

- Projected fees: `=1200000*(1+0.075)` = **\$1,290,000**
- Actual fees came in at \$1,255,000
- Actual percent change: `=(1255000-1200000)/1200000` = **4.58%** — below the 7.5% target

In Excel, always use cell references: $= (B5 - B4) / B4$. Format the result cell as **Percentage** — do not manually multiply by 100.

⚠ Common Mistake: % Change vs. % of Total

These are different formulas.

- **% change** compares a value to its earlier self: $(\text{New} - \text{Old}) / \text{Old}$

- **% of total** compares a part to a whole at one point in time: $\text{Part} / \text{Whole}$

Ask yourself: am I measuring how something **changed over time**, or what **share** it represents of a bigger number?

3 · FORMULA REFERENCE

Operation	Excel Formula	What It Calculates
SUM range	$=\text{SUM}(B2:B10)$	Add all values in a range
Arithmetic ops	$=B2+B3-B4$	Add/subtract individual cells
Multiply	$=B3 * C3$	Qty × Price, Accounts × Fee
Divide	$=B7 / C7$	Total ÷ Count = per-unit
Fraction → Decimal	$=B2 / C2$	Numerator ÷ Denominator
% of Whole	$=\text{whole} * \text{rate}$	e.g., $=B4 * 0.15$
% Change	$= (\text{new} - \text{old}) / \text{old}$	Base = older number always
Ratio as %	$=\text{part} / \text{whole}$	Format result as Percentage
Grow by rate	$=B4 * (1 + \text{rate})$	e.g., $=B4 * (1 + 0.075)$

4 · CHECK YOUR UNDERSTANDING

Answer each question before looking at the answer key below. Show your Excel formula approach, not just the arithmetic result.

1. Meridian starts a quarter with 57 accounts and loses 4. What is the new account total? Write the Excel formula.
1. Total quarterly fee revenue was \$239,400. Tax services generated \$85,000 of that. What fraction — and what percent — of total fees came from Tax?
1. Meridian is targeting a 12% increase in total AUM fees from a base of \$239,400/quarter. What is the quarterly fee target? Write the Excel formula.
1. Actual fees came in at \$268,200. The prior quarter was \$239,400. What was the percent change? Which formula case does this represent?
1. Meridian's Investment team of 2 advisors represents what fraction of a 9-person total team? Express as a decimal rounded to 4 places.
1. True or False: the denominator in a percent change calculation should always be the newer (larger) number.

- Meridian runs 4 service lines. Two lines each have 15 clients, one has 18 clients, one has 9 clients. What percent of total clients does the line with 18 clients represent?

Answer Key · Check Your Understanding

#	Answer
1	53 accounts. Excel: <code>=57-4</code> or <code>=B2-B3</code> . Unit check: both are counts of accounts.
2	Fraction: $85,000/239,400 \approx 0.3551$. Percent: $\approx 35.51\%$. Excel: <code>=85000/239400</code> → format as Percentage. This is Case 3 (ratio as %).
3	\$268,128. Excel: <code>=239400*(1+0.12)</code> or <code>=B4*(1+0.12)</code> . This is Case 1 (% of whole, grow-by variant).
4	Percent change = $(268,200 - 239,400) / 239,400 \approx 12.03\%$. Excel: <code>=(268200-239400)/239400</code> . This is Case 2 (% change). The base is the older number (\$239,400).
5	$2/9 \approx 0.2222$. Excel: <code>=2/9</code> . Note: the denominator is 9 (total team), not 6 (old team).
6	False. The denominator in a percent change calculation should always be the older (earlier) number.
7	18 clients out of $(15+15+18+9) = 57$ total. $18/57 \approx 31.58\%$. Excel: <code>=18/57</code> → format as Percentage. This is Case 3.

5 · KEY VOCABULARY

Term	Definition
Numerator	The top number in a fraction; represents the part being measured.
Denominator	The bottom number in a fraction; represents the total or whole.
Percent	A ratio expressed as a fraction out of 100. Written with the % symbol.
Percent Change	A measure of how much a value has grown or declined relative to its starting point. Formula: $(\text{New} - \text{Old}) / \text{Old}$.
Percent of Total	A ratio comparing a part to the whole at a single point in time. Formula: $\text{Part} / \text{Whole}$.
Cell Reference	In Excel, using a cell address (e.g., B4) in a formula rather than typing a number directly, so the formula recalculates automatically when input changes.
Unit Mismatch	An error that occurs when two values being multiplied or compared use different units or time periods (e.g., mixing annual and quarterly rates).
AUM Fees	Assets Under Management fees — revenue a financial advisory firm earns based on the total value of client assets it manages.

BUS 123 — MATH-M02-L01 — Solving Equations

BUS 123 · MATH-M02-L01 · SOLVING EQUATIONS

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M02 · **Lesson:** L01

Case Study Company: Anchor & Oak Events

1 · CONNECT TO PRIOR KNOWLEDGE

In the previous module you built basic formulas — translating business sentences like *Revenue = Price × Quantity* into symbolic expressions. Every formula you wrote was an equation with a known output. That's a powerful starting point.

But in real business decisions, **the output is often what you know and one of the inputs is what you need to find**. If you know an event earned \$4,080 in catering revenue from 60 guests, you can solve for the per-person rate. If you know Anchor & Oak can afford \$7,300 in fixed costs, you can check whether a venue is in budget.

Solving equations is the technique that makes formulas work in both directions — and it underpins every financial model you will build this semester.

2 · CORE CONCEPTS

The Balanced Scale Model

An equation is a statement that two expressions are equal. The equal sign is a fulcrum — a perfectly balanced scale. The **Golden Rule** of algebra follows directly from this image: whatever operation you apply to one side, you must apply to the other. If you add 50 to the left side without adding 50 to the right, the scale tips and the equation is no longer true.

Golden Rule: Whatever you do to one side of an equation, you MUST do to the other side. No exceptions.

Type 1 — One-Step Equations

A one-step equation has exactly one operation separating the variable from its value. The solve strategy is a single **inverse operation** — the opposite of what was done to the variable.

Form	Inverse Operation	Result
$x + a = b$	Subtract a from both sides	$x = b - a$
$x - a = b$	Add a to both sides	$x = b + a$
$a * x = b$	Divide both sides by a	$x = b / a$
$x / a = b$	Multiply both sides by a	$x = b * a$

Worked Example — Anchor & Oak Events:

Anchor & Oak budgeted \$4,200 for floral arrangements. After booking three weddings, \$1,650 remains. How much did the three weddings cost?

- The unknown is the total wedding floral cost (x).
- It was subtracted from the budget to leave the remainder: $x + 1,650 = 4,200$
- Subtract 1,650 from both sides: $x = 2,550$

The three weddings cost \$2,550 combined.

Type 2 — Two-Step Equations

Two-step equations require two inverse operations. The standard form is $ax + b = c$. The solve order is always: **undo addition or subtraction first, then undo multiplication or division**. This is the reverse of the standard order of operations — because building the equation applied multiplication before addition, solving it undoes addition before multiplication.

Step	Operation	Result
1	Subtract b from both sides	$ax = c - b$
2	Divide both sides by a	$x = (c - b) / a$

Worked Example — Anchor & Oak Events:

A coordinator earned \$1,240 total in a week. The week included \$120 in base pay and 8 overtime hours at an unknown rate.

- Equation: $8x + 120 = 1,240$
- Step 1 — subtract 120: $8x = 1,120$
- Step 2 — divide by 8: $x = 140$

The overtime hourly rate is \$140.

Type 3 — Variables on Both Sides

When the unknown variable appears on both sides of the equal sign, the first task is to **collect all variable terms on one side** before solving. The standard move is to subtract the smaller variable term from both sides — this keeps the remaining coefficient positive and reduces arithmetic errors. Once the variable is on one side only, the problem reduces to a two-step equation.

Step	Operation	Result
1	Subtract smaller variable term from both sides	$(a - c)x + b = d$
2	Subtract b from both sides	$(a - c)x = d - b$
3	Divide by $(a - c)$	$x = (d - b) / (a - c)$

Worked Example — Anchor & Oak Events:

Comparing two tent vendors: Vendor A charges \$500 flat plus \$25 per guest; Vendor B charges \$200 flat plus \$40 per guest. At how many guests are the costs equal?

- Equation: $500 + 25g = 200 + 40g$
- Subtract 25g from both sides: $500 = 200 + 15g$
- Subtract 200: $300 = 15g$
- Divide by 15: $g = 20$

At exactly 20 guests, both vendors charge the same amount (\$1,000).

Type 4 — Literal Equations and Formula Rearrangement

A **literal equation** contains two or more variables, like $P = R - C$ (Profit = Revenue minus Costs). Solving a literal equation means isolating one specific variable while treating all others as constants. The exact same inverse operation rules apply — the only difference is that letters stand in for numbers you haven't specified yet.

This skill is the foundation of every formula rearrangement you will do in TVM, break-even analysis, markup/markdown, and loan math.

Formula	Solve For	Steps	Result
$R = VC + FC$	FC	Subtract VC from both sides	$FC = R - VC$
$R = VC + FC$	VC	Subtract FC from both sides	$VC = R - FC$
$P = R - C$	R	Add C to both sides	$R = P + C$
$P = R - C$	C	Add P to both sides, swap	$C = R - P$

Worked Example — Anchor & Oak Events:

A corporate gala has projected revenue of \$18,500. Variable costs (catering, staffing, linens) total \$11,200.

- Using $FC = R - VC$: $FC = 18,500 - 11,200 = \$7,300$
- Anchor & Oak can spend up to \$7,300 on fixed costs (venue, permits, equipment) and still break even. Any fixed cost above \$7,300 generates a loss.

3 · FORMULA REFERENCE

Equation Type	Standard Form	Solve Strategy	Result
One-Step (add)	$x + a = b$	Subtract a from both sides	$x = b - a$
One-Step (multiply)	$a \cdot x = b$	Divide both sides by a	$x = b / a$
Two-Step	$a \cdot x + b = c$	1. Subtract b · 2. Divide by a	$x = (c - b) / a$
Both Sides	$ax + b = cx + d$	1. Collect var terms · 2. Two-step	$x = (d - b) / (a - c)$
Literal	$R = VC + FC$	Isolate target variable	$FC = R - VC$

4 · CHECK YOUR UNDERSTANDING

Answer each question before looking at the answer key below. Show your equation setup — graders check the setup, not just the final number.

1. What is the Golden Rule of equation solving?
2. Solve: $x + 480 = 1,250$. What is x ?
3. Solve: $12x = 900$. What is x ?
4. Solve: $5x + 75 = 450$. What is x ?
5. Solve: $400 + 30n = 100 + 60n$. What is n ?
6. Using $FC = R - VC$, if $R = \$22,400$ and $VC = \$14,900$, what is FC ?
7. Anchor & Oak earned \$5,400 on a private dinner. Labor cost \$28/hour and the event lasted 15 hours. The remaining revenue covers venue and supplies. Write and solve the equation to find non-labor revenue.

Answer Key · Check Your Understanding

#	Answer
1	Whatever operation you apply to one side of an equation, you must apply to the other side.
2	$x = 1,250 - 480 = 770$
3	$x = 900 / 12 = 75$
4	Step 1: $5x = 450 - 75 = 375$. Step 2: $x = 375 / 5 = 75$
5	Subtract $30n$ from both sides: $400 = 100 + 30n$. Subtract 100: $300 = 30n$. Divide: $n = 10$
6	$FC = 22,400 - 14,900 = \$7,500$
7	Total revenue = \$5,400. Labor = $\$28 \times 15 = \420 . Equation: $x + 420 = 5,400$. $x = \$4,980$ (non-labor revenue)

5 · KEY VOCABULARY

Term	Definition
Equation	A mathematical statement that two expressions are equal, connected by an equal sign.
Variable	A letter (e.g., x, n, g) representing an unknown quantity to be solved for.
Inverse Operation	The operation that undoes another: addition undoes subtraction; multiplication undoes division.
One-Step Equation	An equation requiring exactly one inverse operation to isolate the variable.
Two-Step Equation	An equation requiring two inverse operations, applied in reverse order of operations.
Literal Equation	An equation containing two or more variables; solved by isolating one variable while treating others as constants.
Break-Even	The point at which total revenue equals total costs, yielding zero profit.
Coefficient	The numerical factor multiplied by a variable (e.g., in $8x$, the coefficient is 8).



Before Next Class

During class, use the **Live You Try It** tab for the short Excel pauses after worked examples. Before the next checkpoint, complete the **Class Challenge** tab as the graded or next-level activity. Write your equation setup before solving so your process is visible, not just the final number.

BUS 123 — MATH-M03-L01 — Discounts: Trade and Cash

BUS 123 · MATH-M03-L01 · DISCOUNTS: TRADE AND CASH

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M03 · **Lesson:** L01
Case Study Company: Tidal Goods Co.

1 · CONNECT TO PRIOR KNOWLEDGE

Earlier in the math track, you worked with percentages — converting between decimals and percents, calculating percentage change, and finding amounts from percentages. Today's material builds directly on those M01 skills while continuing the equation thinking from M02. **Discounts are simply percentages applied to prices**, and once you can work with percentages confidently, discount math follows naturally.

You have probably encountered discounts as a consumer: "30% off" a jacket, or "buy within 10 days and save 2%." This pre-reading gives you the vocabulary and formulas to work with discounts the way businesses do — precisely and systematically.

2 · CORE CONCEPTS

Part 1 — Trade Discounts

When a manufacturer or wholesaler publishes a **list price** (also called a catalog price), they rarely expect every buyer to pay that full amount. Businesses at different points in the supply chain receive different reductions from the list price, called **trade discounts**. A retailer might receive 30% off list. A large chain might receive 40%. The actual amount the buyer pays is called the **net price**.

To find the net price, we use the **complement** of the discount rate. The complement is simply (1 minus the discount rate). If the trade discount is 30%, the complement is 0.70 — meaning the buyer pays 70% of the list price.

$$\text{Net Price} = \text{List Price} \times (1 - \text{Discount Rate})$$

Worked Example — Tidal Goods Co.:

Tidal Goods Co. orders wetsuits from a supplier. The list price is \$189.99 per unit, and Tidal Goods receives a 30% trade discount as a retail partner.

- Complement: $1 - 0.30 = 0.70$
- Net Price: $\$189.99 \times 0.70 = \132.99
- Discount Amount: $\$189.99 - \$132.99 = \$57.00$

✓ Why the Complement Works

If you receive 30% off, you pay the remaining 70%. Multiplying by 0.70 gives you the net price in one step, rather than calculating the discount amount first and then subtracting it. Both approaches give the same answer — the complement method is faster.

Part 2 — Chain Discounts

Sometimes a supplier offers more than one trade discount on the same item. For example, a standard trade discount might be 20%, with an additional 10% for large orders and another 5% for preferred accounts. When multiple discounts apply, they are called **chain discounts** and written as "20/10/5."

The critical rule: **each discount applies to the result of the previous discount, not to the original list price.** This means you cannot simply add the rates together. 20/10/5 is not the same as a single 35% discount.

$$\text{Net Price} = \text{List Price} \times (1 - d1) \times (1 - d2) \times (1 - d3)$$

Worked Example — Surfboard Fins:

Tidal Goods orders surfboard fins with a list price of \$64.00 per set. The supplier offers a 20/10/5 chain discount.

- Step 1: $\$64.00 \times (1 - 0.20) = \$64.00 \times 0.80 = \$51.20$
- Step 2: $\$51.20 \times (1 - 0.10) = \$51.20 \times 0.90 = \$46.08$
- Step 3: $\$46.08 \times (1 - 0.05) = \$46.08 \times 0.95 = \mathbf{\$43.78}$

⚠ Common Mistake — Adding Rates

Adding $20 + 10 + 5 = 35\%$ would give $\$64 \times 0.65 = \mathbf{\$41.60}$ — a different (lower) answer. The chain method correctly gives **\$43.78**. These are not equal because each discount is applied to an already-reduced base, not the original list price.

Part 3 — Cash Discounts and Payment Terms

While trade discounts reduce the price of goods, **cash discounts** reward buyers who pay their invoices early. Suppliers offer cash discounts to improve their own cash flow — getting paid sooner is worth a small reduction in the amount received.

Cash discount terms are written in standardized shorthand. The most common format is:

Term component	Meaning
2	Cash discount rate: 2%
10	Discount period: pay within 10 days to earn the discount
net 30	Full payment is due within 30 days

Worked Example — Tidal Goods Invoice:

Tidal Goods receives an invoice for \$4,200.00 from their wetsuit supplier. Terms: **2/10 net 30**.

- Discount Amount: $\$4,200 \times 0.02 = \mathbf{\$84.00}$

- Payment if paid within 10 days: $\$4,200 \times (1 - 0.02) = \$4,200 \times 0.98 = \mathbf{\$4,116.00}$
- If Tidal Goods waits past 10 days: full **\$4,200** is due by day 30.

The True Cost of Skipping a Cash Discount

Not taking a 2/10 net 30 discount is equivalent to borrowing money at roughly **37% annual interest**. The \$84 "savings" from waiting buys only 20 extra days of payment deferral. For most businesses, it nearly always makes financial sense to take the cash discount when cash is available.

3 · FORMULA REFERENCE

Concept	Formula	Example
Net Price (single)	$= \text{List} \times (1 - d)$	$\$189.99 \times 0.70 = \132.99
Discount Amount	$= \text{List} \times d$	$\$189.99 \times 0.30 = \57.00
Complement	$= 1 - d$	$1 - 0.30 = 0.70$
Net Price (chain)	$= \text{List} \times (1-d_1) \times (1-d_2) \times (1-d_3)$	$\$64 \times 0.80 \times 0.90 \times 0.95 = \43.78
Single Equivalent Rate	$= 1 - [(1-d_1)(1-d_2)(1-d_3)]$	$1 - 0.684 = 31.6\%$
Cash Discount Payment	$= \text{Invoice} \times (1 - \text{cash rate})$	$\$4,200 \times 0.98 = \$4,116$
Cash Discount Savings	$= \text{Invoice} \times \text{cash rate}$	$\$4,200 \times 0.02 = \84

4 · CHECK YOUR UNDERSTANDING

Answer these questions before class. Bring your work — we'll use these as discussion starters.

1. A surfboard has a list price of \$525.00. A retailer receives a 22% trade discount. What is the net price?
2. Tidal Goods receives a chain discount of 25/10 on a \$380 order of accessories. What is the net price?
3. An item has a list price of \$75.00 and is subject to a 30/15/5 chain discount. Calculate the net price.
4. What is the single equivalent discount rate for a 30/15/5 chain discount?
5. An invoice for \$6,500 has terms of 2/10 net 30. How much is saved by paying within 10 days?
6. Tidal Goods receives invoice terms of "3/15 net 45." What does each number represent?
7. A buyer can take a 2/10 net 30 cash discount or pay the full amount in 30 days. If the invoice is \$2,800, what is the annualized cost of NOT taking the discount?

Answer Key · Check Your Understanding

#	Answer
1	$\$525.00 \times (1 - 0.22) = \$525.00 \times 0.78 = \mathbf{\$409.50}$
2	$\$380 \times (1 - 0.25) \times (1 - 0.10) = \$380 \times 0.75 \times 0.90 = \mathbf{\$256.50}$
3	$\$75 \times 0.70 \times 0.85 \times 0.95 = \$42.39375 \approx \mathbf{\$42.39}$
4	$1 - (0.70 \times 0.85 \times 0.95) = 1 - 0.56525 = 0.43475 \approx \mathbf{43.5\%}$
5	$\$6,500 \times 0.02 = \mathbf{\$130.00}$
6	3% discount if paid within 15 days; full amount due in 45 days
7	Cost = $(\$56 \div \$2,744) \times (365 \div 20) \approx \mathbf{37.2\% \text{ annualized}}$

5 · KEY VOCABULARY

Term	Definition
List Price	The published catalog or manufacturer's suggested price before any discounts are applied.
Trade Discount	A reduction from list price offered to buyers in the supply chain (retailers, wholesalers, etc.).
Net Price	The actual price paid after all trade discounts have been subtracted from list price.
Complement	The portion of the price the buyer actually pays; calculated as $(1 - \text{discount rate})$.
Chain Discount	A series of successive trade discounts applied one after the other to the same item.
Single Equivalent Rate	A single discount rate that produces the same net price as a chain of discounts.
Cash Discount	A percentage reduction offered to buyers who pay their invoice within a specified early payment window.
Payment Terms	The notation describing a cash discount — e.g., 2/10 net 30 means 2% off if paid within 10 days, full amount due in 30 days.
Discount Period	The number of days within which a buyer must pay to qualify for the cash discount.
Net Period	The total number of days a buyer has to pay the full invoice amount.

BUS 123 — MATH-M04-L01 — Markup, Markdown & Break-Even

BUS 123 · MATH-M04-L01 · MARKUP, MARKDOWN & BREAK-EVEN

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M04 · **Lesson:** L01

Case Study Company: Anchor & Oak Events

1 · CONNECT TO PRIOR KNOWLEDGE

In the discounts lesson, you started with a list price and reduced it. In this lesson, you often **start with cost and build up to a selling price**. The percentage mechanics are familiar, but the business question changes: will this price cover the cost, reflect the risk, and still make sense to the customer?

2 · CORE CONCEPTS

Part A — Markup

Markup is the amount added to cost to create a selling price. Anchor & Oak Events may know the ingredient and packaging cost for a tasting box, but that cost is not the price. The price must also cover labor, event preparation, waste, and profit.

A 45% markup on cost means the selling price equals cost plus 45% of cost. In a spreadsheet, this is modeled by placing cost in one cell, markup rate in another, and writing a formula that multiplies cost by one plus the markup rate.

Selling Price = Cost × (1 + Markup Rate)

Markup on Cost vs. Markup on Selling Price

A common source of confusion is **the base**. The same dollar amount of markup produces a different percentage depending on what you divide by.

Measure	Formula	Example (\$18 cost, \$26.10 price)
Markup Amount	Selling Price – Cost	\$26.10 – \$18.00 = \$8.10
Markup on Cost	Markup Amount / Cost	\$8.10 / \$18.00 = 45%
Markup on Selling Price	Markup Amount / Selling Price	\$8.10 / \$26.10 = 31%

Same dollars — different percentage. Whenever you read a business problem, **pause and identify the base** before choosing a formula.



Use this five-step routine throughout the lesson:

1. **Name the business question.** Are you setting a price, reducing a price, accounting for waste, or finding break-even volume?
2. **Identify the base.** Name the amount or unit to which the percentage applies.
3. **Predict the direction.** Decide whether the answer should rise or fall before calculating.
4. **Build the formula.** Keep assumptions in labeled spreadsheet cells instead of typing them inside formulas.
5. **Check the result.** Confirm that the units, direction, and business meaning are reasonable.

Part B — Markdowns and Perishables

Markdowns reduce price after the original price has been set. For an event business, markdowns can help fill late seats or sell perishable items before they lose value. **A markdown is not automatically a failure** — it is a tradeoff: lower margin per sale in exchange for more cash recovery, fuller attendance, or less waste.

Markdown Price = Original Price $\times$ (1 – Markdown Rate)

Waste Changes the Pricing Problem

Perishable products require extra care because not every produced unit may become a sellable unit. If Anchor & Oak produces 120 tasting boxes and expects 15% waste, only 102 boxes are expected to be sold.

Key Insight

The cost of **all 120 boxes** still needs to be recovered from the **102 sellable boxes**. This is why a price that looks profitable before waste can become too low after waste is considered.

Sellable Units = Produced Units $\times$ (1 – Waste Rate)

Part C — Break-Even

Break-even is the point where **total revenue equals total cost**. Before break-even, the business is losing money on the event. After break-even, each additional unit contributes toward profit.

The key idea is **contribution margin**: selling price minus variable cost per unit. Contribution margin is not the same as profit, because fixed costs still need to be covered first.

Contribution Margin = Selling Price – Variable Cost per Unit

Break-Even Units = Fixed Costs ÷ Contribution Margin per Unit

💡 The Rounding Rule

Break-even units must always be **rounded up** to the nearest whole unit. Selling to 171 guests when the break-even is 171.43 would still leave a small loss. Use `=ROUNDUP(result, 0)` in Excel — never round down.

3 · BUILD THE MODEL IN EXCEL

Enter Percentages Correctly

Enter a rate as `45%` or `0.45`, then use **Home** → **Number** → **Percent Style** when needed. Do not type `45` and then apply Percent Style; Excel would display `4500%` because the stored value is forty-five, not forty-five hundredths.

Enter percentage assumptions as percentages

File	Home	Insert	Page Layout	Formulas	Data
------	-------------	--------	-------------	----------	------

Number group

%

Percent Style
Home → Number → %

Correct input
45%
Excel stores this as 0.45
**Do not type 45 and then
apply Percent Style**

Separate Inputs From Formulas

Place each assumption in its own labeled cell. This makes the model easier to audit and lets you test a new markup or waste assumption without rewriting formulas. In the model below, yellow cells are inputs and green cells are calculated outputs.

Build the markup-and-waste model from labeled inputs

fx	=B6/B5	
	A	B
1	Produced Units	120
2	Cost per Unit	\$18.00
3	Markup Rate	45%
4	Waste Rate	15%
5	Sellable Units	=B1*(1-B4)
6	Target Revenue	=B1*B2*(1+B3)
7	Required Price	=B6/B5

Modeling habit

Yellow cells = inputs

Green cells = formulas

Keep each assumption
in its own cell

Before accepting the result, predict its direction. Markup should raise price, markdown should lower price, waste should reduce sellable units, and a higher variable cost should increase break-even attendance.

4 · WORKED EXAMPLES

Worked Example 1 — Markup With Waste

Anchor & Oak plans to produce 120 tasting boxes. Each box costs \$18 to prepare, and the target markup is 45% on cost. The business expects 15% waste.

Step 1 — Target Revenue (assuming all boxes sell):

Prediction: adding markup should make target revenue greater than the original production cost.

- $120 \times \$18.00 \times 1.45 = \$3,132$

Step 2 — Sellable Units:

- $120 \times (1 - 0.15) = 102 \text{ boxes}$

Step 3 — Required Price per Sellable Box:

Prediction: waste leaves fewer sellable boxes, so the required price should rise.

- $\$3,132 \div 102 = \30.71

Notice what happened: the price rose because fewer sellable boxes must recover the same production cost.

Excel patterns: Target Revenue is `=ProducedUnits * CostPerUnit * (1 + MarkupRate)`. Required Price is `=TargetRevenue / SellableUnits`.

Worked Example 2 — Break-Even

Anchor & Oak is considering a tasting event with \$3,600 in fixed costs, a \$48 ticket price, and \$27 variable cost per guest.

Step 1 — Contribution Margin:

Prediction: contribution margin must be less than the \$48 selling price because variable cost is subtracted from it.

- $\$48 - \$27 = \$21$ per guest

Step 2 — Break-Even Guests:

- $\$3,600 \div \$21 = 171.43 \rightarrow$ round up to **172 guests**

Excel pattern: `=ROUNDUP(FixedCost / (TicketPrice - VariableCost), 0)`

Use ROUNDUP because Anchor & Oak cannot sell part of a guest

fx =ROUNDUP(B1/(B2-B3),0)		
	A	B
1	Fixed Costs	\$3,600
2	Ticket Price	\$48
3	Variable Cost per Guest	\$27
4	Contribution Margin	=B2-B3
5	Break-Even Guests	=ROUNDUP(B1/B4,0)

Reasonableness check

$$\$48 - \$27 = \$21 \text{ CM}$$

$$\$3,600 \div \$21 = 171.43$$

ROUND UP $\rightarrow$ 172 guests

Check the result from both directions: 171 guests would produce only \$3,591 of contribution margin, which is not enough to cover \$3,600 of fixed costs. At 172 guests, contribution margin reaches \$3,612, so the event has crossed break-even.

5 • FORMULA REFERENCE

Concept	Formula
Markup Amount	<code>Selling Price - Cost</code>
Markup on Cost	<code>Markup Amount / Cost</code>
Selling Price (markup)	<code>Cost × (1 + Markup Rate)</code>
Markdown Price	<code>Original Price × (1 - Markdown Rate)</code>
Sellable Units	<code>Produced Units × (1 - Waste Rate)</code>
Contribution Margin	<code>Selling Price - Variable Cost per Unit</code>
Break-Even Units	<code>Fixed Costs / Contribution Margin per Unit</code>
Break-Even Revenue	<code>Break-Even Units × Selling Price</code>

Looking Ahead — Target Profit

Break-even covers costs but produces no profit. When a manager sets a target profit, add that target to fixed costs before dividing by contribution margin:

```
=ROUNDUP((FixedCosts + TargetProfit) / ContributionMargin, 0)
```

This is an extension of the same model, not a different break-even formula.

6 • CHECK YOUR UNDERSTANDING

Answer each question before looking at the answer key below. Show your formula setup, not just the final number.

1. A dessert box costs \$20 and is marked up 40% on cost. What is the selling price?
2. A \$52 ticket is marked down 20%. What is the new price?
3. Anchor & Oak produces 160 favors and expects 10% waste. How many are sellable?
4. An event has a \$35 ticket price and \$18 variable cost per guest. What is the contribution margin?
5. Fixed cost is \$2,400 and contribution margin is \$16. How many units are needed to break even?

Answer Key • Check Your Understanding

#	Answer
1	$\$20 \times (1 + 0.40) = \$20 \times 1.40 = \mathbf{\$28.00}$
2	$\$52 \times (1 - 0.20) = \$52 \times 0.80 = \mathbf{\$41.60}$
3	$160 \times (1 - 0.10) = 160 \times 0.90 = \mathbf{144 \text{ favors}}$
4	$\$35 - \$18 = \mathbf{\$17 \text{ per guest}}$
5	$\$2,400 / \$16 = \mathbf{150 \text{ units}}$ (exact — no rounding needed here)

7 · KEY VOCABULARY

Term	Definition
Markup	The amount added to cost to create a selling price.
Markup on Cost	Markup measured as a percentage of cost. Formula: $\text{Markup Amount} / \text{Cost}$.
Markup on Selling Price	Markup measured as a percentage of the selling price. Formula: $\text{Markup Amount} / \text{Selling Price}$.
Markdown	A reduction from an original selling price. Formula: $\text{Original Price} \times (1 - \text{Markdown Rate})$.
Perishable Inventory	Goods or capacity that lose value over time and cannot be held indefinitely for sale.
Waste Rate	The percentage of produced units expected to be unsellable. Reduces the denominator when calculating required price.
Variable Cost	Cost that changes with each additional unit produced or guest served.
Fixed Cost	Cost that does not change with production volume in the short run (e.g., venue rental, permits).
Contribution Margin	Selling price minus variable cost per unit. The amount each unit contributes toward covering fixed costs and profit.
Break-Even Point	The sales volume at which total revenue exactly equals total cost; zero profit, zero loss.



Bring to Class

You will use the starter workbook to test the same pricing logic from this reading. Expect short **Live You Try** It pauses for markup, markdown, and break-even formulas before moving into the **Class Challenge**. The challenge is where you combine the formulas and explain whether Anchor & Oak's pricing decision is financially reasonable.

BUS 123 — MATH-M05-L01 — Payroll and Depreciation

BUS 123 · MATH-M05-L01 · PAYROLL AND DEPRECIATION

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M05 · **Lesson:** L01

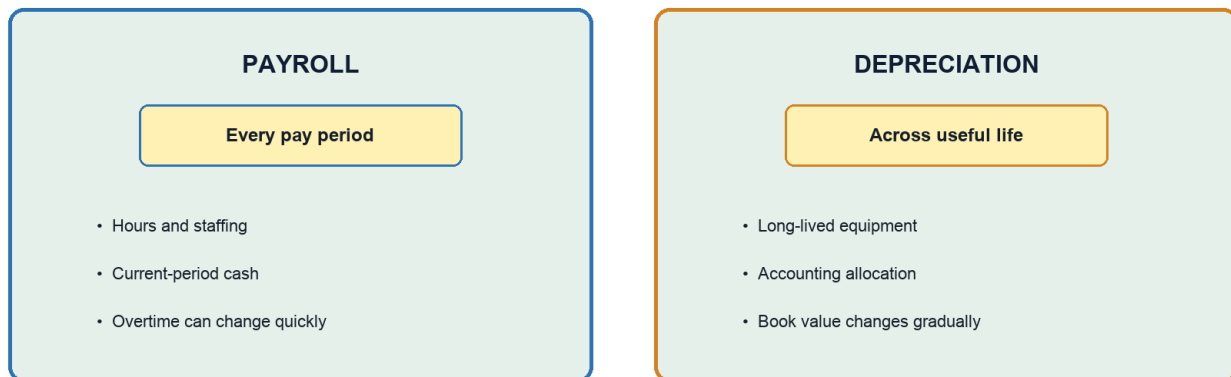
Case Study Company: Harborside Medical Center

1 · CONNECT TO PRIOR KNOWLEDGE

Read this briefing before class so the calculations in the live activity feel like business decisions instead of isolated formulas. Harborside Medical Center is deciding how to manage weekly staffing costs while planning for long-lived medical equipment.

Managers need a clean view of two different cost rhythms: **payroll costs** that repeat every pay period, and **asset costs** that are spread across years. Payroll helps Harborside staff patient care safely. Depreciation helps Harborside recognize that equipment wears out, becomes outdated, or loses value over time.

Payroll and depreciation move on different timelines



Payroll often involves current cash paid to employees. Depreciation is different: it allocates a prior asset purchase across accounting periods and usually does not represent new cash leaving the bank in the year it is recorded.

2 · CORE CONCEPTS

Part A — Payroll

Key Payroll Vocabulary

Term	Meaning for Harborside Medical	Basic Formula
Hourly pay	Employees paid for hours worked, often with overtime eligibility	$\text{Hours} \times \text{Rate}$
Overtime pay	Extra pay for hours beyond the regular threshold (commonly 40 hrs/week)	$\text{OT Hours} \times \text{Rate} \times 1.5$
Salary	Fixed annual pay converted into periodic checks	$\frac{\text{Annual Salary}}{\text{Pay Periods}}$
Commission	Pay based on a percentage of sales, referrals, or measurable revenue	$\text{Sales} \times \text{Commission Rate}$
Gross pay	Total earned before deductions	$\text{Regular} + \text{OT} + \text{Commission}$

Payroll Cycles

The pay cycle changes the size and timing of a paycheck — not the annual salary. A \$52,000 salary can be paid as:

Cycle	Periods/Year	Paycheck Amount
Weekly	52	\$1,000.00
Biweekly	26	\$2,000.00
Semimonthly	24	\$2,166.67
Monthly	12	\$4,333.33

Formula: $\text{Salary per Period} = \frac{\text{Annual Salary}}{\text{Number of Pay Periods}}$

In Excel, place the annual salary and number of pay periods in separate labeled cells, then use `=AnnualSalary/PayPeriods`.

Pay frequency changes check size, not annual salary

fx =AnnualSalary/PayPeriods			
Pay Cycle	Annual Salary	Periods	Pay per Period
Weekly	\$52,000	52	= \$1,000.00
Biweekly	\$52,000	26	= \$2,000.00
Semimonthly	\$52,000	24	= \$2,166.67
Monthly	\$52,000	12	= \$4,333.33

Do not interchange

Biweekly = 26

Semimonthly = 24

Both total \$52,000

Biweekly and semimonthly are not interchangeable. Biweekly payroll normally has 26 periods per year; semimonthly payroll has 24. Both still distribute the same annual salary.

Overtime Example — Harborside Medical Center

A Harborside nurse earns \$44/hour and works 43 hours in one week. The first 40 hours are paid at the regular rate; the 3 overtime hours are paid at time-and-a-half.

- Regular pay: $40 \times \$44 = \$1,760$
- Overtime pay: $3 \times \$44 \times 1.5 = \198
- Gross pay: **\$1,958**

Build Overtime Safely in Excel

Use formulas that handle employees both above and below the overtime threshold:

- Regular pay: `=MIN(Hours,40)*Rate`
- Overtime pay: `=MAX(Hours-40,0)*Rate*1.5`
- Gross pay: `=RegularPay+OvertimePay`

`MIN` caps regular hours at 40. `MAX` prevents a negative overtime result when an employee works fewer than 40 hours. Gross pay is earnings before deductions; it is not take-home pay.

Use MIN and MAX to separate regular and overtime pay

fx `=MIN(B2,40)*C2 + MAX(B2-40,0)*C2*D2`

Employee	Hours	Rate	OT Mult.	Gross Pay
Harborside Nurse	43	\$44	1.5	= \$1,958

Regular pay

`=MIN(Hours,40)*Rate`

MIN caps regular hours at 40

Overtime pay

`=MAX(Hours-40,0)*Rate*1.5`

MAX prevents negative overtime

For the eight-nurse team used in class, one nurse earns \$1,958, so total weekly gross pay is 8×1958 , or **\$15,664**. Keep the employee count in its own input cell so a manager can test a different staffing level without rewriting the pay formula.



Manager Question

If many employees are working overtime every week, is Harborside solving a temporary staffing issue — or hiding a permanent headcount problem?

Pay System Comparison

Pay System	Best Fit	Risk to Watch
Hourly	Patient-care shifts, front desk, variable schedules	Overtime can rise quickly
Salary	Managers, billing leaders, ongoing responsibility	Workload may become invisible
Piece rate / per visit	Task-based work, contract coverage, repeatable services	Volume may be rewarded over quality
Commission	Outreach, referrals, optional service growth	Revenue incentive may need guardrails
Salary + commission	Stable role with a growth target	Formula must be transparent

Part B — Depreciation

Key Depreciation Vocabulary

Term	Definition
Asset cost	The purchase price plus costs needed to get the asset ready for use.
Residual value	Expected value at the end of useful life.
Useful life	How long the company expects to use the asset.
Accumulated depreciation	Total depreciation recorded so far.
Book value	Asset cost minus accumulated depreciation.

Depreciation is an accounting estimate of the cost of a long-lived asset used up during a period. It is **not** the same as cash leaving the bank this year — it helps Harborside match the cost of equipment to the years that benefit from that equipment.

Straight-Line Depreciation

Straight-line depreciation spreads the depreciable cost evenly across useful life. Managers like it because it is predictable, easy to explain, easy to audit, and useful when an asset provides steady service.

$$\text{Annual Depreciation} = (\text{Asset Cost} - \text{Residual Value}) / \text{Useful Life}$$

Worked Example — Harborside Medical Center

Harborside buys a diagnostic ultrasound unit for \$84,000. Expected residual value: \$12,000 after 6 years.

- Annual depreciation: $(\$84,000 - \$12,000) / 6 = \$12,000/\text{year}$

Year	Cost	Depreciation Expense	Accumulated Depreciation	Book Value
1	\$84,000	\$12,000	\$12,000	\$72,000
2	\$84,000	\$12,000	\$24,000	\$60,000
3	\$84,000	\$12,000	\$36,000	\$48,000
4	\$84,000	\$12,000	\$48,000	\$36,000
5	\$84,000	\$12,000	\$60,000	\$24,000
6	\$84,000	\$12,000	\$72,000	\$12,000

Build the Depreciation Schedule in Excel

Use three connected formulas:

- Annual depreciation: `=(Cost-ResidualValue)/UsefulLife`
- Accumulated depreciation: `=AnnualDepreciation*Year`
- Book value: `=Cost-AccumulatedDepreciation`

For Year 3, the formulas return \$12,000 annual depreciation, \$36,000 accumulated depreciation, and \$48,000 book value.

Build Year 3 depreciation as a three-formula sequence

fx =AssetCost-AccumulatedDepreciation		
Input or Output	Value	Excel Formula
Asset Cost	\$84,000	input
Residual Value	\$12,000	input
Useful Life	6 years	input
Year	3	input
Annual Depreciation	\$12,000	<code>=(B1-B2)/B3</code>
Accumulated Depreciation	\$36,000	<code>=B5*B4</code>
Book Value	\$48,000	<code>=B1-B6</code>

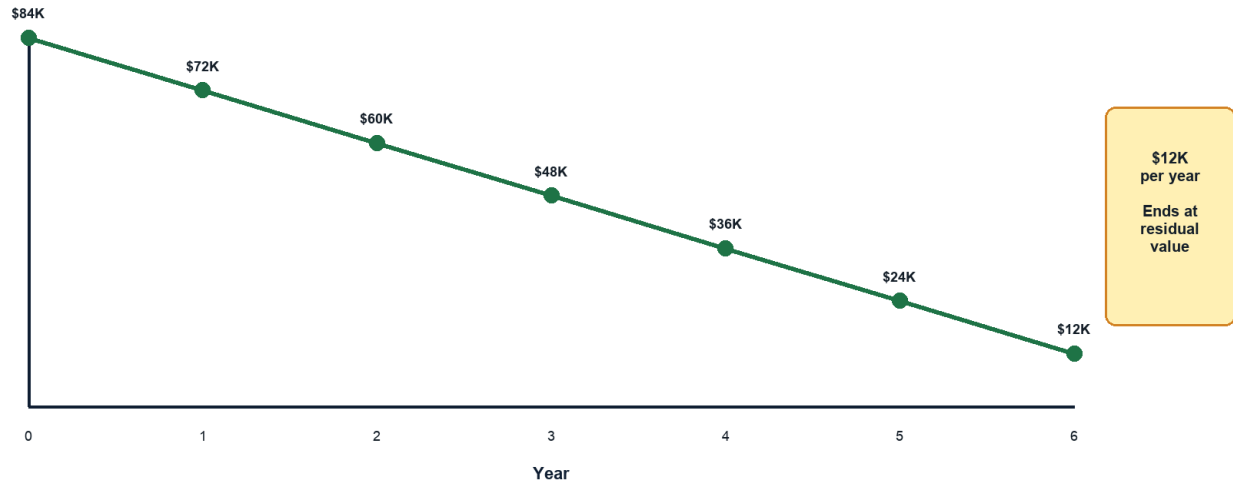
Ending check

After Year 6:

Book value = \$12,000

Residual value, not \$0

Straight-line book value falls evenly to residual value



Straight-line book value falls evenly and ends at residual value, not zero. Book value is an accounting amount and may differ from the equipment's current market price.

3 · PREDICT BEFORE YOU CALCULATE

Use direction checks to catch formulas that return a number but do not make business sense:

1. If hours rise above 40, total payroll should rise faster because overtime hours receive premium pay.
2. If the hourly rate rises, both regular and overtime pay should rise.
3. If useful life increases while cost and residual value stay fixed, annual depreciation should fall.
4. If residual value increases while cost and useful life stay fixed, annual depreciation should fall.
5. At the end of useful life, book value should equal residual value rather than zero.

4 · CHECK YOUR UNDERSTANDING

Answer these questions before class. Be ready to discuss your reasoning — not just the number.

1. Which payroll type gives Harborside the most flexibility when patient volume changes week to week?
2. Why does overtime make total payroll rise faster than regular hours?
3. If an employee earns a fixed annual salary, why do weekly and monthly paychecks have different dollar amounts?
4. What is the difference between depreciation expense and accumulated depreciation?
5. Why might straight-line depreciation be a good fit for a medical device used steadily throughout the year?
6. What Excel formula calculates weekly pay for a \$52,000 annual salary?
7. Using Excel logic, calculate regular pay, overtime pay, and gross pay for 43 hours at \$44 per hour.
8. What is the total weekly gross pay for 8 nurses working those same hours?
9. For the ultrasound unit, calculate Year 3 accumulated depreciation and book value.
10. A six-year depreciation schedule ends at \$0. What assumption or formula was probably omitted?

Self-Check

#	Expected result
6	$=52000/52$ gives \$1,000 per week .
7	Regular pay is \$1,760 , overtime pay is \$198 , and gross pay is \$1,958 .
8	$=8*1958$ gives \$15,664 total weekly gross pay.
9	Accumulated depreciation is \$36,000 and book value is \$48,000 .
10	Residual value was probably omitted; ending book value should be \$12,000 .

5 • KEY VOCABULARY

Term	Definition
Gross Pay	Total earnings before any deductions; includes regular pay, overtime, and commission.
Overtime Pay	Compensation for hours worked beyond the standard threshold (typically 40 hrs/week), calculated at 1.5× the regular hourly rate.
Pay Period	The recurring interval at which employees are paid: weekly, biweekly, semimonthly, or monthly.
Depreciation	An accounting allocation of an asset's cost over its useful life; not a cash outflow in the period it is recorded.
Straight-Line Depreciation	A depreciation method that spreads the depreciable cost evenly across all years of useful life.
Residual Value	The estimated value of an asset at the end of its useful life; subtracted from cost before computing depreciation.
Book Value	The remaining recorded value of an asset: Asset Cost – Accumulated Depreciation.
Accumulated Depreciation	The running total of all depreciation expense recorded on an asset since it was placed in service.

Bring to Class

Be ready to use the starter workbook as a payroll and depreciation management system. The **Live You Try It** pauses will use the same salary, overtime, team payroll, and ultrasound depreciation numbers from the slides so you can self-check each formula before moving on.

The **Class Challenge** asks you to combine payroll cost, staffing pressure, depreciation, and business interpretation. Your goal is not only to calculate the new payroll or asset amount, but also to explain whether the decision is **operationally reasonable** for a medical practice.

BUS 123 — MATH-M06-L01 — Inventory, Overhead & Cost of Goods Sold

BUS 123 · MATH-M06-L01 · INVENTORY, OVERHEAD & COST OF GOODS SOLD

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M06 · **Lesson:** L01

Case Study Company: Tidal Goods Co.

1 · CONNECT TO PRIOR KNOWLEDGE

Earlier in the math track, you learned how retailers use markup to set selling prices from a known cost — how a 50% markup on a \$40 item produces a \$60 retail price. But markup math assumes you already know the cost of that item with precision.

This module goes one level deeper: **what exactly is the cost of the item you just sold?** When a business carries hundreds of units bought at different prices across different weeks, the answer is not obvious. The method you choose — FIFO, LIFO, or Weighted Average Cost — changes reported COGS, profit, and the asset value on the balance sheet. It can also affect taxable income and taxes under the applicable reporting and tax rules.

Layered on top of product cost are **overhead expenses**: rent, utilities, insurance, and indirect labor that keep the business running but cannot be traced to any single unit sold. Together, these elements feed the **Cost of Goods Sold (COGS)** calculation — the single most important cost line on a product business's income statement.

2 · CORE CONCEPTS

Part 1 — Inventory Valuation

Why Valuation Matters

Tidal Goods Co. is a coastal outdoor retailer selling paddleboards, surf gear, and accessories. Over a single quarter, Tidal places three purchase orders for the same paddleboard model as supplier prices shift due to tariff pressures and shipping costs:

Purchase Order	Unit Cost	Quantity
PO-101 (Feb)	\$210/unit	10 units
PO-102 (Mar)	\$230/unit	10 units
PO-103 (Apr)	\$250/unit	10 units

Tidal sells 20 paddleboards during the quarter. The question every accountant must answer: which 20 units' costs flow to the income statement, and which 10 remain on the balance sheet? This is a **formal accounting policy decision** with real financial consequences.

The Four Valuation Methods

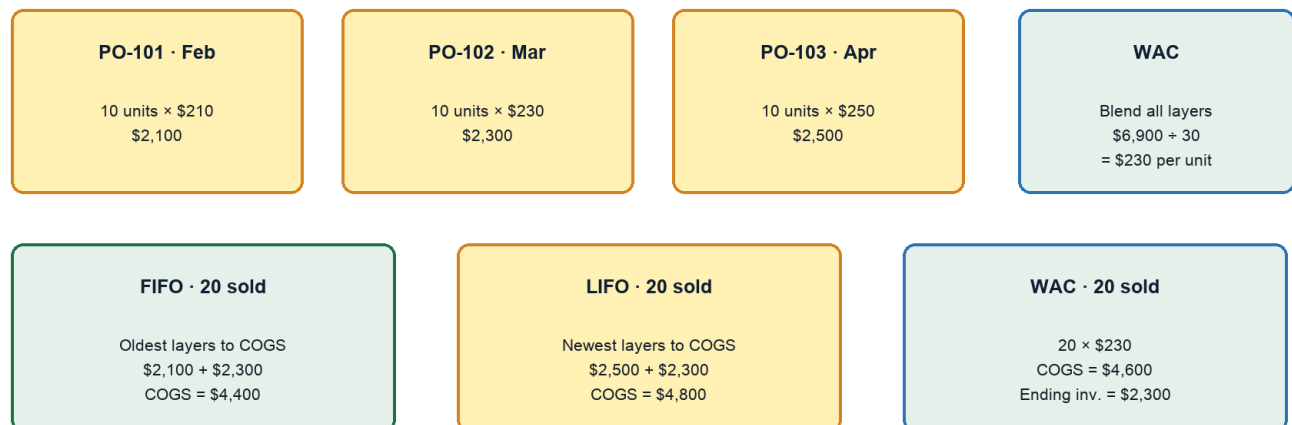
Specific Identification traces each physical unit to its exact purchase cost. This works for unique, high-value items — custom boats, fine jewelry, artwork — but is impractical for any business selling fungible goods in volume.

First-In, First-Out (FIFO) assumes the oldest inventory is sold first. Tidal's oldest boards (at \$210) flow to COGS before the newer, more expensive ones. During inflation, FIFO produces **lower COGS** and **higher gross profit**. Ending inventory reflects the most recent, higher prices.

Last-In, First-Out (LIFO) assumes the newest inventory is sold first. The most recently purchased boards (\$250) flow to COGS first. During inflation, LIFO produces **higher COGS** and **lower taxable income** — a cash flow advantage. However, LIFO is **prohibited under IFRS**, making it unavailable to companies that report under international rules.

Weighted Average Cost (WAC) calculates a single blended unit cost by dividing total cost available by total units available. It smooths out price fluctuations and is popular for commodity-type goods.

The inventory method decides which cost layers flow to COGS



✓ WAC Unit Cost Formula

WAC Unit Cost = Total Cost of Goods Available / Total Units Available

Tidal: $(\$2,100 + \$2,300 + \$2,500) / 30 \text{ units} = \mathbf{\$230.00/unit}$

COGS (WAC): $20 \text{ units} \times \$230.00 = \mathbf{\$4,600}$

The Numbers Side by Side

Method	COGS	Ending Inventory
FIFO	$10 \times \$210 + 10 \times \$230 = \$4,400$	\$2,500 (10 units $\times$ \$250)
LIFO	$10 \times \$250 + 10 \times \$230 = \$4,800$	\$2,100 (10 units $\times$ \$210)
WAC	$20 \times \$230.00 = \$4,600$	\$2,300 (10 units $\times$ \$230)

The \$400 spread between FIFO and LIFO COGS is not rounding — it is a direct result of the accounting policy chosen. Both methods describe the same 20 boards sold from the same physical stockroom. This is why the choice of inventory method is a **strategic management decision**, not just a bookkeeping detail.

Build the Inventory Methods in Excel

Enter each purchase layer's quantity and unit cost in separate labeled cells. Calculate each layer's cost with `=Quantity*UnitCost`. Then use:

- WAC unit cost: `=SUM(LayerCostCells)/SUM(QuantityCells)`
- WAC COGS: `=UnitsSold*WACUnitCost`
- FIFO COGS for 20 boards: `=10*210+10*230`

Build inventory methods from labeled Excel inputs

fx <code>=SUM(D6:D8)/SUM(B6:B8)</code>				WAC Unit Cost <code>=SUM(LayerCost)</code> <code>+ SUM(Quantity)</code> <code>= \$230</code>	
Purchase Layer	Quantity	Unit Cost	Layer Cost		
PO-101	10	\$210	<code>=B6*C6</code>		
PO-102	10	\$230	<code>=B7*C7</code>		
PO-103	10	\$250	<code>=B8*C8</code>		
FIFO COGS <code>=10*210+10*230</code> \$4,400 · oldest layers first					
WAC COGS <code>=UnitsSold*WACUnitCost</code> \$4,600 · $20 \times \$230$					

Before accepting the result, confirm that units assigned to COGS plus units in ending inventory equal total units available.

LIFO and international reporting: LIFO is permitted under U.S. GAAP but prohibited under IFRS. Changing reporting frameworks may require substantial accounting-policy and comparative-reporting work; exact tax and reporting consequences depend on the rules that apply to the company.

Part 2 — Cost of Goods Sold (COGS)

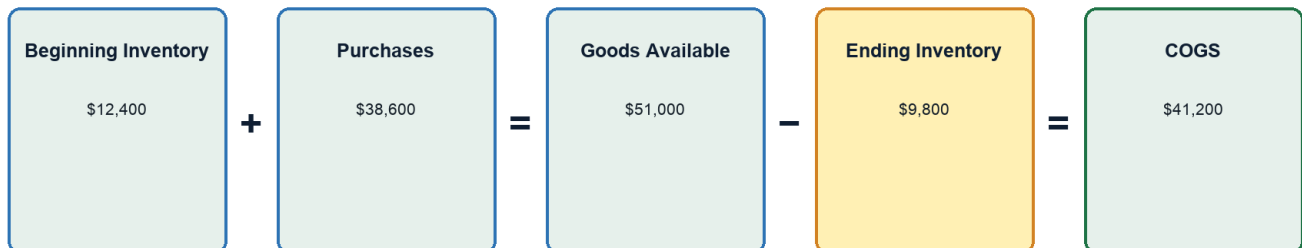
The Core Formula

COGS is the total direct cost of the products a business sold during a period. For a retailer like Tidal, this means the purchase cost of every item that left the store or shipped to a customer.

Beginning Inventory + Purchases – Ending Inventory = COGS

In plain English: start with what you had, add what you bought, subtract what is left, and the result is what you sold.

Ending inventory is the bridge from goods available to COGS



Check: COGS cannot exceed goods available for sale

Tidal Goods Co. — Q1 2026

Line Item	Amount
Beginning Inventory	\$12,400
+ Purchases	\$38,600
= Goods Available	\$51,000
– Ending Inventory	(\$9,800)
= COGS	\$41,200
Revenue	\$110,000
= Gross Profit	\$68,800
Gross Margin %	62.5%

Why Ending Inventory Accuracy Is Critical

Notice that ending inventory **directly reduces COGS**. If the physical count at period end is wrong — even by a small amount — COGS is wrong, which means gross profit is wrong, which means net income is wrong. Worse, that error does not disappear: the ending inventory of this period becomes the beginning inventory of the next period, carrying the error forward.

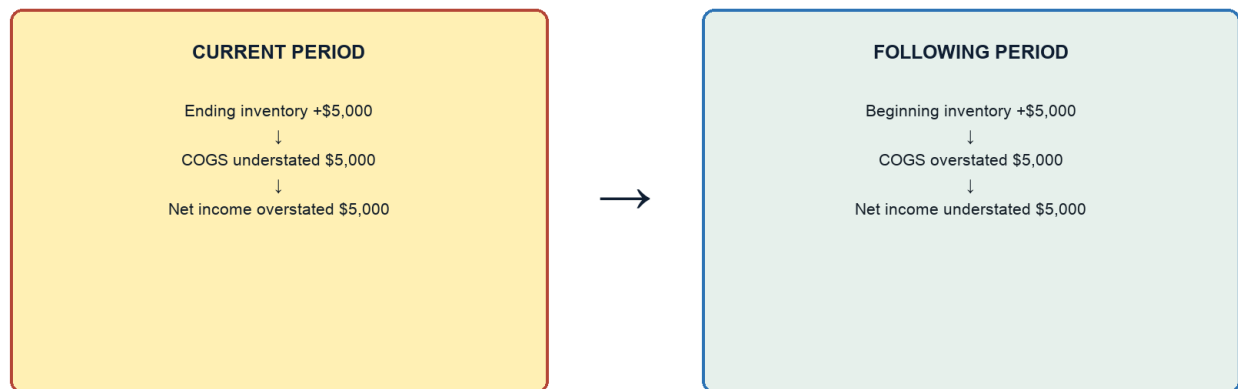
⚠ Common Mistake — Overstated Ending Inventory

If Tidal's count overstates ending inventory by \$5,000:

- **Period 1:** COGS understated by \$5,000 → net income **OVERSTATED** by \$5,000
- **Period 2:** Beginning inventory now too high → COGS overstated → net income **UNDERSTATED**

The errors cancel over two periods — but both periods reported false numbers.

One overstated count creates opposite errors across two periods



Part 3 — Margin vs. Markup

Both margin and markup measure profitability from the same transaction. They use the same inputs — revenue and cost — but **divide by different denominators**. Confusing them is one of the most common and costly errors in retail pricing.

Measure	Formula	Tidal Example (\$80 price, \$48 cost)	Result	Used By
Gross Margin	$(\text{Revenue} - \text{Cost}) / \text{Revenue}$	$(\$80 - \$48) / \$80$	40.0%	Finance, investors
Markup	$(\text{Revenue} - \text{Cost}) / \text{Cost}$	$(\$80 - \$48) / \$48$	66.7%	Buyers, sales teams

A buyer who says "we mark up everything 100%" means the selling price is double the cost. That sounds like 100% profit — but the gross margin is only **50%**, because margin divides by the larger selling price. A pricing team that confuses the two will systematically underprice every product in the catalog.

Margin and markup use the same profit but different bases



In Excel, use $\text{=(Revenue-Cost)/Revenue}$ for margin and $\text{=(Revenue-Cost)/Cost}$ for markup. Format both results as percentages and verify that the denominator matches the business question.

Quick Check — Converting Between Them

- Given markup of 66.7%: $\text{Margin} = \text{Markup} / (1 + \text{Markup}) = 0.667 / 1.667 = \mathbf{40\%}$
- Given margin of 40%: $\text{Markup} = \text{Margin} / (1 - \text{Margin}) = 0.40 / 0.60 = \mathbf{66.7\%}$

Part 4 — Overhead Allocation

What Is Overhead?

Overhead costs are indirect, recurring expenses required to keep the business operational that **cannot be traced to any single unit sold**.

Tidal's monthly overhead:

Category	Monthly Amount
Rent	\$9,000
Utilities	\$4,200
Insurance	\$3,800
Indirect labor	\$3,500
Maintenance	\$1,500
Total	\$22,000

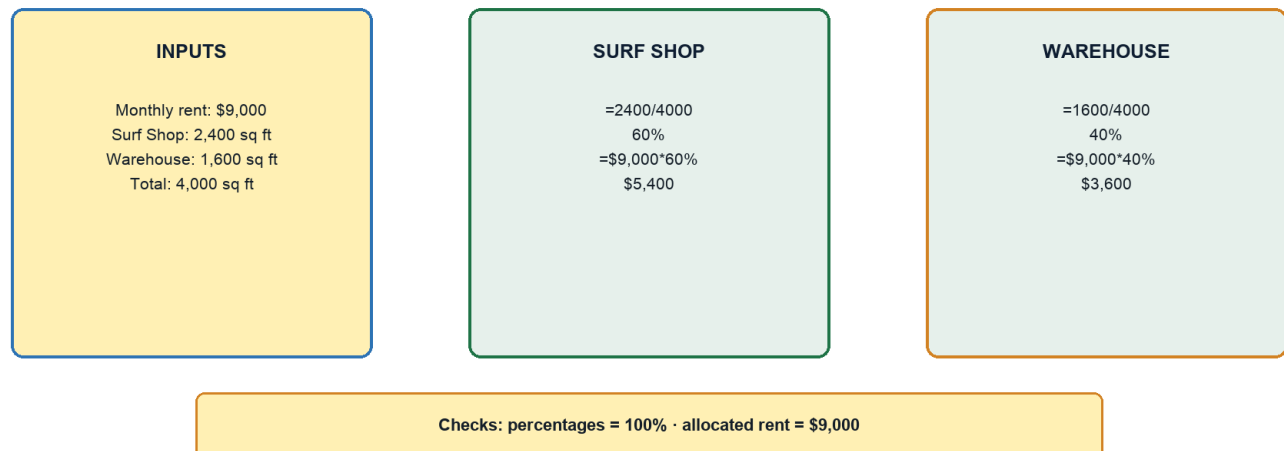
Whether Tidal sells one board or one thousand, these costs exist.

Allocating Across Departments

Tidal operates two departments: a retail Surf Shop (2,400 sq ft) and an Online Warehouse (1,600 sq ft). Most overhead categories are allocated by **square footage**, because those costs increase with physical space.

Department	Sq Ft	Allocation Basis	Overhead Allocated
Surf Shop	$2,400 / 4,000 = 60\%$	Sq footage	$60\% \times \$9,000 = \mathbf{\$5,400}$
Warehouse	$1,600 / 4,000 = 40\%$	Sq footage	$40\% \times \$9,000 = \mathbf{\$3,600}$
Total	100%		\$9,000

Allocate overhead using a driver that explains the cost



Always verify that department percentages total **100%** and allocated amounts total the original cost pool. Here, allocated rent must total **\$9,000**. A complete-looking allocation that does not reconcile is not ready for a management report.

The Overhead Rate

$$\text{Overhead Rate} = \text{Total Overhead} / \text{Revenue} = \$22,000 / \$110,000 = 20.0\%$$

A 20% overhead rate means that for every dollar of revenue Tidal earns, **20 cents goes to keeping the lights on** before COGS is even considered. When this rate rises unexpectedly, management investigates: Did rent increase? Did revenue drop while fixed costs stayed flat?

✓ Choosing the Right Allocation Basis

- **Square footage** — works well for space-based costs (rent, utilities, insurance)
- **Revenue share** — works better for costs driven by sales volume (commissions, processing fees)
- **Direct labor hours** — works best for labor-intensive production overhead

The key: the basis should reflect **what actually drives the cost**.

3 · PREDICT BEFORE YOU CALCULATE

Use these direction checks before trusting an Excel result:

1. During rising prices, FIFO should produce lower COGS than LIFO because older costs are lower.
2. If ending inventory rises while beginning inventory and purchases stay fixed, COGS should fall.
3. If revenue falls while overhead remains fixed, the overhead rate should rise.
4. For the same profitable sale, markup percentage should be larger than margin percentage because cost is the smaller denominator.
5. Department allocation percentages must total 100%, and allocated dollars must total the original overhead.

4 • FORMULA REFERENCE

Formula	Expression	Tidal Goods Example
COGS	=BeginningInventory+Purchases-EndingInventory	\$12,400 + \$38,600 - \$9,800 = \$41,200
Goods Available	=BeginningInventory+Purchases	\$12,400 + \$38,600 = \$51,000
Gross Profit	=Revenue-COGS	\$110,000 - \$41,200 = \$68,800
Gross Margin %	=GrossProfit/Revenue	\$68,800 / \$110,000 = 62.5%
Markup %	=(Revenue-Cost)/Cost	(\$80 - \$48) / \$48 = 66.7%
WAC Unit Cost	=TotalCostAvailable/TotalUnitsAvailable	\$6,900 / 30 = \$230.00
FIFO COGS	Oldest units × oldest cost	10×\$210 + 10×\$230 = \$4,400
Dept OH %	Dept Sq Ft / Total Sq Ft	2,400 / 4,000 = 60%
OH Allocated	=TotalOverhead*DeptPercent	\$9,000 × 60% = \$5,400
OH Rate	=TotalOverhead/Revenue	\$22,000 / \$110,000 = 20.0%

5 • CHECK YOUR UNDERSTANDING

Answer all seven questions before class. Answers appear at the end of this document.

1. Tidal Goods Co. had beginning inventory of \$14,000, made purchases of \$32,000 during the quarter, and ended with inventory of \$11,500. Calculate COGS.
 1. Using the paddleboard purchase orders in Part 1 (10 units at \$210, 10 at \$230, 10 at \$250), calculate the COGS if Tidal sells 15 units using the **FIFO** method.
 1. Using the same purchase orders, calculate the Weighted Average Cost per unit and the COGS for selling 15 units under **WAC**.
 1. Tidal sells a wetsuit for \$120. The landed cost is \$72. Calculate both the **gross margin percentage** and the **markup percentage**.
 1. Tidal's overhead for the month is \$18,000. Revenue is \$90,000. Calculate the overhead rate. If revenue drops to \$75,000 next month but overhead stays flat, what is the new overhead rate?
 1. Tidal's Surf Shop occupies 3,000 sq ft and the Warehouse occupies 2,000 sq ft. Monthly rent is \$10,000. How much rent should be allocated to each department?
 1. Tidal's ending inventory count is accidentally overstated by \$3,000 this quarter. Explain what happens to COGS and net income this period, and what happens in the following period as a result.
-

Answer Key · Check Your Understanding

#	Answer
1	COGS = \$14,000 + \$32,000 - \$11,500 = \$34,500
2	FIFO — sell oldest first: 10 units × \$210 + 5 units × \$230 = \$2,100 + \$1,150 = \$3,250
3	WAC unit cost = (\$2,100 + \$2,300 + \$2,500) / 30 = \$230.00 . COGS = 15 × \$230.00 = \$3,450
4	Gross Margin = (\$120 - \$72) / \$120 = 40.0% . Markup = (\$120 - \$72) / \$72 = 66.7%
5	Current: \$18,000 / \$90,000 = 20.0% . If revenue drops: \$18,000 / \$75,000 = 24.0% — same overhead, less revenue to absorb it.
6	Surf Shop: 3,000 / 5,000 = 60% × \$10,000 = \$6,000 . Warehouse: 2,000 / 5,000 = 40% × \$10,000 = \$4,000 .
7	This period: ending inventory overstated by \$3,000 → COGS understated → net income overstated by \$3,000. Next period: beginning inventory too high → COGS overstated → net income understated by \$3,000. The errors offset over two periods, but both periods reported incorrect figures.

6 · KEY VOCABULARY

Term	Definition
Cost of Goods Sold (COGS)	The direct cost of products sold during a period. Calculated as: Beginning Inventory + Purchases - Ending Inventory.
FIFO	First-In, First-Out. Inventory method that assumes the oldest inventory is sold first. Produces lower COGS and higher profit during inflationary periods.
LIFO	Last-In, First-Out. Inventory method that assumes the newest inventory is sold first. Produces higher COGS and lower taxable income. Prohibited under IFRS.
Weighted Average Cost (WAC)	Inventory method that calculates a single blended unit cost by dividing total cost available by total units available.
Gross Margin	Profit expressed as a percentage of revenue: $(\text{Revenue} - \text{Cost}) / \text{Revenue}$. Used by finance teams and investors.
Markup	Profit expressed as a percentage of cost: $(\text{Revenue} - \text{Cost}) / \text{Cost}$. Used by purchasing and sales teams to set prices from known costs.
Overhead	Indirect, recurring costs that keep the business operational but cannot be traced to any single unit sold.
Overhead Allocation	The process of distributing shared overhead costs across departments using a practical basis such as square footage or revenue share.
Overhead Rate	Total overhead costs as a percentage of revenue. A rising rate signals cost control issues or falling revenue.
Goods Available for Sale	Beginning Inventory + Purchases. The total inventory that could have been sold; COGS cannot exceed this figure.



Bring to Class

Be ready to turn Tidal Goods Co.'s raw transaction information into a management report. The class activity will ask you to build a COGS schedule, compare inventory methods, calculate margin versus markup, and allocate overhead in a way a manager could explain.

BUS 123 — MATH-M07-L01 — Sales, Excise & Property Taxes

BUS 123 · MATH-M07-L01 · SALES, EXCISE & PROPERTY TAXES

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M07 · **Lesson:** L01

Case Study Companies: Tidal Goods Co. · Meridian Advisory Group

1 · CONNECT TO PRIOR KNOWLEDGE

In the previous module, you learned how businesses set selling prices — applying trade discounts to reduce the retail price before quoting a buyer, and marking up costs to establish a profitable price point. You built those calculations in Excel using cell references so that any change in an input automatically updated the entire chain.

This lesson adds a critical layer: **government taxes**. Once a price is set, taxes are applied at specific points in a defined order — and getting that order wrong produces incorrect totals, incorrect revenue reporting, and potential legal liability. Today we examine three distinct tax types — sales, excise, and property — each of which places a different obligation on the business.

2 · CORE CONCEPTS

Part 1 — Sales Tax

What Is Sales Tax?

Sales tax is a broad-based consumption tax levied by state governments on the retail sale of goods and certain services. Every state sets its own rate; some municipalities add a local surcharge on top. As of 2024, state rates range from **0%** (Oregon, Montana, New Hampshire, Delaware, Alaska) to **over 7%** before local additions.

The Merchant's Role

When a business collects sales tax from a customer, those funds are **NOT revenue**. They are a **liability held in trust** for the state government. The business is acting as a collection agent. Misclassifying collected tax as earned revenue is a serious accounting error.

The Order of Operations

The calculation sequence is strictly defined. Trade discounts reduce the taxable base — they **must** come before tax is calculated. Shipping charges are added **after** tax has been computed. Reversing these steps produces an incorrect tax figure.

Step	Operation	Formula
1	Subtract trade discount from retail price	$\text{Taxable} = \text{Price} - \text{Discount}$
2	Apply tax rate to taxable subtotal	$\text{Tax} = \text{Taxable} \times \text{Rate}$
3	Add shipping to get total price	$\text{Total} = \text{Taxable} + \text{Tax} + \text{Shipping}$

Worked Example — Tidal Goods Co.

Tidal Goods Co. sells a bulk supply order to a wholesale partner. Retail price: \$480.00. Trade discount: \$80.00. Sales tax: 6%. Flat shipping: \$15.00.

Item	Amount
Retail Price	\$480.00
Less: Trade Discount	-\$80.00
Taxable Subtotal	\$400.00
× Sales Tax Rate (6%)	× 0.06
Tax Owed	\$24.00
Taxable Subtotal + Tax	\$424.00
+ Shipping	+\$15.00
Total Price	\$439.00

Back-Calculating Actual Sales

When a retailer's register total includes both sales revenue and collected tax, the actual pre-tax revenue must be isolated:

$$\text{Actual Sales} = \text{Total Receipts} / (1 + \text{Tax Rate})$$

Example: Tidal Goods Co.'s register shows \$52,000 in total daily receipts at a 6% tax rate.

- Actual sales = $\$52,000 \div 1.06 = \mathbf{\$49,056.60}$
- The remaining **\$2,943.40** belongs to the state.

Part 2 — Excise Tax

What Is Excise Tax?

Excise tax is a targeted, selective tax applied to specific categories of goods considered luxuries or nonessentials — fuel, alcohol, tobacco, certain appliances, firearms, and air travel. Unlike sales tax, which is collected at the retail register, excise tax is typically **paid upstream by manufacturers or wholesalers** and is already embedded in the sticker price by the time it reaches a consumer.

Strategic Impact

Because excise tax raises the cost of acquiring inventory, it directly compresses gross margin. Managers must evaluate **demand elasticity** — whether the market will absorb a price increase — before passing the cost to customers.

When Both Taxes Apply — Never Compound Them

When both sales tax and excise tax apply to the same item, each is calculated **independently from the original base cost**. The rates are not compounded (do not apply the second tax to the already-taxed amount).

Worked Example — Meridian Advisory Group

Meridian Advisory Group purchases a commercial appliance for \$6,200. Sales tax: 7%. Excise tax: 8%.

Tax Type	Calculation	Amount
Base Cost	Given	\$6,200.00
Sales Tax (7%)	$\$6,200 \times 0.07$	\$434.00
Excise Tax (8%)	$\$6,200 \times 0.08$	\$496.00
Total Cost	$\$6,200 + \$434 + \$496$	\$7,130.00

✔ **Shortcut**

Add the two rates first ($7\% + 8\% = 15\%$), then multiply: $\$6,200 \times 1.15 = \mathbf{\$7,130}$. This works because both rates apply to the same base — they can be combined before multiplying.

Part 3 — Property Tax

What Is Property Tax?

Property tax is collected by local municipalities to fund community services: schools, roads, fire and police protection, public libraries, and infrastructure. It applies to both **real property** (land and buildings) and **business personal property** (movable assets such as computers, furniture, and machinery).

Municipalities do not tax property at its full market value. Instead, they apply an **assessment rate** to calculate a lower **assessed value**, which is the taxable base.

⚠ **Triple Net Lease (NNN) Warning**

In a NNN lease, the **tenant** — not the landlord — pays property taxes, building insurance, and maintenance. This makes the tenant's effective rent variable, tied directly to the municipality's annual budget. A town that approves a larger budget raises every NNN tenant's costs automatically.

The Three-Step Calculation

Step	What You Calculate	Formula
1	Assessed Value	$\text{Assessment Rate} \times \text{Market Value}$
2	Tax Rate	$\text{ROUND}(\text{Town Budget} / \text{Total Assessed Value}, 4)$
3	Annual Tax Due	$\text{Tax Rate} \times \text{Assessed Value}$

Worked Example — Meridian Advisory Group

Given: Market Value \$820,000 · Assessment Rate 40% · Town Budget \$310,000 · Total Assessed Value in Town \$4,750,000

Step	Calculation	Result
1 — Assessed Value	$0.40 \times \$820,000$	\$328,000.00
2 — Tax Rate (raw)	$\$310,000 / \$4,750,000$	0.0652631...
2 — Tax Rate (rounded)	<code>=ROUND(rate, 4)</code>	0.0653
3 — Annual Tax Due	$0.0653 \times \$328,000$	\$21,418.40

3 · BUILD THE TAX MODELS IN EXCEL

Excel should mirror the order of the business calculation. Keep every assumption in its own labeled input cell, then build formulas that reference those cells. In the examples below, **yellow cells are inputs** and **green cells are formulas**.

Windows and Mac Excel

The images show Windows Excel. Mac Excel uses the same commands and formula syntax, although a button may appear in a slightly different position on the ribbon.

Format Dollar Amounts and Rates

Use **Home** → **Number** to format prices, taxes, and totals as Accounting or Currency. Format tax and assessment rates as percentages. Enter `6%` or `0.06`—do not enter `6` and then click Percent Style, because Excel would display `600%`.

Format tax models from the Home tab

File
Home
Insert
Page Layout
Formulas
Data

Number group

\$

Accounting

Dollar inputs and outputs

%

Percent Style

Tax and assessment rates

.00

Decimals

Cents or four rate decimals

Before formatting

Enter 6% or 0.06

Do not enter 6 and then click Percent Style

That would display 600%

Walkthrough 1 — Build the Sales Tax Formula Chain

Set up this model in a blank area of Excel. The cell addresses below are an example; the important requirement is that your formulas reference the cells containing the inputs.

- Set up:** In column A, enter the labels Retail Price, Trade Discount, Sales Tax Rate, Shipping, Taxable Subtotal, Sales Tax Owed, and Total Price. In cells B1:B4, enter `$480`, `$80`, `6%`, and `$15`.

2. **Select and type:** Select B5 and type $=B1-B2$. Select B6 and type $=B5*B3$. Select B7 and type $=B5+B6+B4$.
3. **Format:** Format B1:B2 and B4:B7 as Accounting or Currency. Format B3 as Percentage.
4. **Check:** B5 should equal **\$400.00**, B6 should equal **\$24.00**, and B7 should equal **\$439.00**. The total must be greater than the taxable subtotal because tax and shipping were added.
5. **Try:** Change the trade discount in B2 from $\$80$ to $\$100$. The taxable subtotal, tax, and total should all update automatically.

Build sales tax in the required order

B7	fx	=B5+B6+B4
----	----	-----------

	A	B
1	Retail Price	\$480.00
2	Trade Discount	\$80.00
3	Sales Tax Rate	6%
4	Shipping	\$15.00
5	Taxable Subtotal	=B1-B2
6	Sales Tax Owed	=B5*B3
7	Total Price	=B5+B6+B4

Order check

- 1 Subtract discount
 - 2 Calculate tax
 - 3 Add shipping last
- Expected total: \$439.00

⚠ Order Matters

Subtract the trade discount first, calculate tax second, and add shipping last. Do not calculate tax from the original retail price when a trade discount reduces the taxable base.

Walkthrough 2 — Keep Sales and Excise Taxes on the Same Base

1. **Set up:** Enter Base Cost in A1, Sales Tax Rate in A2, and Excise Tax Rate in A3. Enter $\$6,200$, 7% , and 8% in B1:B3.
2. **Select and type:** In B4, calculate sales tax with $=B1*B2$. In B5, calculate excise tax with $=B1*B3$. In B6, calculate total cost with $=B1+B4+B5$.
3. **Check:** B4 should equal **\$434.00**, B5 should equal **\$496.00**, and B6 should equal **\$7,130.00**.
4. **Confirm with a second method:** In another cell, type $=B1*(1+B2+B3)$. It should also return **\$7,130.00**.
5. **Try:** Change the excise tax rate to 9% . Both the excise tax and total cost should rise, but the sales tax should remain unchanged.

Keep sales and excise taxes tied to the same base cost

B5

fx =B1*B3

Sales tax

=B1*B2

$\$6,200 \times 7\% = \434

Excise tax

=B1*B3

$\$6,200 \times 8\% = \496

Both
formulas
start with
B1

If the excise formula begins with the sales-tax-inclusive total instead of B1, the taxes are being compounded incorrectly. Both tax formulas must point back to the original base cost.

Walkthrough 3 — Round the Property Tax Rate First

- Set up:** Enter Market Value, Assessment Rate, Town Budget, and Total Assessed Value in A1:A4. Enter \$820,000, 40%, \$310,000, and \$4,750,000 in B1:B4.
- Select and type:** In B5, type `=B1*B2` for assessed value. In B6, type `=ROUND(B3/B4,4)` for the tax rate. In B7, type `=B5*B6` for annual tax due.
- Format:** Format B1, B3:B5, and B7 as Accounting or Currency. Format B2 as Percentage. Display B6 as a number with four decimal places so you can verify the required rounding.
- Check:** B5 should equal **\$328,000.00**, B6 should equal **0.0653**, and B7 should equal **\$21,418.40**.
- Try:** Calculate a monthly NNN planning reserve in B8 with `=B7/12`. The result should be **\$1,784.87** when displayed to two decimal places.

Round the property-tax rate before calculating tax due

B6

fx =ROUND(B3/B4,4)

	A	B
1	Market Value	\$820,000
2	Assessment Rate	40%
3	Town Budget	\$310,000
4	Total Assessed Value	\$4,750,000
5	Assessed Value	=B1*B2
6	Rounded Tax Rate	=ROUND(B3/B4,4)
7	Annual Tax Due	=B5*B6

Use the rounded rate

B6 stores 0.0653

B7 uses =B5*B6

Expected annual tax:
\$21,418.40

Optional monthly reserve:
=B7/12



Workbook Connection

In class, use the same modeling habits on the **Live You Try It** tab of the starter workbook. Use the **FormulaReferenceCard** tab when you need a syntax reminder. Then begin the graded **Class Challenge** tax compliance audit in class and finish it for homework: diagnose each supplied filing, repair the liability formula, calculate the adjustment, and reconcile the corrected total to the control total.

4 • FORMULA REFERENCE

Formula Name	Expression
Sales Tax — Taxable Subtotal	= Price - Trade Discount
Sales Tax — Tax Owed	= Taxable × Tax Rate
Sales Tax — Total Price	= Taxable + Tax + Shipping
Back-Calculate Actual Sales	= Total Receipts / (1 + Tax Rate)
Excise — Each tax separately	= Base × SalesRate and = Base × ExciseRate
Excise — Combined shortcut	= Base × (1 + SalesRate + ExciseRate)
Property — Assessed Value	= Assessment Rate × Market Value
Property — Tax Rate	= ROUND(Town Budget / Total Assessed Value, 4)
Property — Annual Tax Due	= Tax Rate × Assessed Value

5 • CHECK YOUR UNDERSTANDING

Answer all seven questions before class. Answers appear at the end of this document.

1. **[Calculation]** Tidal Goods Co. sells an order priced at \$600. The buyer receives a \$100 trade discount. Sales tax is 5.5%. Shipping is \$18. What is the taxable subtotal and the total price?
1. **[Concept]** A merchant collects \$4,200 in sales tax over a week. Is that \$4,200 the merchant's revenue? Explain why or why not.
1. **[Calculation]** Register receipts total \$78,000 for the day, inclusive of 7% sales tax. What were the actual pre-tax sales?
1. **[Calculation]** Meridian Advisory Group buys office equipment for \$8,500. Sales tax is 6.5% and excise tax is 9%. Calculate the total cost using both methods (separate and combined rate).
1. **[Concept]** Why is it incorrect to apply excise tax to the sales-tax-inclusive amount rather than the base cost?
1. **[Calculation]** A property has a market value of \$450,000. Assessment rate: 35%. Town budget: \$198,000. Total assessed value in town: \$2,600,000. Calculate assessed value, tax rate (to 4 decimal places), and annual tax due.
1. **[Strategy]** Meridian Advisory Group is considering a Triple Net (NNN) lease for new office space. Why should they research the municipality's budget trend before signing? What specific financial risk does an

NNN lease create?

Answer Key · Check Your Understanding

#	Answer
1	Taxable = \$600 – \$100 = \$500. Tax = \$500 × 0.055 = \$27.50. Total = \$500 + \$27.50 + \$18 = \$545.50
2	No. Collected sales tax is a liability held in trust for the state government, not earned revenue. Recording it as revenue overstates income.
3	\$78,000 / 1.07 = \$72,897.20 . Collected tax = \$78,000 – \$72,897.20 = \$5,102.80.
4	Method 1: ST = \$552.50; ET = \$765.00. Total = \$9,817.50 . Method 2: \$8,500 × 1.155 = \$9,817.50 . Both match.
5	Both excise and sales tax are calculated from the original base price — not compounded. Applying the second tax to an already-taxed amount overstates the total and inflates cost of goods.
6	Assessed Value = 0.35 × \$450,000 = \$157,500. Tax Rate = ROUND(\$198,000/\$2,600,000, 4) = 0.0762 . Annual Tax = 0.0762 × \$157,500 = \$12,001.50 .
7	Under NNN, the tenant bears property tax, insurance, and maintenance. If the town's budget increases, the tax rate rises and the tenant's costs increase with no cap. An NNN lease converts a fixed rent into variable overhead — the tenant assumes the risk of local government spending.

6 · KEY VOCABULARY

Term	Definition
Sales Tax	A state-imposed consumption tax on the retail sale of goods and services; collected by merchants on behalf of the government.
Trade Discount	A reduction from the retail price given to a business buyer; subtracted before sales tax is applied.
Excise Tax	A targeted tax on specific goods (fuel, alcohol, tobacco, luxury appliances) typically paid upstream by manufacturers and embedded in the price.
Assessed Value	The dollar value assigned to a property for tax purposes: $\text{Assessment Rate} \times \text{Market Value}$.
Assessment Rate	The percentage of a property's market value that a municipality designates as taxable.
Tax Rate (Property)	The rate applied to assessed value to calculate tax due: $\text{Town Budget} \div \text{Total Assessed Value}$; rounded to four decimal places.
Nexus	The legal connection (physical presence or economic activity) that requires a business to collect and remit sales tax in a given jurisdiction.
Triple Net Lease (NNN)	A lease where the tenant pays property taxes, insurance, and maintenance in addition to base rent — shifting operating cost risk from landlord to tenant.
Business Personal Property	Movable, tangible operating assets (computers, furniture, machinery) subject to local property tax.
Demand Elasticity	The sensitivity of customer demand to price changes; relevant to excise tax when businesses must raise prices to maintain margin.

BUS 123 — MATH-M08-L01 — Simple Interest, Discounting & Present Value

BUS123 · MATH-M08 · L01 PRE-READING

SIMPLE INTEREST, DISCOUNTING & PRESENT VALUE

Course: Solving Business Problems with Technology · BUS123 **Track:** MATH · Module 08 · Lesson 01

Semester: Fall 2026 · Gerrish School of Business, Endicott College **Case Study Company:** Meridian Advisory Group (*fictional — all data simulated for instructional purposes*)

CONNECT TO PRIOR KNOWLEDGE

In the modules leading up to M08, you worked with percentages and proportional change — markup rates, markdown percentages, and ratio comparisons. A rate was a label: it described a change relative to a base amount at a single point in time.

Starting today, a rate becomes an **engine**. When a business borrows money or holds cash in a reserve account, the rate no longer just changes a price — it generates a dollar amount that grows the longer time passes. This module builds the first and simplest version of that relationship: **Simple Interest**, the foundation for all of business finance.

We also introduce the concept that makes financial advisors indispensable: **discounting**. Money promised in the future is worth less than money in hand today, and the difference can be calculated precisely. By the end of this lesson, you will be able to quantify that gap and apply it to real client scenarios.

CORE CONCEPTS

Part A — Why a Dollar Today Is Worth More Than a Dollar Tomorrow

The **Time Value of Money (TVM)** is the principle that money available today is worth more than the same dollar amount promised in the future. Three economic forces explain why:

Force	Explanation
Opportunity Cost	Money today can be invested immediately and begin earning a return. A dollar promised next year cannot earn anything until it arrives.
Inflation	Prices tend to rise over time. The same dollar amount will buy slightly less in the future than it does today.
Risk	Future promises can fail. A payment guaranteed today carries no uncertainty; a future payment might not arrive at all.

The So What

When a financial advisor at Meridian evaluates a client's future payment stream, they don't just add up the dollar amounts. They ask: *what are these future dollars worth **today**, given the client's opportunity cost and the risk of waiting?* That calculation — discounting — is what you will build in today's activity.

The interest rate used in TVM calculations is not arbitrary. It is a sum:

Total Discount Rate = Inflation Rate + Real Interest Rate + Risk Premium

Each component has economic meaning. A 7% discount rate might represent 3% inflation, 2% real return for waiting, and 2% compensation for risk. Choosing the right rate is one of the most consequential judgments in financial analysis.

Part B — Simple Interest: Moving Money Forward

Simple interest calculates interest earned on a **principal amount** using a flat annual rate applied to the **original principal only** — interest does not earn interest (that is compound interest, covered in M09). Simple interest is common in short-term loans, notes payable, and bridge financing.

The Formula: $I = P \times R \times T$

Variable	Stands For	Notes
I	Interest earned	The dollar amount of interest generated
P	Principal	The original amount borrowed or invested
R	Annual interest rate	Always enter as a decimal: 6% = 0.06
T	Time in years	Must be in years — convert: months ÷ 12, days ÷ 365

Ending Balance (Maturity Value / Future Value):

$$FV = P + I \quad \text{or equivalently:} \quad FV = P \times (1 + R \times T)$$

The term $(1 + R \times T)$ is called the **growth factor**. It tells you how many times larger your money becomes. At 5.5% for 9 months, every dollar grows to \$1.04125.

The T Trap — Never Use Days or Months Directly

The most common error on simple interest problems is plugging in time without converting to years first.

- A 90-day loan does **not** use $T = 90$. It uses $T = 90 \div 365 = \mathbf{0.247 \text{ years}}$.

- A 6-month loan uses $T = 6 \div 12 = \mathbf{0.5 \text{ years}}$.

If you forget to convert, your interest will be 90× or 12× too large — a result that should immediately signal something is wrong.

Part C — Two Methods for Day-Based Interest

When a loan term is expressed in days, two conventions are in use:

Exact Interest (÷ 365): $T = \text{exact days} \div 365$. Used by the Federal Reserve and US government. Every calendar day counts as 1/365 of a year.

Ordinary Interest — Banker's Rule ($\div 360$): $T = \text{exact days} \div 360$. Used by most commercial banks. Dividing by the smaller number produces a slightly larger T , which produces slightly more interest — in the bank's favor. This is intentional.

Practical impact: On a \$50,000 loan at 5% for 124 days:

Method	T	Interest
Exact ($\div 365$)	0.3397	\$849.32
Ordinary ($\div 360$)	0.3444	\$861.11
Difference		\$11.79

Across a large commercial loan portfolio, the Banker's Rule generates meaningfully more revenue. It is not an error — it is a deliberate convention.

Part D — Solving for the Unknown Variable

The simple interest formula rearranges cleanly to solve for any one variable when the other two are known:

Solve for	Formula	Excel
Principal	$P = I \div (R \times T)$	<code>=B5 / (B3*B4)</code>
Rate	$R = I \div (P \times T)$	<code>=B5 / (B2*B4)</code>
Time	$T = I \div (P \times R)$	<code>=B5 / (B2*B3)</code>

Worked Example — Meridian Advisory Group: A client's account shows \$19.48 in interest charged at 9.5% for 90 days (ordinary interest). What was the principal?

$$T = 90 \div 360 = 0.25 \text{ years}$$

$$P = \$19.48 \div (0.095 \times 0.25) = \$19.48 \div 0.02375 = \$820.21$$

Real analysts solve for unknowns constantly. A client paid \$ X in interest — what was the original loan? The formula rearranges cleanly; you just need to know which variable is missing.

Part E — Discounting: Running the Math Backward

Future Value asks: *what will this grow to?* Present Value asks the reverse: *what is a future promise worth right now?*

To get Present Value, divide the Future Value by the discount factor instead of multiplying:

Direction	Question	Formula
Forward →	What will P grow to?	$FV = P \times (1 + R \times T)$
← Backward	What is FV worth today?	$PV = FV \div (1 + R \times T)$

The term $(1 + R \times T)$ in the denominator is the **discount factor**. It is always greater than 1 (when R and T are positive), so dividing by it always produces a Present Value **smaller** than the Future Value. The higher the rate or the longer the wait, the larger the discount factor — and the smaller the PV .

Worked Example — Meridian Advisory Group: A corporate client is owed \$25,000 in 18 months from a contract receivable. Discount rate: 7%.

$$\begin{aligned}T &= 18 \div 12 = 1.5 \text{ years} \\ \text{Discount Factor} &= 1 + (0.07 \times 1.5) = 1.105 \\ \text{PV} &= \$25,000 \div 1.105 = \$22,624.43\end{aligned}$$

The gap: $\$25,000 - \$22,624 = \$2,376$. That \$2,376 is the **cost of waiting** 18 months at 7% — the combined effect of opportunity cost and risk measured in dollars.

Part F — The Excel =PV() Function

Excel's `=PV()` function discounts future cash flows using a **compound periodic model**. It is a standard finance function and an important preview of M09, but it does **not** use the simple-interest formula $FV \div (1 + R \times T)$. Therefore, a manual simple-interest PV and `=PV()` will not generally match when the time spans more than one period.

Syntax:

`=PV(rate, nper, pmt, [fv], [type])`

Argument	Meaning	For the Excel PV Model
rate	Compound interest rate per period	Annual rate when <code>nper</code> is measured in years
nper	Total number of compounding periods	Years or fractional annual periods in this lesson
pmt	Recurring payment	Enter 0
fv	Future amount to discount	Enter as a positive number
Result	Present value	Returns negative — see below

The sign convention: Excel treats outflows (money leaving your pocket) as negative and inflows (money coming to you) as positive. The `=PV()` result is negative because it represents the amount you would need to *invest today* to receive the future FV. This is not an error — it is the correct sign.

`=PV(7%, 1.5, 0, 25000) → -$22,587.30`

By comparison, the manual simple-interest calculation is $\$25,000 \div (1 + 0.07 \times 1.5) = \$22,624.43$. The values differ because the formulas use different growth assumptions, not because either Excel calculation is broken.

Verify the Excel function with a round trip: `=FV(7%, 1.5, 0, -22587.30) → $25,000 ✓`

Always run this check. If `=FV()` does not reproduce the original future value, something is wrong with your inputs.

⚠ Common Mistake — The Negative PV Panic

Students see a negative PV and assume it is an error. They manually add a minus sign to flip it positive. Now the sign meaning is inverted and the round-trip check will fail. The negative sign is meaningful: it represents *outflow*, the amount you must put in today. Do not flip it.

BUILD THE MODELS IN EXCEL

Excel should mirror the structure of the business calculation. Place each assumption in a separate labeled input cell, convert time before calculating interest, and use cell references rather than typing the numbers directly into every formula. In the examples below, **yellow cells are inputs** and **green cells are formulas**.

🖨 Windows and Mac Excel

The worksheet images use the standard Excel layout. Windows and Mac Excel use the same formulas and function names, although ribbon buttons may appear in slightly different positions.

Walkthrough 1 — Convert Time to Years

An annual rate must be paired with time measured in years. Choose the conversion that matches the wording of the problem:

- For months, use `=Months/12`.
- For exact-interest days, use `=Days/365`.
- For ordinary-interest days, use `=Days/360`.

Convert time before using an annual interest rate



1. **Set up:** Enter Months in A1 and 9 in B1. Enter Months per Year in A2 and 12 in B2.
2. **Select and type:** Select B3 and type `=B1/B2`.
3. **Format:** Display B3 as a Number with at least two decimal places—not as Currency or Percentage.
4. **Check:** B3 should equal **0.75 years**. Because nine months is less than one year, the result must be less than 1.
5. **Try:** Change B1 to 18. The result should update to **1.5 years**.

Walkthrough 2 — Build Simple Interest and Future Value

1. **Set up:** In A1:A3, enter Principal, Annual Rate, and Months. In B1:B3, enter \$8,000 , 5.5% , and 9 .
2. **Convert time:** In B4, type $=B3/12$.
3. **Calculate interest:** In B5, type $=B1*B2*B4$.
4. **Calculate future value:** In B6, type $=B1*(1+B2*B4)$.
5. **Format:** Format B1, B5, and B6 as Accounting or Currency. Format B2 as Percentage. Display B4 as a Number.
6. **Check:** B4 should equal 0.75, B5 should equal \$330.00, and B6 should equal \$8,330.00. Future value must be greater than principal when the rate and time are positive.
7. **Try:** Change the term from 9 months to 12 months. Interest and future value should increase automatically.

Build simple interest and future value with cell references

B6	fx =B1*(1+B2*B4)	
	A	B
1	Principal	\$8,000.00
2	Annual Rate	5.5%
3	Months	9
4	Time in Years	=B3/12
5	Simple Interest	=B1*B2*B4
6	Future Value	=B1*(1+B2*B4)

Expected results

Time = 0.75 years

Interest = \$330.00

Future Value = \$8,330.00

FV must be greater than principal

⚠ Do Not Use FV() for This Step

This is a simple-interest model, so build $=\text{Principal}*(1+\text{Rate}*\text{Time})$ directly. Excel's `FV()` function uses compound periodic growth.

Walkthrough 3 — Compare Exact and Ordinary Interest

1. **Set up:** Enter Principal, Annual Rate, and Days in A1:A3. Enter \$50,000 , 5% , and 124 in B1:B3.
2. **Exact interest:** In B4, type $=B1*B2*(B3/365)$.
3. **Ordinary interest:** In B5, type $=B1*B2*(B3/360)$.
4. **Check:** Exact interest should equal \$849.32 and ordinary interest should equal \$861.11. The 360-day method should be \$11.79 higher.
5. **Try:** Calculate the difference in B6 with $=B5-B4$. If the result is negative, the formulas or labels were reversed.

Compare exact and ordinary interest from the same loan inputs

B5

fx =B1*B2*(B3/360)

Exact interest

=B1*B2*(B3/365)

\$50,000 x 5% x (124/365)

\$849.32

Ordinary interest

=B1*B2*(B3/360)

\$50,000 x 5% x (124/360)

\$861.11

+\$11.79

360 gives more interest

Walkthrough 4 — Compare Manual PV With Excel PV()

Use the same inputs to compare two different models, not to prove that they return the same answer.

1. **Set up:** Enter Future Value, Annual Rate, and Years in A1:A3. Enter \$25,000, 7%, and 1.5 in B1:B3.
2. **Manual simple-interest PV:** In B4, type =B1/(1+B2*B3). The result should be **\$22,624.43**.
3. **Excel compound-periodic PV:** In B5, type =PV(B2,B3,0,B1). The result should be **-\$22,587.30**.
4. **Read the sign:** The negative result is Excel's cash-flow convention. A positive future inflow requires a present outflow.
5. **Round-trip check:** In B6, type =FV(B2,B3,0,B5). The result should return approximately **\$25,000.00**.
6. **Try:** Increase the rate to 8%. The absolute value of both present-value results should decrease because a higher discount rate makes future money worth less today.

Identify the discounting model before choosing the Excel formula

Manual simple-interest PV

=B1/(1+B2*B3)

\$25,000 / (1 + 7% x 1.5)

= \$22,624.43

Linear simple-interest model

Excel PV() periodic model

=PV(B2,B3,0,B1)

=PV(7%, 1.5, 0, 25000)

= -\$22,587.30

Compound periodic model and sign convention

Do not expect the two answers to match

✓ Workbook Connection

In class, apply these habits on the **Live You Try It** tab of the starter workbook. Use the **FormulaReferenceCard** tab for syntax reminders. Complete the **Class Challenge** only when your instructor directs you to begin the graded work.

FORMULA REFERENCE TABLE

Formula	Use
$I = P \times R \times T$	Calculate simple interest
$FV = P \times (1 + R \times T)$	Calculate maturity value / ending balance
$P = I \div (R \times T)$	Solve for unknown principal
$R = I \div (P \times T)$	Solve for unknown rate
$T = I \div (P \times R)$	Solve for unknown time
$T = \text{months} \div 12$	Convert months to years
$T = \text{days} \div 365$	Exact interest time conversion
$T = \text{days} \div 360$	Ordinary interest (Banker's Rule)
$PV = FV \div (1 + R \times T)$	Discount a future payment — manual
<code>=PV(rate, nper, 0, fv)</code>	Discount a future payment using Excel's compound periodic model

CHECK YOUR UNDERSTANDING

Answer these questions before class. Show your work on questions 2–6.

1. In one or two sentences, explain why a financial advisor would tell a client that \$15,000 promised in two years is not worth \$15,000 today. Do not use a formula — explain the economic logic.
2. A Meridian client borrows \$10,000 at 6% simple interest for 2 years. Calculate (a) the interest earned and (b) the ending balance.
3. A 180-day commercial note has a principal of \$7,500 and an annual rate of 4.8%. Calculate the interest earned using the **exact interest method** (365-day year). Show the time conversion step.
4. A Meridian client will receive \$18,000 in 2 years. The discount rate is 7%. Calculate the present value using the manual formula. Show the discount factor calculation.
5. Using the same scenario as Question 4, calculate the PV using Excel's compound-periodic `=PV()` function. Write out the formula exactly as you would type it in a cell. What sign does the result carry, what does it mean, and why does its absolute value differ slightly from the manual simple-interest answer?
6. Using a 6% discount rate, calculate the present value of each payment below, then find the total present value of the two-payment stream:
 - Payment A: \$12,000 due in 1 year

- Payment B: \$20,000 due in 3 years

7. A student calculates the present value of a future payment and gets an answer *larger* than the future value. Without doing the math, explain what error the student must have made.

ANSWER KEY

1. Three forces reduce the value of future money: opportunity cost (the \$15,000 today could be invested to earn a return), inflation (prices will likely rise, so \$15,000 buys less in two years), and risk (the future payment might not arrive). Any two of these, explained clearly, is a complete answer.

2. (a) $I = \$10,000 \times 0.06 \times 2 = \mathbf{\$1,200.00}$. (b) Ending Balance = $\$10,000 + \$1,200 = \mathbf{\$11,200.00}$ (or: $\$10,000 \times (1 + 0.06 \times 2)$).

3. $T = 180 \div 365 = 0.4932$ years. $I = \$7,500 \times 0.048 \times 0.4932 = \mathbf{\$177.53}$.

4. Discount Factor = $1 + (0.07 \times 2) = 1.14$. $PV = \$18,000 \div 1.14 = \mathbf{\$15,789.47}$.

5. $=PV(7\%, 2, 0, 18000) \rightarrow \mathbf{-\$15,721.90}$. The result is negative because it represents the amount you must *invest today* (an outflow) to have \$18,000 in 2 years at 7%. Its absolute value differs from the manual answer because $PV()$ uses compound periodic discounting while Question 4 uses the simple-interest discount factor $1 + R \times T$.

6. Payment A: $PV = \$12,000 \div (1 + 0.06 \times 1) = \$12,000 \div 1.06 = \mathbf{\$11,320.75}$. Payment B: $PV = \$20,000 \div (1 + 0.06 \times 3) = \$20,000 \div 1.18 = \mathbf{\$16,949.15}$. Total PV = $\mathbf{\$28,269.90}$.

7. PV must always be *less* than FV (because the discount factor is always greater than 1). If $PV > FV$, the student almost certainly **multiplied** by the discount factor instead of dividing — running the formula in the wrong direction.

KEY VOCABULARY

Term	Definition
Time Value of Money (TVM)	The principle that a dollar today is worth more than a dollar in the future, because today's dollar can be invested to earn a return.
Principal (P)	The original amount of money borrowed or invested — before interest is added.
Simple Interest	Interest calculated only on the original principal. Interest does not earn interest on itself.
Future Value (FV)	The value a present sum will reach at a specified future date, given a specific interest rate and time. Also called maturity value or ending balance.
Present Value (PV)	The current worth of a future sum, calculated by discounting at a specific rate.
Discount Rate	The interest rate used to convert future cash flows back to present value. Reflects opportunity cost plus a risk premium.
Discount Factor	The denominator in the PV formula: $(1 + R \times T)$. Always greater than 1; shrinks future value down to its present equivalent.
Exact Interest	Simple interest using a 365-day year for the time conversion.
Ordinary Interest	Simple interest using a 360-day year (Banker's Rule) — produces slightly more interest, favoring the lender.
Growth Factor	The term $(1 + R \times T)$ in the FV formula. Tells you how many times larger your money becomes.
Discounted Cash Flow (DCF)	A method that finds the present value of a stream of future cash flows by discounting each one back to today and summing the results.

BUS 123 — MATH-M09-L01 — Compound Interest & Future Value

BUS123 · MATH-M09 · L01 PRE-READING

COMPOUND INTEREST & FUTURE VALUE

Course: Solving Business Problems with Technology · BUS123 **Track:** MATH · Module 09 · Lesson 01
Semester: Fall 2026 · Gerrish School of Business, Endicott College **Case Study Company:** Meridian Advisory Group (*fictional — all data simulated for instructional purposes*)

CONNECT TO PRIOR KNOWLEDGE

In M08 we worked with simple interest — a flat, linear amount earned only on the original principal. The ending balance grew by the same dollar amount every period: if \$1,000 earns 5% simple interest, it adds exactly \$50 per year, every year. The growth is predictable and constant.

Today that changes. **Compound interest** earns interest on the accumulated balance, not just the original principal. In year one the result is identical. In year two, the interest from year one has itself started earning — so the growth amount is slightly larger. By year 30, the difference is dramatic. This is the mathematical engine behind every savings account, investment portfolio, and retirement fund — and it is why starting to save early matters far more than most people realize.

CORE CONCEPTS

Part A — Simple vs. Compound: The Structural Difference

The formulas look similar but behave completely differently over time:

	Simple Interest	Compound Interest
Interest earns on	Original principal only	Accumulated balance (principal + prior interest)
Growth shape	Linear — same dollar amount every period	Exponential — growing amount every period
Formula	$FV = P \times (1 + R \times T)$	$FV = PV \times (1 + i)^n$
Excel function	No built-in function — build directly	<code>=FV(rate, nper, 0, -pv)</code>

The key structural change is the **exponent** n . In simple interest, time multiplies: $(1 + R \times T)$. In compound interest, time is an exponent: $(1 + i)^n$. That one change — from multiplication to exponentiation — is what creates exponential growth.

\$1,000 at 5%: how the two methods diverge over time

Year	Simple Interest	Compound Interest	Difference
1	\$1,050.00	\$1,050.00	\$0.00
2	\$1,100.00	\$1,102.50	\$2.50
3	\$1,150.00	\$1,157.63	\$7.63
5	\$1,250.00	\$1,276.28	\$26.28
10	\$1,500.00	\$1,628.89	\$128.89
30	\$2,500.00	\$4,321.94	\$1,821.94

Year 1: identical. Year 30: compound interest produces 73% more than simple interest — with the same rate and the same principal. This is the **snowball effect**: each year's interest rolls forward and increases the base for the next calculation.

Part B — The Compound Interest Formula

Formula: $FV = PV \times (1 + i)^n$

Variable	Name	Definition
FV	Future Value	The ending balance after compounding
PV	Present Value	The starting principal today
i	Rate per period	Annual rate ÷ compounding periods per year
n	Number of periods	Years × compounding periods per year

Critical Rule: i and n must refer to the same unit of time.

If you compound annually, i = annual rate and n = years. If you compound monthly, i = annual rate ÷ 12 and n = years × 12. Mixing units — using an annual rate with monthly periods — produces wildly inflated results. This is the single most common compound interest error.

Worked Example — Meridian Advisory Group: A client deposits \$40,000 at 4% compounded annually for 3 years.

- $FV = \$40,000 \times (1.04)^3$
- $FV = \$40,000 \times 1.124864$
- $FV = \$44,994.56$

Simple interest over the same period: $\$40,000 \times (1 + 0.04 \times 3) = \$44,800$. Compound interest adds an extra \$194.56 — earned entirely from interest-on-interest over three years.

Part C — Non-Annual Compounding

When interest compounds more frequently than once per year, **both rate and nper must be adjusted** to match the compounding period. The annual rate alone is not sufficient.

Compounding	Rate per period	Number of periods
Annually	$R \div 1$	Years $\times 1$
Semiannually	$R \div 2$	Years $\times 2$
Quarterly	$R \div 4$	Years $\times 4$
Monthly	$R \div 12$	Years $\times 12$

\$25,000 at 6% for 5 years — what compounding frequency does:

Frequency	rate arg	nper arg	FV
Annually	6%	5	\$33,455.64
Quarterly	1.5%	20	\$33,671.38
Monthly	0.5%	60	\$33,721.25

More frequent compounding produces a higher ending balance when the same nominal annual rate is divided across more compounding periods. The practical difference between annual and monthly compounding on a \$25,000 investment over 5 years is \$265.61 — not enormous, but meaningful across a large portfolio.

Part D — The Excel =FV() Function

Excel calculates Future Value automatically. You supply the inputs; Excel handles the exponent.

Syntax:

`=FV(rate, nper, pmt, [pv], [type])`

Argument	Meaning	For lump sum problems
rate	Interest rate per period	Annual rate $\div$ periods per year
nper	Total number of periods	Years $\times$ periods per year
pmt	Recurring payment per period	Enter 0 (lump sums only)
pv	Starting principal	Enter as a negative number
type	0 = end of period (default)	Omit or enter 0

The Sign Convention: Excel's financial functions assign direction to every cash flow. When you invest \$40,000, that money leaves your pocket — so PV = **-\$40,000**. The FV result comes back positive, representing the money returning to you.

- `=FV(4%, 3, 0, -40000)` → \$44,994.56 ✓
- `=FV(4%, 3, 0, 40000)` → -\$44,994.56 ✗ (wrong sign — confusing, not an error)

Getting the sign wrong does not produce a formula error — it produces a plausible-looking answer with the wrong sign. Always sanity-check: FV should be **larger** than the absolute value of PV when the rate is positive.

Manual verification habit: After every `=FV()` result, run: `=ABS(pv)*(1+rate)^nper`. Both must produce the same number. If they don't, your inputs are inconsistent.

Build a Labeled FV Model in Excel

Keep inputs separate from calculated outputs. In a blank worksheet, type the labels in column A and values in column B:

Cell	Label	Enter	Format
B4	Starting principal	25000	Currency, 2 decimals
B5	Annual rate	6%	Percentage, 2 decimals
B6	Years	5	Number, 0 decimals
B7	Periods per year	4	Number, 0 decimals
B8	Future value	formula	Currency, 2 decimals

1. Select cell **B8**.
2. Type `=FV(B5/B7,B6*B7,0,-B4)` in the formula bar and press **Enter**.
3. The expected result is **\$33,671.38**.
4. On Windows Excel, use **Home > Number** to apply Currency, Percentage, and Number formats. Mac Excel may position these commands differently.
5. Reasonableness check: because the rate and time are positive, FV must be greater than the \$25,000 starting principal.
6. Recalculation test: change **B5 from 6% to 7%**. B8 should automatically increase to about **\$35,369.45**. Change B5 back to 6% before continuing.

Excel for Windows - Home tab

Formula bar: =FV(B5/B7,B6*B7,0,-B4)

Starting principal	\$25,000
Annual rate	6.00%
Years	5
Periods per year	4
Future value	\$33,671.38

Build with cell references

1. Select B8.
 2. Type the formula shown above.
 3. Press Enter.
 4. Format B8 as Currency.
- Mac commands may be positioned differently.

Excel for Windows - select, then format

Home tab > Number group

\$ Currency

% Percent

Increase Decimal

Select rate cells for Percentage; select FV for Currency.

Change 6% to 7%: FV should increase automatically.

This model matches the workbook habit: enter the yellow inputs first, then build the output from cell references. Use the same approach in **Live You Try It**. In **Class Challenge**, follow the prompts and hints, but build your own formulas; the pre-reading does not provide the graded answers.

Part E — The Rule of 72

Before reaching for Excel, use this mental math shortcut to estimate how long it takes an investment to double:

Rule of 72: $\text{Years to Double} \approx 72 \div \text{Annual Rate (\%)}$

Rate	Approximate doubling time
4%	$72 \div 4 = 18$ years
6%	$72 \div 6 = 12$ years
8%	$72 \div 8 = 9$ years
12%	$72 \div 12 = 6$ years

This is an approximation — use `=FV()` for precise answers. The Rule of 72 is a planning tool: useful in quick client conversations when you need a fast, credible estimate before opening a spreadsheet.

Verification: At 7%, the Rule of 72 estimates $72 \div 7 \approx 10.3$ years. `=FV(7%, 10, 0, -1000)` = \$1,967.15 — very close to \$2,000. The rule is accurate.

FORMULA REFERENCE TABLE

Formula	Use
<code>FV = PV × (1 + i)ⁿ</code>	Compound interest — manual calculation
<code>=FV(rate, nper, 0, -pv)</code>	Compound FV — Excel (lump sum)
<code>=ABS(pv)*(1+rate)^nper</code>	Manual verify — must match <code>=FV()</code>
<code>rate per period = annual ÷ periods/yr</code>	Non-annual compounding adjustment
<code>nper = years × periods/yr</code>	Non-annual compounding adjustment
<code>72 ÷ rate (%) ≈ years to double</code>	Rule of 72 estimate

CHECK YOUR UNDERSTANDING

Answer these questions before class. Show your work on questions 2–6.

1. What is the structural difference between simple interest and compound interest? Your answer should explain *why* compound interest grows exponentially while simple interest grows linearly — not just state that they use different formulas.
2. \$1,000 is invested at 6% compounded annually for 5 years. Calculate the FV using the formula $FV = PV \times (1 + i)^n$. Show your work including the exponent calculation.
3. Using the same scenario as Question 2, write the Excel `=FV()` formula exactly as you would type it into a cell. Include the correct sign on pv.
4. Use the Rule of 72 to estimate how long it takes to double \$5,000 at 8% annual interest. Then verify with `=FV()` at your estimated number of years — is the result close to \$10,000?
5. A Meridian client invests \$25,000 at 6% for 5 years. Calculate the FV under (a) annual compounding and (b) monthly compounding. For monthly, show the adjusted rate and nper values before calculating.
6. True or False: In Excel's `=FV()` function, pv should always be entered as a negative number for investment problems. Explain why in one or two sentences.
7. A Meridian client considers two options: (a) invest \$15,000 today at 5% compounded annually for 10 years, or (b) invest the same \$15,000 at 5% compounded monthly for 10 years. Which produces more? Calculate both and explain the difference.

ANSWER KEY

1. Simple interest applies the rate to the original principal only — the same dollar amount is added every period, producing linear growth. Compound interest applies the rate to the accumulated balance, so each period's interest is slightly larger than the last because prior interest is now itself earning a return. The exponent n is the structural cause: $(1+i)^n$ raises the growth factor to a power, producing exponential rather than linear growth.
2. $FV = \$1,000 \times (1.06)^5 = \$1,000 \times 1.3382 = \mathbf{\$1,338.23}$.
3. `=FV(6%, 5, 0, -1000)` → \$1,338.23. PV is negative because the \$1,000 is money leaving the investor's pocket (an outflow).
4. $72 \div 8 = \mathbf{9 \text{ years}}$ (estimate). Verification: `=FV(8%, 9, 0, -5000)` = \$9,993.50 — very close to \$10,000. The Rule of 72 is accurate here.
5. (a) Annual: rate = 6%, nper = 5. `=FV(6%, 5, 0, -25000)` = **\$33,455.64**. (b) Monthly: rate = $6\%/12 = 0.5\%$, nper = $5 \times 12 = 60$. `=FV(0.5%, 60, 0, -25000)` = **\$33,721.25**. Monthly produces **\$265.61** more.
6. **True**. pv represents the starting investment — money leaving the investor's pocket on day one. Excel's sign convention treats outflows as negative. Entering pv as positive reverses the sign of the FV result, producing a negative FV that is correct in magnitude but wrong in directional meaning.
7. (a) Annual: `=FV(5%, 10, 0, -15000)` = **\$24,433.42**. (b) Monthly: `=FV(5%/12, 120, 0, -15000)` = **\$24,706.71**. Monthly produces **\$273.29** more. The difference arises because monthly compounding applies the rate more frequently, so interest begins earning on interest 12 times per year instead of once.

KEY VOCABULARY

Term	Definition
Compound Interest	Interest calculated on the accumulated balance — principal plus all prior interest earned.
Future Value (FV)	The value of an investment at a specific future date, after compounding at a given rate and for a given number of periods.
Present Value (PV)	The current worth of a future amount. In =FV(): the starting principal, entered as negative.
i (rate per period)	The interest rate per compounding period. Annual rate ÷ compounding periods per year.
n (number of periods)	Total number of compounding periods. Years × compounding periods per year.
Compounding	The process of earning interest on previously earned interest. The source of exponential growth.
Non-annual Compounding	Compounding more than once per year (monthly, quarterly, semiannual). Requires adjusting both rate and nper.
Sign Convention	In Excel TVM functions, outflows (money paid out) are negative; inflows (money received) are positive.
Rule of 72	Mental math shortcut: $72 \div \text{annual rate (\%)} \approx \text{years to double an investment}$. An approximation — verify with =FV().
Snowball Effect	The accelerating growth dynamic of compound interest: each period's interest increases the base for the next calculation.
Growth Factor	The term $(1 + i)^n$. Tells you how many times larger your principal becomes after n periods at rate i.

BUS 123 — MATH-M09-L02 — PMT Function & Annuities

BUS123 · MATH-M09 · L02 PRE-READING

PMT FUNCTION & ANNUITIES

Course: Solving Business Problems with Technology · BUS123 **Track:** MATH · Module 09 · Lesson 02

Semester: Fall 2026 · Gerrish School of Business, Endicott College **Case Study Company:** Meridian Advisory Group (*fictional — all data simulated for instructional purposes*)

CONNECT TO PRIOR KNOWLEDGE

In M09-L01 we used `=FV()` to grow a single lump sum forward in time. In every problem, the `pmt` argument was set to zero — because there were no recurring payments, only a one-time deposit.

Today that changes. In the real world, most financial decisions involve **regular payments**: you pay your mortgage every month, contribute to your retirement account every paycheck, and collect a pension every quarter. These payment streams are called **annuities**, and they are the most common financial structure in everyday life.

The good news: the same three Excel functions — `=PV()`, `=FV()`, and the new `=PMT()` — handle everything. The only change is that `pmt` is no longer zero. The sign rules and the period-matching rule from L01 carry forward unchanged.

CORE CONCEPTS

Part A — What Is an Annuity?

An **annuity** is a series of equal payments made at equal time intervals. Every mortgage payment, car loan installment, monthly retirement contribution, and lease payment is an annuity. Annuities are the most common financial structure in personal and business finance.

Two types based on payment timing:

Type	When payments occur	Excel <code>type</code> argument	Common uses
Ordinary Annuity	End of each period	0 (default — omit)	Loans, mortgages, most savings plans
Annuity Due	Beginning of each period	1	Lease payments, insurance premiums

Annuity due always produces a slightly higher FV or lower required PMT than ordinary annuity, because each payment earns one extra period of interest. Unless a problem specifically states "beginning of period" or uses the word "due," assume ordinary annuity (`type = 0`).

Two types based on what you're solving for:

- **Loan amortization:** You know the loan amount (PV) — you want to know the required periodic payment (PMT).
- **Savings goal:** You know the target balance (FV) — you want to know the required periodic contribution (PMT).

Both use `=PMT()`. The difference is which argument holds the target value.

Part B — The `=PMT()` Function

`=PMT()` calculates the periodic payment required to pay off a loan or reach a savings goal over a fixed number of periods at a fixed interest rate.

Syntax:

`=PMT(rate, nper, pv, [fv], [type])`

Argument	Meaning	For loan problems	For savings goal problems
rate	Rate per period	Annual rate ÷ 12 (monthly)	Annual rate ÷ 12 (monthly)
nper	Total payment periods	Years × 12 (monthly)	Years × 12 (monthly)
pv	Present value (loan amount)	Loan amount — enter positive	Enter 0
fv	Future value (savings target)	Enter 0 (loan paid off)	Target balance — enter positive
Result	Required payment	Returns negative	Returns negative

The result is always **negative** — it represents money leaving your pocket each period (an outflow). When communicating the monthly payment to a client, use the absolute value: `=ABS(PMT(...))`.

Worked Example — Meridian Advisory Group Mortgage: A client purchases a home for \$320,000 at 6.5% annual interest, 30-year term, monthly payments.

- `rate = 6.5% ÷ 12 = 0.5417% per month`
- `nper = 30 × 12 = 360 months`
- `=PMT(6.5%/12, 360, 320000, 0) → -$2,022.62/month`

The client pays **\$2,022.62 per month**. Over 30 years:

- `Total paid = ABS(-$2,022.617675...) × 360 = $728,142.36`
- `Original PV = $320,000`
- `Total interest paid = $728,142.36 - $320,000 = $408,142.36`

The client will pay more than \$400,000 in interest over the life of the loan — more than the original principal. Keep the unrounded PMT in the formula cell and round only the display; multiplying the displayed \$2,022.62 by 360 creates a small rounding difference.

Build a Labeled PMT Model in Excel

Use a compact input-and-output model instead of typing numbers directly into formulas:

Cell	Label	Enter	Format
B4	Loan amount	320000	Currency, 2 decimals
B5	Annual rate	6.5%	Percentage, 2 decimals
B6	Years	30	Number, 0 decimals
B7	Payments per year	12	Number, 0 decimals
B8	Monthly payment	formula	Currency, 2 decimals
B9	Total paid	formula	Currency, 2 decimals
B10	Total interest	formula	Currency, 2 decimals

1. Select **B8**, type `=PMT(B5/B7,B6*B7,B4,0)`, and press **Enter**. Expected result: **-\$2,022.62**.
2. Select **B9**, type `=ABS(B8)*B6*B7`, and press **Enter**. Expected result: **\$728,142.36**.
3. Select **B10**, type `=B9-B4`, and press **Enter**. Expected result: **\$408,142.36**.
4. On Windows Excel, use **Home > Number** and apply Currency with 2 decimal places to B4 and B8:B10. Mac Excel may position these commands differently.
5. Reasonableness checks: PMT should be negative because it is a payment; total paid must exceed the loan amount; total interest must be positive.
6. Recalculation test: change **B5 from 6.5% to 7.0%**. The payment should become more negative and total interest should increase automatically. Then restore B5 to 6.5%.

Excel for Windows - mortgage model

Formula bar: `=PMT(B5/B7,B6*B7,B4,0)`

Loan amount	\$320,000
Annual rate	6.50%
Years	30
Payments/year	12
Monthly payment	-\$2,022.62
Total interest	\$408,142.36

Output formulas

B8: `=PMT(B5/B7,B6*B7,B4,0)`

B9: `=ABS(B8)*B6*B7`

B10: `=B9-B4`

Use Currency with 2 decimals.

Negative PMT means cash paid out.

Cash-flow signs and automatic recalculation

Money received = positive

Loan proceeds: $PV = +320000$

Settlement checks: $PMT = +1500$

Excel balances with the opposite sign.

Input-change test

Change 6.5% to 7.0%.

PMT becomes more negative.

Total interest increases.

The layout mirrors the starter workbook: use labeled yellow input cells, enter formulas in the answer/output cells, and check the feedback in **Live You Try It**. The **FormulaReferenceCard** is a syntax reminder, not a completed solution. Build the graded **Class Challenge** formulas yourself.

⚠ The Most Common PMT Error: Wrong Period Match

`=PMT(6.5%, 30, 320000)` returns $-\$27,006$ — a large *annual* payment that looks plausible. But the monthly payment is $\$2,023$. The formulas are not wrong; the *inputs* are mismatched. Always ask: what is the payment frequency? Rate and nper must match that frequency.

Part C — Solving for Required Savings Contribution

`=PMT()` also works in reverse: given a savings goal (FV), it tells you how much to contribute each period.

Worked Example — Retirement Goal: A Meridian client wants $\$500,000$ in 25 years. Their account earns 7% annual, compounded monthly. How much must they contribute each month?

`=PMT(7%/12, 25*12, 0, 500000)` → $-\$576.57/\text{month}$

They must contribute **$\$576.57$ per month**. Over 25 years:

- Total contributed = $\$576.57 \times 300 = \$172,971$
- Account balance at goal = $\$500,000$
- Growth from compounding = $\$327,029$

They put in $\$172,971$ and compounding nearly doubled it to $\$500,000$. The gap — $\$327,029$ — came entirely from interest earning interest over 25 years.

Part D — `=FV()` with Regular Contributions

When a client makes regular monthly contributions, `=FV()` projects the ending balance. The `pmt` argument is no longer zero.

Syntax (same as before, pmt now filled in):

`=FV(rate, nper, pmt, [pv], [type])`

For a savings account with monthly contributions and no starting balance:

- `rate` = annual rate ÷ 12
- `nper` = years × 12
- `pmt` = monthly contribution — enter as **negative** (money leaving your pocket)
- `pv` = 0 (no starting lump sum)

Worked Example: A Meridian client contributes \$500/month to a retirement account earning 7% annual for 30 years.

`=FV(7%/12, 30*12, -500, 0)` → \$609,985.50

They contributed $\$500 \times 360 = \$180,000$ total. Compounding grew it to **\$609,985.50** — the extra **\$429,985.50** is entirely from compound interest on prior contributions.

Starting with both a lump sum and regular contributions:

If the client already has \$10,000 saved and also contributes \$500/month:

`=FV(7%/12, 360, -500, -10000)` → \$691,150.47

Both `pmt` and `pv` are negative because both are outflows — money leaving the client's pocket.

Part E — =PV() with Regular Payments

`=PV()` with a `pmt` value answers: *what is a stream of equal future payments worth today?*

Worked Example: A Meridian client will receive \$1,500/month for 10 years from a structured settlement. Discount rate: 5% annual. What is that stream worth in today's dollars?

`=PV(5%/12, 10*12, 1500, 0)` → -\$141,422.03

The stream pays out $\$1,500 \times 120 = \$180,000$ total in nominal dollars. In today's dollars at 5%, it is worth only **\$141,422.03**. The difference — **\$38,577.97** — represents the time value eroded by 10 years of waiting and discounting.

The result is negative because it represents the **cost to acquire this stream today** (outflow). If a buyer offers \$130,000 for the settlement, the client should hold — it is worth more. If the offer is \$155,000, the client should accept — the offer exceeds the present value.

Part F — The Sign Convention Across All Three Functions

The same cash-flow rule applies in every TVM function:

- **Money leaving your pocket** (investing, paying a loan, making a deposit) → **negative**
- **Money coming to you** (receiving a loan, collecting a payment, withdrawing savings) → **positive**

Scenario	pvt sign	pmt sign	fv sign
Taking out a loan	+ (received)	- (paying back)	0
Regular savings contributions	0 or - (initial deposit)	- (monthly deposit)	+ (balance received)
Valuing received annuity payments	0	+ (payments received)	0

Getting any one of these signs wrong does not cause a formula error — it produces a plausible-looking answer with the wrong sign or direction. Always verify: does the result make economic sense given what you entered?

FORMULA REFERENCE TABLE

Formula	Use
<code>=PMT(rate/12, years×12, pv, 0)</code>	Monthly loan payment (pv = loan amount, positive)
<code>=PMT(rate/12, years×12, 0, fv)</code>	Required monthly savings contribution (fv = target)
<code>Total interest = ABS(PMT) × nper - PV</code>	Total interest cost over loan life
<code>=FV(rate/12, years×12, -pmt, 0)</code>	Future savings balance from regular contributions
<code>=FV(rate/12, years×12, -pmt, -pv)</code>	Future savings with starting balance + contributions
<code>=PV(rate/12, years×12, pmt, 0)</code>	Present value of a payment stream (pmt positive = received)

CHECK YOUR UNDERSTANDING

Answer these questions before class. Show your work on questions 2–6.

1. In your own words, what is an annuity? Give two examples from everyday life — one involving money flowing out (a loan) and one involving money flowing in (a payment received).
 2. A Meridian client takes out a \$200,000 mortgage at 5.5% annual interest for 15 years, with monthly payments. (a) Write the `=PMT()` formula. (b) What is the monthly payment? (c) What is the total interest paid over the life of the loan?
 3. A client wants to accumulate \$300,000 in 20 years. Their account earns 6% annual, compounded monthly. How much must they contribute each month? Write the `=PMT()` formula and identify which argument holds the savings goal.
 4. A Meridian client contributes \$400/month to an account earning 6% annual for 25 years, starting with no existing balance. Use `=FV()` to project the ending balance. What is the total amount contributed, and how much of the balance came from compounding?
 5. A client will receive \$2,000/month for 12 years from a structured settlement. The current discount rate is 6% annual. Use `=PV()` to find the present value of this stream. Is a lump-sum offer of \$170,000 a good deal for the client — or should they collect the payments?
 6. True or False: In `=PMT()`, the result is always negative. Explain what the negative sign means economically.
 7. A client is weighing two options over 10 years at 5% annual (monthly compounding): (a) invest \$40,000 today as a lump sum, or (b) contribute \$350/month with no starting balance. Which produces a higher account balance at year 10? Show both calculations.
-

ANSWER KEY

1. An annuity is a series of equal payments made at equal time intervals. Examples: outflow — monthly car loan or mortgage payments (money leaves your pocket each month). Inflow — monthly pension payments or a structured settlement (money arrives each month).

2. (a) $\text{=PMT}(5.5\%/12, 15*12, 200000, 0)$. (b) Monthly payment = **\$1,634.17**. (c) Total paid = $\$1,634.17 \times 180 = \$294,150.60$. Total interest = $\$294,150.60 - \$200,000 = \text{\$94,150.60}$.
3. $\text{=PMT}(6\%/12, 20*12, 0, 300000)$ → **-\$671.93/month**. The savings goal lives in the **fv** argument (the fourth argument). **pv** is 0 because the client is not starting with a lump sum.
4. $\text{=FV}(6\%/12, 300, -400, 0)$ → **\$401,806.46**. Total contributed = $\$400 \times 300 = \$120,000$. Growth from compounding = $\$401,806 - \$120,000 = \text{\$281,806}$. Compounding produced more than twice the total contributions.
5. $\text{=PV}(6\%/12, 144, 2000, 0)$ → **-\$200,904.37**. The stream is worth \$200,904 in today's dollars. A lump-sum offer of \$170,000 is **below** present value — the client should decline and collect the payments. If offered more than \$200,904, the client should consider accepting.
6. **True.** The $\text{=PMT}()$ result is always negative because it represents money leaving the borrower's or saver's pocket each period — an outflow. For a loan, you are paying the lender. For a savings goal, you are making a deposit. In both cases the cash flows in the negative direction from your perspective. Report the payment to clients as $\text{ABS}(\text{PMT}())$.
7. (a) Lump sum: $\text{=FV}(5\%/12, 120, 0, -40000)$ = **\$65,926.88**. (b) Monthly contributions: $\text{=FV}(5\%/12, 120, -350, 0)$ = **\$54,428.77**. The lump sum wins by **\$11,498** because the full \$40,000 compounds for all 10 years, while the \$350/month contributions average only about 5 years of compounding. Starting early with a larger amount generally beats smaller regular contributions over the same horizon at the same rate.

KEY VOCABULARY

Term	Definition
Annuity	A series of equal payments made at equal time intervals.
Ordinary Annuity	Payments made at the end of each period. The Excel default (type = 0). Used for most loans and savings plans.
Annuity Due	Payments made at the beginning of each period (type = 1). Used for leases and some insurance products.
=PMT()	Excel function that calculates the required periodic payment given rate, nper, and either PV (loan) or FV (savings goal).
Loan Amortization	The process of paying off a loan through regular equal payments, each of which covers some interest and some principal.
Total Interest	The total cost of borrowing: $\text{ABS}(\text{PMT}) \times \text{nper} - \text{PV}$.
Savings Contribution	A periodic deposit toward a future savings goal. In $\text{=FV}()$ and $\text{=PMT}()$, entered as a negative number (outflow).
pmt (argument)	The recurring payment per period in Excel TVM functions. Negative for money leaving your pocket; positive for money received.
Sign Convention	In Excel TVM functions, outflows are negative and inflows are positive. Applies to pv, pmt, and fv.
Structured Settlement	A financial arrangement that pays a fixed amount at regular intervals — a real-world annuity commonly valued using $\text{=PV}()$.

BUS 123 — MATH-M10-L01 — Foundations of Investing: Stocks, Bonds, Mutual Funds & ETFs

BUS123 · MATH-M10-L01 PRE-READING

FOUNDATIONS OF INVESTING: STOCKS, BONDS, MUTUAL FUNDS & ETFS

Meridian Advisory Group Case Study Gerrish School of Business · Endicott College · Fall 2026

CURRENT EVENTS HOOK

When a closely watched company prepares to enter the public markets, headlines often focus on its expected valuation and possible offering price. Investors still must ask a question that has no easy answer: *Is this security worth buying at that price?* The names and numbers change; the need to understand ownership, return, risk, and valuation does not.

That question — *how do I decide whether a security is worth owning?* — is exactly what this module is about. By the time you finish MATH-M10, you will be able to read a stock quote, calculate the return on a bond, and determine the net asset value of a mutual fund. Those are the building blocks of every serious investment decision.

CONNECT TO PRIOR KNOWLEDGE

In MATH-M09, you worked with the time value of money — specifically compound interest and present value. You learned that a dollar today is worth more than a dollar in the future because money can earn a return over time.

That exact principle drives everything in this lesson. When you buy a stock, you are paying today for a share of future profits. When you buy a bond, you are lending money today in exchange for interest payments and a return of principal in the future. When you invest in a mutual fund, you are pooling your present-day dollars with others to capture future market growth. The TVM logic does not disappear — it runs underneath every calculation you will do today.

PART 1 · STOCKS — OWNERSHIP IN A BUSINESS

What a Stock Actually Is

A **stock** is a share of ownership in a company. When Meridian Advisory Group recommends that a client buy 200 shares of a company, that client becomes a partial owner of the business — entitled to a share of its profits (paid as dividends) and a claim on its assets if the company is ever liquidated.

Companies issue stock to raise capital. Rather than taking out a bank loan, a business can sell pieces of itself to thousands of investors, using the proceeds to expand, hire, or pay down debt. In return, investors hope the company grows and that their shares become more valuable.

Common vs. Preferred Stock

There are two main types of stock, and they come with very different rights.

Common stock is what most people mean when they say "buying stock." Common shareholders have voting rights — they can vote on major corporate decisions such as who sits on the board of directors. However, they are last in line when dividends are distributed and last in line if the company goes bankrupt.

Preferred stock generally does not come with voting rights, but preferred shareholders receive declared dividends before common shareholders, and they have a higher claim than common shareholders on remaining assets if the company fails. With **cumulative preferred stock**, a skipped dividend accumulates as **dividends in arrears**. The arrears normally must be paid before common dividends resume, but they generally are not corporate debt unless the board has formally declared the dividend.

Key rule: Preferred shareholders are paid first. Common shareholders receive whatever is left.

Reading a Stock Quote

Every publicly traded company is identified by a **ticker symbol** — a short code used on exchanges for speed and precision. Coca-Cola is KO. Apple is AAPL. Meridian Advisory Group Holdings, in our case study, uses the symbol MAGH.

When you look up a stock quote, you will see several key data points:

Term	What It Means
Last Price	The most recent price at which a buyer and seller completed a trade
Bid Price	The highest price any buyer is currently willing to pay
Ask Price	The lowest price any seller is currently willing to accept
Bid-Ask Spread	Ask minus Bid — the immediate cost of executing a trade
52-Week High / Low	The highest and lowest traded prices over the past year
P/E Ratio	Price per share divided by earnings per share — how much investors pay per \$1 of earnings
Dividend Yield	Annual dividend divided by current price — income return expressed as a percentage

The **bid-ask spread** matters more than most beginners realize. Because you buy at the Ask and sell at the Bid, you are effectively at a small loss the moment a trade is executed. Understanding this gap is one mark of a disciplined investor.

Calculating Dividends

When a company declares dividends, preferred shareholders collect first. The calculation always follows the same steps:

1. Calculate total preferred dividends: **Rate × Par Value × Number of Preferred Shares**
2. Subtract from total dividends declared to find the **remainder for common shareholders**
3. Divide the remainder by the number of common shares to find the **dividend per common share**

Worked Example — Meridian Advisory Group

Meridian's model portfolio company has declared \$10,000 in dividends. The capital structure is: 8% preferred stock, \$50 par value, 1,000 shares; and 4,000 common shares outstanding.

- Preferred dividend per share: $8\% \times \$50 = \4.00
- Total preferred dividend: $\$4.00 \times 1,000 \text{ shares} = \$4,000$
- Remainder for common: $\$10,000 - \$4,000 = \$6,000$
- Common dividend per share: $\$6,000 \div 4,000 \text{ shares} = \1.50 per share

Calculating Return on Investment (ROI)

ROI measures how much you made (or lost) on an investment, expressed as a percentage of what you originally put in. It accounts for the price change, commissions, and any dividends received.

Formula:

$$\text{ROI} = (\text{Net Proceeds} - \text{Total Buy Cost} + \text{Dividends Received}) \div \text{Total Buy Cost}$$

Worked Example — Meridian Advisory Group

An analyst at Meridian bought 200 shares of MAGH at \$39.09 and sold them one year later at \$41.10. The commission rate was 1% on both transactions. MAGH paid a \$1.21 annual dividend per share.

Step	Calculation	Result
Buy cost (before commission)	$200 \times \$39.09$	\$7,818.00
Buy commission	$\$7,818 \times 1\%$	\$78.18
Total buy cost	$\$7,818 + \78.18	\$7,896.18
Sale proceeds (before commission)	$200 \times \$41.10$	\$8,220.00
Sell commission	$\$8,220 \times 1\%$	\$82.20
Net sale proceeds	$\$8,220 - \82.20	\$8,137.80
Total dividends received	$200 \times \$1.21$	\$242.00
ROI	$(\$8,137.80 - \$7,896.18 + \$242.00) \div \$7,896.18$	6.12%

PART 2 · BONDS — LENDING TO A COMPANY

What a Bond Actually Is

When you own a stock, you own a piece of the company. When you own a bond, you are a **creditor** — you have lent money to the company. The company promises to pay you interest on a schedule and return the full face amount (the **par value**, typically \$1,000) at a future date called the **maturity date**.

If a company goes bankrupt, bondholders are creditors and rank ahead of stockholders. Their exact priority depends on the bond's terms and other claims: secured creditors may rank ahead of unsecured bondholders. This creditor status generally makes a particular company's bonds less risky than its stock, though bonds still carry default, interest-rate, inflation, and liquidity risk.

How Bond Prices Work

Bond prices are always quoted as a **percent of par value**. This is the single most important rule in bond math, and the most common source of errors.

A bond quoted at **99.59** does not cost \$99.59. It costs **99.59% of \$1,000 = \$995.90**.

Quoted Price	Dollar Price	Status
Above 100 (e.g., 105.25)	Above \$1,000 (e.g., \$1,052.50)	Premium — bond's rate > market rate
Exactly 100	Exactly \$1,000	Par — bond's rate = market rate
Below 100 (e.g., 98.00)	Below \$1,000 (e.g., \$980.00)	Discount — bond's rate < market rate

Calculating Current Bond Yield

The **current yield** tells you what percentage return you earn annually based on the price you actually paid — not the par value.

Formula:

$$\text{Current Yield} = \text{Annual Interest} \div \text{Dollar Price of Bond}$$

Note that annual interest is always calculated on the **par value**, not the purchase price:

$$\text{Annual Interest} = \text{Stated Interest Rate} \times \text{Par Value (\$1,000)}$$

Worked Example — Meridian Advisory Group

Meridian purchased 5 bonds at a quoted price of 99.59. The stated interest rate is 4%.

Step	Calculation	Result
Dollar price per bond	$99.59\% \times \$1,000$	\$995.90
Total purchase cost (5 bonds)	$\$995.90 \times 5$	\$4,979.50
Annual interest per bond	$4\% \times \$1,000$	\$40.00
Total annual interest	$\$40.00 \times 5$	\$200.00
Current yield	$\$40.00 \div \995.90	4.02%

Notice that the current yield (4.02%) is slightly higher than the stated rate (4%) because the bond was purchased at a slight discount. When you pay less than par, every dollar of interest represents a higher percentage of your actual investment.

PART 3 · MUTUAL FUNDS & ETFs — POOLED INVESTING

The Case for Diversification

Consider a street vendor who sells both umbrellas and sunglasses. On rainy days, umbrella sales are strong. On sunny days, sunglass sales are strong. By carrying both, the vendor earns consistent revenue regardless of

the weather. This is the logic of **diversification** — spreading investments across assets that do not all move in the same direction at the same time.

Research suggests you need at least a dozen carefully selected individual stocks across multiple sectors to be meaningfully diversified. For a beginning investor, building and monitoring that portfolio takes considerable time, skill, and capital. That is where **pooled investment vehicles** come in.

Mutual Funds vs. ETFs

Both mutual funds and ETFs are "baskets" of investments — one purchase gives you exposure to many underlying securities. Their key differences:

Feature	Mutual Fund	ETF (Exchange-Traded Fund)
Pricing	Once per day, at market close	Real-time throughout the trading day
Trading	Priced and traded at end of day only	Trades like a stock — any time during market hours
Minimum Investment	Often \$500–\$3,000 minimum	As low as one share
Tax Efficiency	May distribute capital gains to all holders	More tax-efficient in taxable accounts via "in-kind" exchanges

One important warning: buying a single-sector ETF (technology only, for example) does not provide true diversification. If the entire tech sector declines, a tech-only ETF declines with it. True diversification requires exposure across multiple sectors — technology, healthcare, energy, financials, consumer goods, and more.

Net Asset Value (NAV)

Every mutual fund calculates its **Net Asset Value (NAV)** at the end of each trading day. NAV is the dollar value of one share of the fund.

Formula:

$$\text{NAV} = (\text{Total Market Value of Fund Holdings} - \text{Total Liabilities}) \div \text{Shares Outstanding}$$

Worked Example — Meridian Advisory Group Growth & Income Fund

Input	Value
Market value of fund holdings	\$5,550,000
Current liabilities	\$770,000
Shares outstanding	600,000

- Net assets: $\$5,550,000 - \$770,000 = \$4,780,000$
- NAV per share: $\$4,780,000 \div 600,000 = \7.97 per share

If an investor wanted to purchase 80 shares, they would pay: $\$7.97 \times 80 = \637.60 .

FORMULA REFERENCE

Concept	Formula
Bid-Ask Spread	Ask Price – Bid Price
Dividend Yield	Annual Dividend ÷ Stock Price
Preferred Dividend (total)	Rate × Par Value × Preferred Shares
Common Dividend per Share	(Total Dividends – Preferred Dividends) ÷ Common Shares
Return on Investment (ROI)	(Net Proceeds – Total Buy Cost + Dividends) ÷ Total Buy Cost
Bond Dollar Price	Quoted % × Face Value (\$1,000)
Bond Current Yield	Annual Interest ÷ Dollar Price
Mutual Fund NAV	(Total Assets – Total Liabilities) ÷ Shares Outstanding
Budget Test	=IF(Cost<=Budget,1,0)
Income or Return Test	=IF(OR(Yield>=MinYield,ROI>=MinROI),1,0)
Risk-Proxy Test	=IF(OR(Type="Bond",Type="Diversified mutual fund"),1,0)
Overall Eligibility	=IF(AND(BudgetTest=1,PerformanceTest=1,RiskTest=1),1,0)

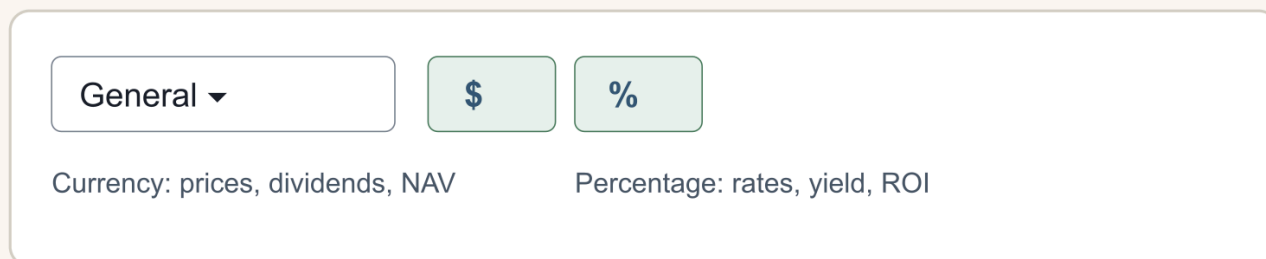
EXCEL SETUP: INPUTS FIRST, OUTPUTS SECOND

Open the starter workbook and select **Live You Try It**. Each practice block uses the same beginner-friendly model:

1. Column **B** names the business item.
2. Yellow cells in column **C** hold either a given input or your formula.
3. Column **D** gives self-check feedback; it is not an answer cell.
4. Column **E** provides a hint.

Use the **Name Box** to confirm the selected cell, then look at the **Formula Bar** to confirm what you typed. These screenshots show Windows Excel. Mac Excel includes the same core tools, but commands may appear in a different position.

Windows Excel · Home tab · Number group



Mac Excel has the same formats, but the controls may be positioned differently.

Build the Bond Model in Cells C51:C60

Select **C51** and enter `1000`. Select **C52** and enter `4%`. Select **C53** and enter `99.59` — not `99.59%`. Select **C54** and enter `5`.

Labeled worksheet model · Live You Try It

Row	B · Item	C · Student cell
51	Face value	<code>1000</code>
52	Stated interest rate	<code>4%</code>
53	Quoted price	<code>99.59</code>
54	Number of bonds	<code>5</code>
56	Dollar price per bond	<code>=C53/100*C51</code>
60	Current yield	<code>=C58/C56</code>

Then enter these formulas one cell at a time:

Select this cell	Type this exact formula	Format	Expected result
C56	=C53/100*C51	Currency, 2 decimals	\$995.90
C57	=C56*C54	Currency, 2 decimals	\$4,979.50
C58	=C52*C51	Currency, 2 decimals	\$40.00
C59	=C58*C54	Currency, 2 decimals	\$200.00
C60	=C58/C56	Percentage, 2 decimals	4.02%

Important distinction · bond quote vs. Excel percentage

Quoted price input

99.59

Type the market quote as a number.

Rate input

4%

Type the coupon rate as a percentage.

Formula: =C53/100*C51 · dividing the quote by 100 converts it to 99.59%.

Reasonableness checks: Because the quote is slightly below 100, the dollar price should be slightly below \$1,000. Because the bond sells below par, current yield should be slightly above the 4% stated rate. If C56 shows \$99,590 or \$9.96, recheck the /100 conversion.

Automatic-recalculation test: Change C53 from 99.59 to 98. C56 should change to \$980.00, C57 to \$4,900.00, and C60 to about 4.08%. Undo the change or type 99.59 again before continuing.

Read an ROI Formula from the Formula Bar

For the ROI practice block, enter the slide givens in C33:C37. Build the helper calculations in C39:C45. Finally, select C46 and type $=(C44-C41+C45)/C41$, then apply Percentage with two decimals. The expected ROI is 6.12%.

Formula bar · active cell C46

C46

$=(C44-C41+C45)/C41$

Read the references aloud:

net sale proceeds – total buy cost + dividends, all divided by total buy cost

Reasonableness check: the selling price is higher than the purchase price and dividends are positive, so ROI should be positive. It should not be enormous because commissions reduce the gain. Change C37 from 1.21 to 0 as a quick test; ROI should fall to about 3.06%, proving the result is linked to the dividend input. Restore 1.21 afterward.

Class Challenge: Begin in Class, Finish for Homework

Keep **Class Challenge** separate from Live You Try It. The challenge is one connected Meridian client model rather than a list of isolated answers:

1. Build the linked calculations for the MAGH stock, Meridian bond, and Meridian Balanced Fund.
2. Link each result into the comparison table instead of retyping calculated values.
3. Test the \$8,000 budget, the income-or-return threshold, and the lesson-supported lower-risk proxy.
4. Select an eligible recommendation and explain the cost, performance evidence, risk proxy, and why the other alternatives fail.

Begin the model during class and finish it as homework. Use the **FormulaReferenceCard** for syntax reminders after you decide which metric or test belongs in a cell. This pre-reading does not provide the graded formulas, completed comparison table, or recommendation.

CHECK YOUR UNDERSTANDING

Answer these seven questions before class. Bring your work — these may be discussed.

1. Meridian recommends a stock currently trading at \$52.40 with a bid of \$52.33 and an ask of \$52.47. What is the bid-ask spread?
2. A company has 500 shares of 6% preferred stock with a par value of \$100, and 2,000 shares of common stock. Total dividends declared: \$8,000. How much does each common shareholder receive per share?
3. An analyst at Meridian bought 150 shares at \$28.50 and sold them one year later at \$31.00. Commission rate: 1% on both sides. No dividend was paid. What is the ROI?
4. A bond is quoted at 103.75. What is the dollar price per bond? Is this a premium, discount, or par bond?
5. A 5% bond is quoted at 97.50 (\$975.00 per bond). What is the current yield?
6. Meridian's Balanced Fund has \$8,400,000 in holdings, \$950,000 in liabilities, and 1,200,000 shares outstanding. What is the NAV per share?
7. An investor wants to buy 100 shares of the fund in Question 6. How much will they pay at NAV?

Answer Key

Q	Answer	Key calculation
1	\$0.14	$\$52.47 - \52.33
2	\$2.50/share	Preferred = $6\% \times \$100 \times 500 = \$3,000$; Remainder = $\$5,000$; $\$5,000 \div 2,000$
3	7.75%	Buy cost = $\$4,346.25$; Sell proceeds = $\$4,611.50$; ROI = $\$265.25 \div \$4,346.25$
4	\$1,037.50; Premium	$103.75\% \times \$1,000$
5	5.13%	Annual interest = $5\% \times \$1,000 = \50 ; $\$50 \div \975.00
6	\$6.21/share	$(\$8,400,000 - \$950,000) \div 1,200,000$
7	\$621.00	$\$6.21 \times 100$

KEY VOCABULARY

Term	Definition
Stock	A share of ownership in a company, entitling the holder to a portion of assets and profits
Common Stock	Stock that provides voting rights; paid dividends after preferred shareholders
Preferred Stock	Stock without voting rights but with priority on dividends and asset claims
Dividends in Arrears	Unpaid dividends owed to cumulative preferred shareholders that accumulate over time
Ticker Symbol	A unique letter code identifying a publicly traded security on an exchange
Bid-Ask Spread	The difference between the highest buyer price (bid) and lowest seller price (ask)
P/E Ratio	Price-to-Earnings ratio — stock price divided by earnings per share
Bond	A debt instrument; the investor lends money to a company in exchange for interest and repayment of principal
Par Value	The face value of a bond, typically \$1,000; the amount repaid at maturity
Current Yield	Annual bond interest divided by the bond's current dollar price
Net Asset Value (NAV)	The per-share value of a mutual fund's holdings after subtracting liabilities
Diversification	Spreading investments across different assets to reduce the impact of any single loss
ETF	Exchange-Traded Fund — a basket of securities that trades on an exchange like a stock

BUS 123 — MATH-M11-L01 — Reading and Analyzing Financial Reports

BUS123 · MATH-M11-L01 · PRE-READING

READING AND ANALYZING FINANCIAL REPORTS

Harborside Medical Center · Fall 2026

Before You Begin

Open `bus123-math-m11-l01-starter.xlsx`. You will use four tabs:

- **START HERE** explains the color key and workflow.
- **Live You Try It** is self-graded practice. Enter inputs in yellow cells and build formulas that reference those cells.
- **Class Challenge** is the graded integrated statement tie-out. The business facts are provided, while the classification, formula, control-check, and explanation cells begin blank. Start it in class and finish it as homework.
- **FormulaReferenceCard** is a syntax reminder after you decide what the business question is asking.

The screenshots in this reading show **Windows Excel**. Mac Excel has the same core commands, but a command may appear in a different position.

PART 1 · THE BALANCE SHEET

The Accounting Equation

The balance sheet is a snapshot at one date. It reports what a business controls and who has claims on those resources.

Formula: `Assets = Liabilities + Owner's Equity`

- **Assets** include cash, accounts receivable, supplies, and equipment.
- **Liabilities** include accounts payable, salaries payable, and loans.
- **Owner's equity** is the residual claim: `Assets - Liabilities`.

If Harborside buys a \$400,000 MRI machine with cash, cash decreases and equipment increases by the same amount. Total assets do not change. If Harborside borrows the money, equipment and liabilities both increase by \$400,000. In both cases, the equation remains balanced.

Harborside check: Total assets are \$3,027,000. Total liabilities are \$1,305,000 and owner's equity is \$1,722,000. Therefore, $\$1,305,000 + \$1,722,000 = \$3,027,000$.

Build a Labeled Excel Model

Put labels in one column, inputs in the next column, and calculated outputs in a separate result area. Do not type numbers directly inside a formula when those numbers already exist in cells.

Balance sheet model — labels, inputs, calculated outputs

A · Label	B · Amount
Cash	\$412,000
Accounts Receivable	\$680,000
Medical Supplies	\$95,000
Equipment (net)	\$1,840,000
Total Assets	=SUM(B4:B7)
Total Liabilities	\$1,305,000
Owner's Equity	\$1,722,000

Selected cell: B8

=SUM(B4:B7)

Expected: \$3,027,000

Reasonableness check

Total assets must equal liabilities + equity.

1. In a blank practice area, type the asset labels in **A4:A7** and their amounts in **B4:B7**: Cash \$412,000, Accounts Receivable \$680,000, Medical Supplies \$95,000, and Equipment (net) \$1,840,000.
2. Type **Total Assets** in **A8**. Select **B8**, type **=SUM(B4:B7)**, and press Enter. Expected result: **\$3,027,000**.
3. Type Total Liabilities \$1,305,000 in **B11** and Owner's Equity \$1,722,000 in **B12**. Select **B13**, type **=SUM(B11:B12)**, and press Enter. Expected result: **\$3,027,000**.
4. Select **B4:B13**, then choose **Home > Number > Accounting Number Format** (or Currency). Use zero decimal places.
5. Reasonableness check: **B8 and B13 must match**. If not, check missing lines, signs, and cell references.

Input-change test: Change Cash in **B4** from \$412,000 to \$422,000. B8 should automatically become **\$3,037,000**. B13 will remain \$3,027,000, revealing a \$10,000 imbalance until the financing side is updated. Press Undo when finished.

PART 2 · THE INCOME STATEMENT

The income statement covers a period of time and explains whether operations produced a profit. For this service-business model:

- Net Revenue – Cost of Services = Gross Profit
- Gross Profit – Operating Expenses = Net Income

Harborside reports net revenue of \$4,250,000, cost of services of \$2,380,000, and operating expenses of \$1,420,000. Gross profit is **\$1,870,000** and net income is **\$450,000**.

Retail businesses often use $\text{COGS} = \text{Beginning Inventory} + \text{Net Purchases} - \text{Ending Inventory}$. That inventory formula is useful context, but Harborside's workbook uses **Cost of Services**, not retail inventory.

Income statement model — formulas reference labeled inputs

fx **=B8-B6**

Selected cell: B9

Label	Amount
Net Revenue	\$4,250,000
Cost of Services	\$2,380,000
Operating Expenses	\$1,420,000
Gross Profit	=B4-B5
Net Income	=B8-B6

Modeling distinction

Inputs: typed amounts

Outputs: formulas

Profit ≠ cash

Expected net income: \$450,000

1. Type **Net Revenue**, **Cost of Services**, and **Operating Expenses** in **A4:A6**. Enter the three amounts in **B4:B6**.
2. Type **Gross Profit** in **A8**. Select **B8**, type **=B4-B5**, and press Enter. Expected result: **\$1,870,000**.
3. Type **Net Income** in **A9**. Select **B9**, type **=B8-B6**, and press Enter. Expected result: **\$450,000**.
4. Format **B4:B9** as Accounting or Currency with zero decimals.
5. Reasonableness checks: gross profit should be less than revenue; net income should be less than gross profit when operating expenses are positive.

Input-change test: Change Operating Expenses in **B6** to **\$1,500,000**. B9 should recalculate to **\$370,000**. Press Undo when finished.

PART 3 · THE STATEMENT OF CASH FLOWS

Profit and cash are not the same. Under accrual accounting, revenue is recorded when earned and expenses when incurred, even if cash moves later. Accounts receivable can therefore increase profit before cash arrives.

The statement of cash flows groups cash movements into:

Section	What It Covers	Harborside
Operating activities	Day-to-day collections and payments	\$310,000
Investing activities	Long-term asset purchases and sales	(\$580,000)
Financing activities	Borrowing, repayment, and owner transactions	\$220,000

Cash-flow model — signs matter

Cash-flow section		Amount
Operating		\$310,000
Investing		(\$580,000)
Financing		\$220,000
Net change	=SUM(B4:B6)	(\$50,000)
Closing cash	=B9+B8	\$412,000

Outflow ≠ positive expense

Enter investing cash flow as
-580000

Reconciliation

Opening cash + net change
must equal closing cash.

1. Enter Operating, Investing, and Financing Cash Flow in **A4:A6** and their values in **B4:B6**. Enter the investing outflow as **-580000**, not as a positive number.
2. Select **B8**, type **=SUM(B4:B6)**, and press Enter. Expected net change in cash: **-\$50,000**.
3. Enter Opening Cash **\$462,000** in **B9**. Select **B10**, type **=B9+B8**, and press Enter. Expected closing cash: **\$412,000**.
4. Use Accounting format with zero decimals. Negative cash flows should display with a minus sign or parentheses.
5. Reasonableness check: closing cash must equal opening cash plus the net change, and it should match the balance-sheet cash line for the same date.

Input-change test: Change Financing Cash Flow in **B6** from **\$220,000** to **\$250,000**. Net change should become **-\$20,000**, and closing cash should become **\$442,000**. Press Undo when finished.

How the Three Statements Tie Together

The Class Challenge keeps the statement build intentionally simplified, but it requires the same connections managers use to test a model:

- **Net income flows into equity.** With owner distributions entered as a negative amount, a simplified roll-forward is **Beginning Equity + Net Income + Owner Distributions = Ending Equity**.
- **Ending cash flows into the balance sheet.** The closing cash calculated in the cash-flow section should be linked to the balance-sheet cash line instead of typed again.
- **The accounting equation must balance.** **Total Assets - Total Liabilities and Equity** should equal zero.
- **The cash movement must reconcile.** **Cash-Flow Ending Cash - Balance-Sheet Cash** should equal zero.

These checks do not create the answer. They tell you whether the links, signs, and subtotals work together.

PART 4 · TREND AND RATIO ANALYSIS

Trend Analysis

Trend analysis expresses each year relative to one base year:

Excel pattern: `=CurrentYearCell/BaseYearCell`

The base year is 100%. With 2024 net income of \$320,000 and 2026 net income of \$450,000, `450000/320000` = 140.625%, displayed as **141%** with zero decimal places. This means 2026 income is 41% above the base-year amount—not that it increased by 141%.

Ratio Analysis

Ratios standardize financial relationships. Their interpretation depends on the organization, industry, accounting choices, and comparison period; there is no universal “good” ratio.

Ratio	Excel pattern	What it asks
Current ratio	<code>=CurrentAssets/CurrentLiabilities</code>	Can current assets cover current obligations?
Quick ratio	<code>=(Cash+AccountsReceivable)/CurrentLiabilities</code>	Can the most liquid assets cover current obligations?
Debt to assets	<code>=TotalLiabilities/TotalAssets</code>	What share of assets is financed by liabilities?
Return on equity	<code>=NetIncome/OwnerEquity</code>	How much income was generated per dollar of equity?
Profit margin	<code>=NetIncome/NetRevenue</code>	How much profit remains per revenue dollar?
Asset turnover	<code>=NetRevenue/TotalAssets</code>	How efficiently are assets generating revenue?

Windows Excel · Home tab · Number group

Accounting ▼

%

, 000

.0 → .00

Currency / Accounting

Percentage

Decimal places

Select the result cells first, then apply the format.

Mac Excel contains the same number formats, but the controls may be positioned differently.

For **Live You Try It rows 13–17**, place the stated amounts in the yellow input cells, select the yellow result cell in column I, and type a formula using those input-cell references. Use **Percentage** format for trends, profit margin, debt to assets, and ROE. A current or quick ratio is usually shown as a number such as 2.43, while asset turnover is commonly shown as a number such as 1.40.

Expected practice results from the slides are:

- 2026 net revenue trend: $\$4,250,000 \div \$3,800,000 = 111.84\%$ (about **112%**).
- 2026 net income trend: $\$450,000 \div \$320,000 = 140.63\%$ (about **141%**).
- Profit margin: $\$450,000 \div \$4,250,000 = 10.59\%$.
- Debt to assets: $\$1,305,000 \div \$3,027,000 = 43.11\%$.

- ROE: $\$450,000 \div \$1,722,000 = 26.13\%$.

Reasonableness checks: percentages produced by division should be stored as decimals in Excel; do not multiply by 100 and then also apply Percentage format. Compare ratios with prior years or a relevant peer set before calling them strong or weak.

COMPLETE THE WORKBOOK SAFELY

For **Live You Try It**, keep using the practice workflow: enter the slide givens in yellow cells, build formulas that reference those cells, read the feedback, and undo any input-change test.

For **Class Challenge**:

1. Classify each provided business fact by its primary statement.
2. Link the fact cell into the matching statement line. Do not retype the amount inside a formula.
3. Build the income-statement, balance-sheet, and cash-flow subtotals using the labeled rows.
4. Link net income into the equity section and cash-flow ending cash into the balance sheet.
5. Confirm the balance check, net income link check, and cash tie-out all equal zero.
6. Calculate the numerical basis for the red flag, then write a concise management interpretation supported by at least one completed result.
7. Begin the activity in class and submit the completed starter workbook after finishing the remaining tie-out and explanation as homework.

The FormulaReferenceCard supplies syntax patterns, but it does not identify the correct source cells or provide completed graded answers.

CHECK YOUR UNDERSTANDING

1. Why must both sides of the accounting equation use amounts from the same date?
 2. Why can a profitable organization still have a cash shortage?
 3. If a trend calculation displays 140.63%, how far above the base year is the current year?
 4. Why should formulas reference input cells instead of repeating numbers inside formulas?
 5. What comparison would you want before deciding whether a 43.11% debt-to-assets ratio is acceptable?
-

KEY VOCABULARY

Term	Definition
Income statement	Revenues, expenses, and profit over a period
Accrual basis	Records revenue when earned and expenses when incurred, regardless of cash timing
Accounts receivable	Amounts customers or payers owe for services already provided
Gross profit	Net revenue minus cost of services or cost of goods sold
Statement of cash flows	Cash inflows and outflows grouped as operating, investing, and financing
Trend analysis	Expresses each period relative to a chosen base period
Ratio analysis	Uses standardized relationships to support comparisons

BUS 123 — MATH-M12-L01 — Business Statistics: Center, Spread, and Correlation

BUS123 · MATH-M12-L01 · PRE-READING

BUSINESS STATISTICS: CENTER, SPREAD, AND CORRELATION

Harborside Medical Center · Fall 2026

BEFORE YOU BEGIN

Open `bus123-math-m12-l01-starter.xlsx`. Its four tabs have different jobs:

- **START HERE** explains the workflow and color key.
- **Live You Try It** is self-graded practice tied to slide examples.
- **Class Challenge** is the graded statistical operations investigation. Its 24 raw shift observations are already populated; the analysis, evidence-link, and recommendation cells begin blank.
- **FormulaReferenceCard** is a syntax reminder after you decide which statistic answers the question.

The screenshots in this reading show **Windows Excel**. Mac Excel has the same functions and number formats, but commands may appear in different positions.

PART 1 · ORGANIZE A STATISTICAL MODEL

Keep labels, inputs, and calculated outputs separate. Labels explain the data, input cells contain observed values, and output cells contain formulas that reference the inputs.

Wait-time model — inputs first, formulas second

Row	C	D	E	F	G
Wait time inputs	12	15	18	11	42
Mean	=AVERAGE(C8:G8)				19.6
Median	=MEDIAN(C8:G8)				15.0
Difference	=C10-C11				4.6

Reasonableness

$11 \leq \text{mean} \leq 42$

Outlier pulls mean up

Outputs are formulas,
not typed answers.

In **Live You Try It**, the first yellow input row is **C8:G8**. Enter the five wait times from the slides: 12, 15, 18, 11, and 42.

1. Select **C10**, type `=AVERAGE(C8:G8)`, and press Enter. Expected result: **19.6 minutes**.
2. Select **C11**, type `=MEDIAN(C8:G8)`, and press Enter. Expected result: **15 minutes**.
3. Select **C12**, type `=C10-C11`, and press Enter. Expected result: **4.6 minutes**.
4. Select **C10:C12**, open **Home > Number**, and use Number format with **1 decimal place**.
5. Reasonableness check: the mean must fall between the minimum (11) and maximum (42). Because 42 is a high outlier, the mean should be above the median.

Input-change test: Change the 42 in **G8** to 22. C10 should recalculate to **15.6**, C11 should remain **15.0**, and C12 should become **0.6**. Press Undo when finished.

Mean, Median, and Mode

- **Mean:** `=AVERAGE(range)` adds all numeric values and divides by their count. It uses every value, so outliers affect it.
- **Median:** `=MEDIAN(range)` returns the middle value after sorting. It is less sensitive to outliers.
- **Mode:** `=MODE.SNGL(range)` returns one most-frequent **numeric** value. If no numeric value repeats, Excel returns `#N/A`.
- **Multiple numeric modes:** `=MODE.MULT(range)` returns all tied modes. In current Microsoft 365 Excel, the results spill into neighboring cells; older versions may require legacy array entry.

For text categories such as insurance type or diagnosis code, Excel's `MODE.SNGL` does not work. Use a frequency table, PivotTable, or `COUNTIF`-based summary to find the most common text category.

PART 2 · STANDARD DEVIATION MEASURES SPREAD

Two groups can have the same mean but very different consistency. Standard deviation summarizes how far observations typically spread around the mean, in the same unit as the original data.

- **Sample standard deviation:** `=STDEV.S(range)` estimates spread when the data is a sample from a larger process or population.
- **Population standard deviation:** `=STDEV.P(range)` describes spread when the data contains every observation in the population you have defined.

The choice depends on the business question—not simply whether the dataset is small. A complete month of transactions can be a population if the question is only about that month, but a sample if the goal is to infer future months.

Standard deviation — define the population before choosing the function

fx `=STDEV.S(C17:G17)`

Sample · STDEV.S

Use observations to estimate a broader process or population.

Expected: 3.16 hours

Population · STDEV.P

Use every observation in the population defined by the question.

Expected: 2.83 hours

Both results use the same inputs; the denominator differs.

In **Live You Try It**, enter 4, 6, 8, 10, and 12 in **C17:G17**.

1. Select **C19**, type `=AVERAGE(C17:G17)`, and press Enter. Expected result: **8.00 hours**.
2. Select **C20**, type `=STDEV.S(C17:G17)`, and press Enter. Expected result: **3.16 hours**.
3. Select **C21**, type `=STDEV.P(C17:G17)`, and press Enter. Expected result: **2.83 hours**.
4. Format **C19:C21** as Number with **2 decimal places**.
5. Reasonableness checks: standard deviation cannot be negative; identical values would produce zero; `STDEV.S` is slightly larger than `STDEV.P` for the same non-identical values.

Input-change test: Change the 12 in **G17** to 20. The mean should become **9.60**, sample SD **6.07**, and population SD **5.43**. Press Undo when finished. The larger spread should match what you see in the values.

Windows Excel · Home tab · Number group

Number ▼

Number format

.0 → .00

Increase decimals

.00 → .0

Decrease decimals

Select calculated output cells before applying the format.

Mac Excel has the same formats, but the controls may be positioned differently.

PART 3 · CORRELATION MEASURES A LINEAR RELATIONSHIP

Correlation describes the direction and strength of a **linear** relationship between two numeric variables. Pearson's correlation coefficient, r , ranges from -1 to +1.

- A positive value means the variables tend to move in the same direction.
- A negative value means they tend to move in opposite directions.
- A value near zero means little linear relationship; a curved relationship may still exist.
- Correlation is undefined if either input range has no variation.

Correlation model — keep paired observations in the same column

Row	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Wait	12	15	10	22	30	8	18
Satisfaction	8.4	7.9	8.8	6.5	5.2	9.1	7.1

=CORREL(C26:I26,C27:I27)

-0.9976 → -1.00

Important

Same day =
same column

Correlation
does not prove
causation.

Result must be between -1 and +1. Format as Number, not Percentage.

In **Live You Try It**, enter the seven wait times in **C26:I26** and matching satisfaction scores in **C27:I27**. Each column must describe the same day in both rows.

1. Select **C29**.
2. Type **=CORREL(C26:I26,C27:I27)** and press Enter.
3. Format **C29** as Number with **2 decimal places**.
4. Expected result from the lesson data: **-1.00** when displayed to two decimals (unrounded value approximately **-0.9976**).
5. Reasonableness checks: the answer must be between -1 and +1; the two ranges must have the same number of cells; wait and satisfaction values must stay paired by day.

Input-change test: Change Friday satisfaction in **G27** from **5.2** to **7.2**. C29 should recalculate to approximately **-0.82**. Press Undo when finished.

Correlation Does Not Prove Causation

A strong correlation does not prove that changing one variable will cause the other to change. Staffing, case severity, day of week, measurement choices, or another variable could influence both. Treat correlation as evidence for a question to investigate, not as proof of a mechanism.

Fixed “weak/moderate/strong” cutoffs are only rough conventions. Context, sample size, data quality, outliers, and decision risk all matter.

USE THE WORKBOOK SAFELY

1. Read the question and identify the requested statistic.
2. Enter only the stated observations in the yellow input cells.
3. Select the yellow answer cell and type a formula beginning with **=** that references the input range.
4. Apply an appropriate format: minutes or hours as Number; correlation as Number, not Percentage.
5. Check the range, units, sign, and plausible size of the result.
6. Change one practice input to confirm automatic recalculation, then undo the test.

CLASS CHALLENGE · START IN CLASS, FINISH AS HOMEWORK

The **Class Challenge** uses 24 Harborside shift observations instead of disconnected mini-problems. Do not edit the raw-data table in **B9:I33**.

During class (about 25 minutes):

1. Inspect the raw operational observations and verify which rows are AM versus PM.
2. Build the descriptive-statistics summary in **E37:E48** using bounded ranges from the raw table.
3. Identify the unusual observation by its ID, compare the mean with and without it, and compare AM versus PM center and spread.

Finish as homework before the next class:

1. In **E49**, use `CORREL` on the complete paired **Average wait** and **Satisfaction** columns. Do not substitute staff count, patient volume, or a partial group range.
2. Link the checked group-gap and correlation results into **E53:E54** with formulas rather than retyping the numbers.
3. Write a 2–4 sentence recommendation in **B59:I61**. Recommend one staffing, service, or process action; reference at least one linked result; and state the relationship carefully without claiming that correlation proves causation.
4. Save and upload the completed workbook to Canvas.

The FormulaReferenceCard is a syntax reminder, but you are responsible for choosing the correct statistic and the correct raw-data range.

FORMULA REFERENCE

Function	Excel syntax	Important check
AVERAGE	<code>=AVERAGE (C8 : G8)</code>	Outliers can pull the mean.
MEDIAN	<code>=MEDIAN (C8 : G8)</code>	Uses the middle after sorting conceptually.
MODE.SNGL	<code>=MODE . SNGL (B2 : B8)</code>	Numeric data only; may return <code>#N/A</code> .
MIN / MAX	<code>=MIN (B2 : B8) / =MAX (B2 : B8)</code>	Useful for checking the observed range.
STDEV.S	<code>=STDEV . S (C17 : G17)</code>	Use when estimating a broader process/population.
STDEV.P	<code>=STDEV . P (C17 : G17)</code>	Use for the entire defined population.
CORREL	<code>=CORREL (C26 : I26 , C27 : I27)</code>	Ranges must be paired and equally sized.

CHECK YOUR UNDERSTANDING

1. Why is the median lower than the mean for the five wait times?
2. Why can `MODE . SNGL` return `#N/A` ?
3. How does the business question determine whether `STDEV . S` or `STDEV . P` is appropriate?
4. What does a negative correlation say about direction?

5. Why is a correlation close to -1 not proof that wait time causes satisfaction to fall?
6. Why should correlation ranges be checked column by column before calculating?
-

KEY VOCABULARY

Term	Definition
Mean	Arithmetic average of numeric values
Median	Middle value after values are ordered
Mode	Most frequently occurring value
Outlier	Observation far from most other values
Standard deviation	Spread around the mean, expressed in the data's unit
Sample	Observations used to learn about a larger process or population
Population	The complete set defined by the business question
Correlation	Direction and strength of a linear relationship
Causation	A change in one variable directly produces a change in another

BUS 123 PRE-READINGS

Excel

4 official readings

Navigating Excel, Formatting & Cell References

NAVIGATING EXCEL, FORMATTING & CELL REFERENCES

Pre-Reading — Complete before your first Excel class session

CONNECT TO PRIOR KNOWLEDGE

In the course introduction you practiced setting up your BUS 123 folder in OneDrive and saving your first file. That habit — save first, save often, save to the right place — carries directly into this module. Every Excel workbook you build this semester will live in that folder structure. Now that your workspace is organized, you are ready to work inside Excel itself. Excel is the most widely used business software in the world. Accountants use it to build financial models. Operations managers use it to track inventory. Marketing teams use it to analyze campaign data. Regardless of your career path in business, fluency with Excel is a baseline expectation. This module builds that foundation.

CONCEPT EXPLANATION

Part A — The Excel Interface

When you open a new Excel workbook you see a grid of rows and columns. Every intersection of a row and column is a cell. Cells are addressed by column letter and row number: A1, B3, C10. The five zones you must know before class are listed below.

Zone	Location	What It Does
Quick Access Toolbar	Top-left corner	Stores your most-used commands for one-click access
Ribbon	Top band	Organizes all commands into tabs: Home, Insert, Formulas, Data, and more
Name Box	Left of formula bar	Shows the address of the selected cell (e.g., C3)
Formula Bar	Right of Name Box	Shows the raw content of the active cell — a value or a formula
Sheet Tabs	Bottom of window	Each tab is a separate worksheet within the same workbook file

Navigation shortcuts save time in large files. Tidal Goods Co., one of our case study businesses, manages a product catalog with hundreds of rows. Three shortcuts you must know:

- **Ctrl+End** jumps to the last used cell.
- **Ctrl+Home** returns to A1.
- **Ctrl+Arrow** jumps to the edge of a contiguous data range.

Practice these before class.

Part B — Formatting Essentials

Formatting does not change the underlying value in a cell — it changes how that value is displayed. Professional formatting serves one purpose: communication. A reader should be able to glance at your spreadsheet and immediately understand what type of data they are looking at.

The three most important formatting decisions are:

1. **Number format** — is this a dollar amount, a percentage, or a plain number? Apply the correct format every time.
2. **Alignment** — numbers always align right; text always aligns left. This is a professional convention you must follow.
3. **Headers and borders** — bold your column headers and add borders to separate data sections from totals.

Common Mistake

0.15 is not the same as 15%. If you type a decimal like 0.15 into a cell without applying Percentage format, your reader has no idea whether that represents 15%, 15 cents, or some other quantity. Always apply number format before sharing any file.

Part C — Cell References: Relative and Absolute

This is the most important concept in the lesson. Every formula you write in Excel contains cell references. A reference tells Excel which cell to pull data from. There are two kinds.

Relative references (written as A1, B3, etc.) update automatically when you copy a formula to a new location. If you copy a formula one row down, every relative reference shifts one row down with it. This is the default behavior — and it is usually exactly what you want when applying the same calculation to multiple rows of data.

Absolute references (written as \$A\$1, \$B\$1, etc.) are locked. The dollar signs tell Excel: do not move this reference when I copy the formula. Use an absolute reference whenever every formula in a range needs to point back to the same single cell — such as a tax rate, a markup percentage, or a fixed cost stored in one place.

Tidal Goods Co. example. Tidal Goods stores its 40% product markup rate in cell B1. Column B holds product costs; column C will hold the selling price calculated as: $\text{cost} \times (1 + \text{markup rate})$. The formula in C3 should be `=B3*(1+$B$1)`.

- `B3` is relative — it moves down to B4, B5, B6 as we copy the formula.
- `$B$1` is absolute — it always reads the 40% markup stored in B1, regardless of which row the formula is copied to.

The F4 shortcut. After typing a cell address inside a formula, press F4 to cycle through the four reference modes: `$A$1` (both locked) → `A$1` (row locked) → `$A1` (column locked) → `A1` (relative, back to start). Professional analysts never type dollar signs manually — they always use F4.

Formula Reference	Example	Copy Behavior
Relative	<code>=B3 * (1+B1)</code>	Reference moves when formula is copied
Absolute	<code>=B3 * (1+\$B\$1)</code>	Reference stays fixed regardless of copy destination
Row locked	<code>=B3 * (1+A\$1)</code>	Row number is locked; column can shift
Column locked	<code>=B3 * (1+\$A1)</code>	Column letter is locked; row can shift

Shortcut

Press F4 after typing a cell address in a formula to cycle through all four modes.

CHECK YOUR UNDERSTANDING

Answer these seven questions on your own before class. Answers appear at the end of this document.

1. Which part of the Excel interface shows the address of the currently selected cell?
2. A cell displays \$14.99. You click the cell and the Formula Bar shows `=B3*1.4`. What does this tell you about what is stored in the cell?
3. You press `ctrl+End` in a large worksheet. What happens?
4. You type 0.15 into a cell that represents a 15% discount rate. What formatting step should you take, and why?
5. A formula in C3 reads `=B3 * ( 1+B1 )`. When you copy this formula to C4, what does the formula in C4 become?
6. You want C4 to always reference the markup rate in B1 regardless of where the formula is copied. How do you rewrite the reference to B1?
7. You have typed `=B3` inside a formula. What does pressing F4 once do?

KEY VOCABULARY

Term	Definition
Cell address	The unique identifier for a cell, expressed as a column letter followed by a row number (e.g., B3, C10).
Name Box	The box to the left of the Formula Bar that displays the address of the currently selected cell.
Formula Bar	The bar above the worksheet grid that displays the raw content — value or formula — of the active cell.
Number format	A display setting that controls how a numeric value appears in a cell (currency, percentage, decimal places, etc.) without altering the underlying value.
Relative reference	A cell reference that automatically adjusts when a formula is copied to a new location.
Absolute reference	A cell reference locked with dollar signs (<code>\$A\$1</code>) that does not change when a formula is copied.
F4	Keyboard shortcut that cycles a cell reference through four modes: <code>\$A\$1</code> , <code>A\$1</code> , <code>\$A1</code> , <code>A1</code> .
Ribbon	The horizontal band of tabs and command groups at the top of the Excel window.

CHECK YOUR UNDERSTANDING — ANSWER KEY

1. The Name Box (located to the left of the Formula Bar).
2. The cell contains a formula (`=B3*1.4`), not a typed value. The cell displays the formula's result (\$14.99).
3. Excel jumps to the last used cell in the worksheet — the bottom-right corner of your data range.
4. Apply Percentage format (Home tab > Number group). Without it, a reader cannot tell whether 0.15 represents 15%, \$0.15, or another quantity.
5. `=B4*(1+B2)` . Both references are relative, so each shifts one row down when copied.
6. Write `=$B$1` . The dollar signs lock both the column and row, so the reference always points to B1 regardless of where the formula is copied.
7. F4 changes `=B3` to `=$B$3` — locking both the column and row with dollar signs.

BUS 123 - EXCEL-M02-L01 - Formatting and Organizing Worksheets

BUS 123 - EXCEL-M02-L01 - FORMATTING AND ORGANIZING WORKSHEETS

Course: Solving Business Problems with Technology - Fall 2026 - **Track:** EXCEL - **Module:** M02 - **Lesson:** L01 **Case Study Company:** Anchor & Oak Events

1 - CONNECT TO PRIOR KNOWLEDGE

In Excel M01, you learned how to navigate a workbook, enter data, and build formulas with cell references. A correct formula is essential, but correctness alone does not make a worksheet useful. Managers also need to understand the result quickly, identify totals, distinguish inputs from formulas, and follow the structure without asking the workbook creator for an explanation.

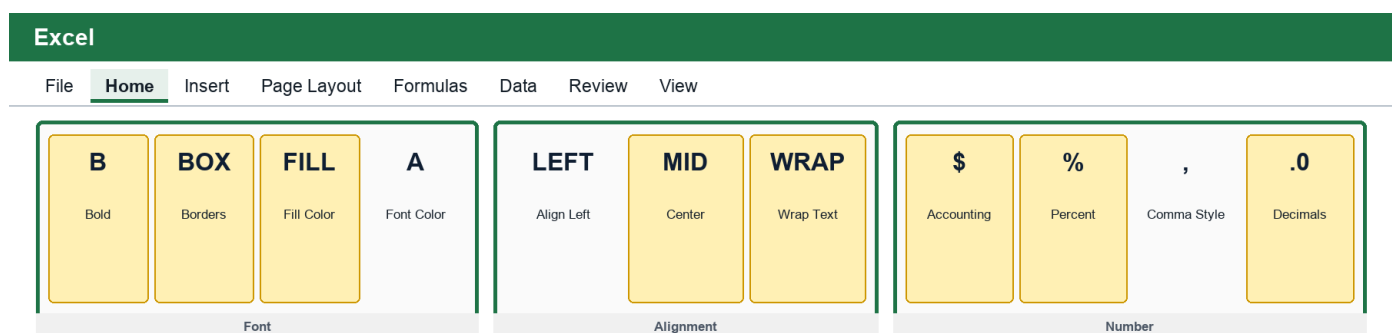
Formatting is therefore part of business communication. A workbook with inconsistent dates, unlabeled currency, random colors, and hidden totals creates doubt even when every calculation is correct. A well-organized workbook makes the same calculations easier to audit and easier to trust.

Anchor & Oak Events tracks event dates, guest counts, revenue, catering costs, staffing costs, and net profit. In this lesson, you will turn that raw booking summary into a manager-ready worksheet.

2 - CORE CONCEPTS

Where the Formatting Tools Are in Windows Excel

Most of the tools in this lesson are on the **Home** tab of the Windows Excel ribbon. Begin every formatting task by selecting the cell or range you want to change. Then choose the command. If you format first and select later, Excel may apply the change somewhere you did not intend.



The exact spacing of the ribbon may vary with screen size or Microsoft 365 updates. If a command is hidden, widen the Excel window or select the small drop-down arrow for its group. The command names and group names remain the best landmarks.

Part A - Number Formats Communicate Meaning

Excel stores a value separately from the way it appears. The number 4800 can be displayed as 4800 , \$4,800 , \$4,800.00 , or another format without changing the underlying value.

Choose a number format based on the business meaning:

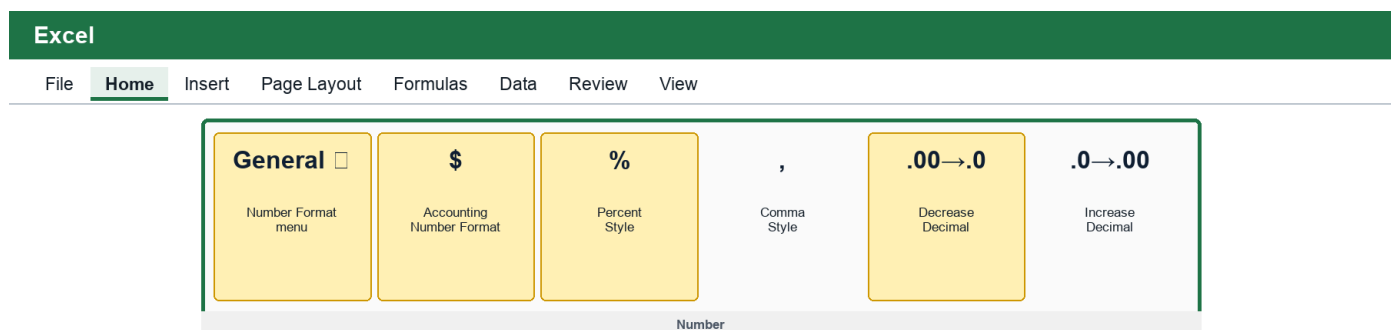
Data type	Recommended display	Why it helps
Event date	15-Mar-26 or another consistent date style	Readers recognize the date immediately and can sort correctly.
Revenue and costs	Currency or Accounting	The dollar unit is visible and decimals are consistent.
Guest count	Whole number	Partial guests would not make sense.
Percentage	Percentage	A stored decimal such as 0.15 displays as 15% .
General count or ID	Number or General	No currency or percentage meaning is implied.

Watch Out - Formatting Does Not Change the Value

Currency formatting does not multiply a number by dollars, and Percentage formatting does not repair an incorrect formula. Always confirm the underlying value and formula first.

Apply a Number Format

1. Select the cells that contain the values.
2. Select the **Home** tab.
3. Find the **Number** group.
4. Open the **Number Format** menu, which usually displays **General** before a format is applied.
5. Choose **Currency**, **Accounting**, **Percentage**, **Short Date**, or another format that matches the meaning of the data.
6. Use **Increase Decimal** or **Decrease Decimal** to control only the number of digits displayed after the decimal point.



For the Anchor & Oak worksheet, select the Revenue, Catering Cost, Staffing Cost, and Net Profit values and apply **Accounting Number Format**. If cents are not useful for the manager's decision, select **Decrease Decimal** until no decimal places are displayed. This changes the display, not the stored values.

Windows Shortcut

Press **Ctrl+1** to open the Format Cells dialog box. Use its **Number** tab when you need more choices, such as a specific date pattern, currency symbol, or negative-number display.

Part B - Alignment and Visual Hierarchy

Alignment helps readers scan a table:

- Text labels are usually left-aligned.
- Numbers are usually right-aligned so place values line up.
- Column headers may be centered when that improves readability.
- A title should be visually distinct from the data table.
- A total row should be more prominent than detail rows.

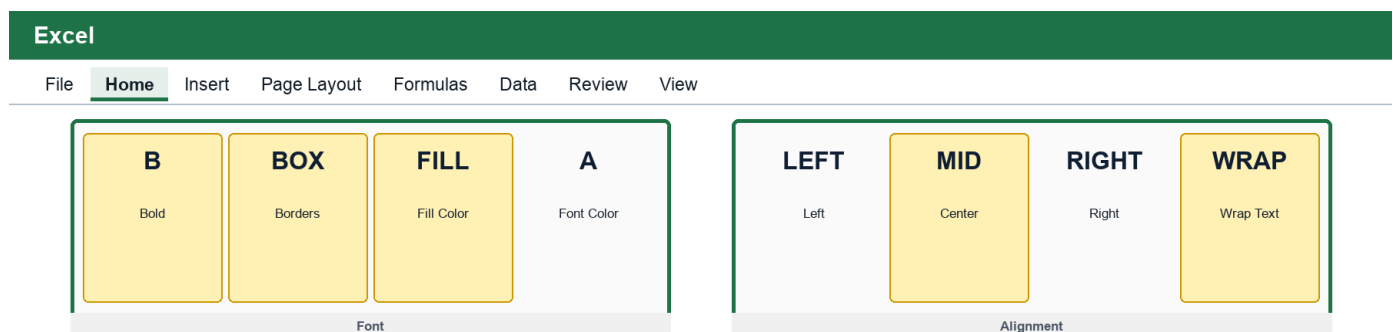
Use bold type, fill color, and borders sparingly. Their purpose is to establish reading order, not to decorate every cell. A quiet header fill and a clear total-row border are more effective than many unrelated colors.

Format a Header and Total Row

To create a clear header row:

1. Select the cells containing the column headings.
2. On **Home**, find the **Font** group and select **Bold**.
3. Open **Fill Color** and choose one quiet, high-contrast fill.
4. Open **Font Color** only if the fill requires a lighter or darker text color for readability.
5. In the **Alignment** group, select **Center** for short headings or **Wrap Text** for headings that should remain readable without making the column extremely wide.

To emphasize a total row, select the entire row of totals, apply **Bold**, and use the **Borders** menu to add a **Top Border** or **Double Bottom Border**. Avoid adding heavy borders around every cell.



Part C - Organize the Worksheet for Scanning

A manager should be able to answer these questions in about 30 seconds:

1. What does this worksheet measure?

2. Where are the input values?
3. Which cells contain calculated results?
4. Where is the total?
5. Are any results unusually high, low, or risky?

Useful organization choices include:

- One clear title and subtitle
- One header row with consistent labels
- Detail rows in a predictable order
- Banded rows for long tables
- A distinct total row
- No merged cells inside the data table
- Consistent formats down each column
- Freeze panes when headers would otherwise scroll off screen

Part D - Freeze Panes and Banded Rows

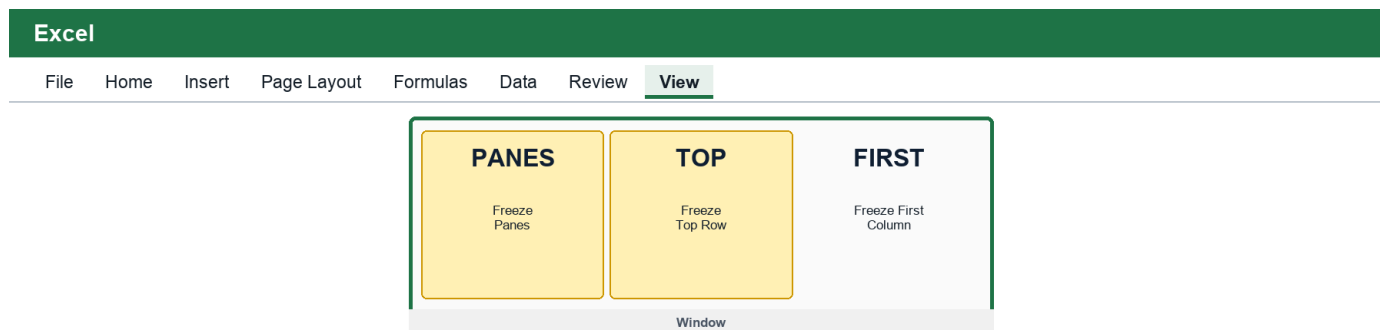
Freeze Panes keeps selected rows or columns visible while the rest of the worksheet scrolls. For a long event list, freezing the header row lets the reader see labels even at row 60.

Banded rows use alternating, subtle fills to help the eye follow one record across a wide table. Banding should be quiet. If every row uses a strong color, the pattern becomes distracting instead of helpful.

Keep the Header Visible While Scrolling

1. Select the **View** tab.
2. Find the **Window** group.
3. Select **Freeze Panes**.
4. Choose **Freeze Top Row** when the headers are in row 1.
5. Scroll downward and confirm that the header remains visible.

If the worksheet has a title above its headers, select the cell immediately below the header row, then choose **View** → **Freeze Panes** → **Freeze Panes**. Excel freezes the rows above and the columns to the left of the selected cell.



Add Banded Rows with Format as Table

1. Select any cell in the data range.

2. Select **Home** → **Format as Table** in the **Styles** group.
3. Choose a light table style with subtle banding.
4. Confirm the full data range in the Create Table dialog box.
5. Select **My table has headers**, and then select **OK**.

Format as Table does more than add color: it converts the range to an Excel table and adds filter buttons. Use it only when those table features fit the worksheet. If the activity calls for a normal cell range, apply light alternating fills manually instead.

Part E - Color With Purpose

Use color only when it communicates a role:

Color role	Possible use
Header	Identifies the table structure.
Input	Shows where a student or analyst should type.
Formula/result	Distinguishes calculated cells from inputs.
Warning	Calls attention to an exception that needs review.

Do not rely on color alone. Pair color with labels, number formats, borders, or text so the workbook remains understandable when printed or viewed by someone with color-vision differences.

Conditional Formatting can make that purpose respond to the data. For example, a manager may need to notice unusually low Net Profit values. Select the Net Profit cells, choose **Home** → **Conditional Formatting**, and apply one rule that highlights the condition you can explain. The rule, range, and business reason matter more than one prescribed color palette.

3 - FORMULA AND FORMATTING REFERENCE

Formatting supports calculations; it does not replace them. The Anchor & Oak booking summary uses these core formulas:

Business question	Excel syntax	Meaning
What is one event's net profit?	<code>=Revenue-Catering-Staffing</code>	Subtract direct event costs from revenue.
What is total revenue?	<code>=SUM(RevenueRange)</code>	Add all event revenue values.
What is total net profit?	<code>=SUM(NetProfitRange)</code>	Add all calculated event profits.
What is revenue per guest?	<code>=Revenue/Guests</code>	Divide event revenue by attendance.

For the Spring Gala example:

- Revenue: \$4,800
- Catering: \$1,820
- Staffing: \$0
- Net profit formula: `=4800-1820-0`
- Net profit: **\$2,980**

For four event profits of \$2,980, \$1,990, \$5,510, and \$3,290:

- Total net profit formula: `=SUM(profit_cells)`
- Total net profit: **\$13,770**

The formulas should reference worksheet cells rather than typing the numbers directly into the calculation. Cell references make the workbook update when an input changes.

4 - FIVE-POINT FORMATTING AUDIT

Before sharing a worksheet, perform this audit:

Check	Question to ask
1. Labels	Are the title, headers, and units clear?
2. Formats	Are dates, currency, percentages, and counts displayed consistently?
3. Alignment	Do text and numbers line up in a way that supports scanning?
4. Structure	Are detail rows, totals, and long-table navigation easy to follow?
5. Purpose	Does every color, border, and emphasis choice communicate something useful?

A formatting audit is not a search for the prettiest worksheet. It is a test of whether the workbook is readable, consistent, and trustworthy.

Before-and-After Practice

Open the `Raw Export` tab only to inspect the original headers and 24 event records; do not edit it. Then work on the separate `Manager View` tab:

1. Convert the full data range into an Excel table with filter buttons.
2. Apply a consistent date format to Event Date.
3. Apply Currency or Accounting to Revenue, Catering Cost, Staffing Cost, and Net Profit; keep Guests as whole numbers.
4. Set readable column widths and alignment so labels and values can be scanned without avoidable clipping.
5. Freeze the table header row and test it by scrolling.
6. Build the four labeled summary formulas with cell or table references.
7. Apply at least one purposeful conditional-formatting rule to Net Profit.
8. Use the `Class Challenge` checklist for a final 30-second manager audit. A professional design is required, but one exact palette or pixel-perfect layout is not.

5 - CHECK YOUR UNDERSTANDING

Answer each question before class.

1. Why can a worksheet be mathematically correct but still difficult to trust?
2. Which number format should be used for event revenue, and why?
3. Why should numeric columns usually be right-aligned?

4. What is the purpose of a distinct total row?
5. When is Freeze Panes useful?
6. Why should color not be the only way a workbook communicates meaning?
7. Anchor & Oak has net profits of \$2,980, \$1,990, \$5,510, and \$3,290. What Excel formula should total the four referenced cells, and what is the result?

Answer Key - Check Your Understanding

#	Answer
1	Correct calculations can still be hard to interpret when labels, formats, structure, or visual hierarchy are inconsistent.
2	Currency or Accounting, because the values represent dollars and should use a consistent display.
3	Right alignment lines up place values and makes comparisons easier.
4	It separates the summary result from detail rows and makes the worksheet easier to scan.
5	Use Freeze Panes when headers or key labels would disappear while scrolling through a long worksheet.
6	Labels and structure must also communicate the meaning so the workbook remains accessible and understandable when printed.
7	Use <code>=SUM(the_four_profit_cells)</code> ; the total is \$13,770 .

6 - KEY VOCABULARY

Term	Definition
Number Format	The display applied to a stored value, such as Currency, Accounting, Date, or Percentage.
Visual Hierarchy	The use of size, weight, spacing, and contrast to show what readers should notice first.
Alignment	The horizontal or vertical position of content within a cell.
Banded Rows	Alternating row fills that help readers track records across a table.
Freeze Panes	An Excel feature that keeps selected rows or columns visible while scrolling.
Total Row	A visually distinct row that summarizes one or more data columns.
Conditional Formatting	Formatting that changes automatically when a cell meets a defined rule.
Audit-Friendly	Organized so another person can trace calculations, identify inputs, and understand results efficiently.

Before Class

Open the Excel M02 starter workbook. Locate the `Live You Try It`, `Raw Export`, `Manager View`, `Class Challenge`, and `FormulaReferenceCard` tabs. Be ready to explain why the `Raw Export` must remain unchanged while the separate `Manager View` becomes the professional deliverable.

Excel Functions for Business Analysis

EXCEL FUNCTIONS FOR BUSINESS ANALYSIS

Field	Details
Course	BUS123 - Solving Business Problems with Technology
Track	Excel
Module	M03
Lesson	L01

This lesson is about using Excel functions to turn messy business data into useful answers. You do not need to memorize hundreds of functions. The goal is to understand the patterns behind the functions that handle most everyday business analysis work.

By the end of this reading, you should be able to explain what a function does, recognize when a formula should spill into multiple cells, and choose a useful function based on the business question.

1. FROM BASIC FORMULAS TO FUNCTIONS

Excel formulas always begin with an equals sign. A basic formula can use arithmetic operators:

Operator	Meaning	Example
+	Add	=B2+C2
-	Subtract	=B2-C2
*	Multiply	=B2*C2
/	Divide	=B2/C2
^	Exponent	=B2^2

For example, Tidal Goods Co. can calculate product revenue with:

```
=Units_Sold*Unit_Price
```

Functions are named formulas that perform a specific task. Instead of writing a long manual formula like:

```
=B2+B3+B4+B5+B6+B7+B8+B9+B10+B11+B12+B13
```

you can use:

```
=SUM(B2:B13)
```

The function version is shorter, easier to read, and safer when rows are inserted inside the range.

Key Idea

A function is not magic. It is a named formula that takes one or more inputs, called arguments, and returns an answer.

Named Ranges: Readable Cell References

A **named range** assigns a meaningful label to a cell or range. Instead of asking a reader to remember what `F7:F18` contains, a formula can use a name such as `Monthly_Revenue`.

Cell-reference formula	Named-range formula	Business meaning
<code>=SUM(F7:F18)</code>	<code>=SUM(Monthly_Revenue)</code>	Total monthly revenue values.
<code>=AVERAGE(G7:G18)</code>	<code>=AVERAGE(Net_Profit)</code>	Average net profit.
<code>=SUMIFS(F7:F18,C7:C18,J3)</code>	<code>=SUMIFS(Sales,Region,J3)</code>	Total sales for the region selected in J3.

Named ranges make formulas easier to read and reuse, but the name must point to the correct cells. A clear name does not repair an incorrect range.

Create a Named Range with the Name Box

1. Select the cell or range you want to name. Do not include a header unless the header is intentionally part of the range.
2. Find the **Name Box** to the left of the formula bar. It normally displays the active cell address, such as `F7`.
3. Select inside the Name Box, type a meaningful name such as `Monthly_Revenue`, and press **Enter**.
4. Open the Name Box drop-down and choose the new name. Confirm that Excel selects the intended cells.
5. Use the name in a formula, such as `=SUM(Monthly_Revenue)`, and press **Enter**.

Excel

FileHomeInsertPage LayoutFormulasDataReviewView

Name Box

Monthly_Revenue

Formula Bar

fx

				\$4,800
				\$3,950
				\$6,225

1. Select the range

2. Type the name

3. Press Enter

Names cannot contain spaces. Begin with a letter, underscore, or backslash, and use underscores between words. Avoid names that look like cell addresses, such as `Q1` or `A10`.

Good Naming Pattern

Prefer descriptive names such as `Sales`, `Region`, `Hourly_Rate`, or `Tax_Rate`. Avoid vague names such as `Data1`, `Numbers`, or `Stuff`.

Create or Edit a Name with Name Manager

1. Select **Formulas** → **Name Manager** in the **Defined Names** group.
2. Select **New** to create a name, or select an existing name and choose **Edit**.
3. Enter the name and confirm the **Refers to** range.
4. Keep the scope set to **Workbook** unless the name should work on only one worksheet.
5. Select **OK**, then **Close**.
6. Test the name by selecting it from the Name Box or pressing **F3** while editing a formula to open the Paste Name list.

Use Name Manager when a range grows, moves, or produces **#REF!**. The **Refers to** field should point to the current worksheet and intended cells, for example `=BookingSummary!$F$7:$F$18`.

Named-Range Example: Reusable Analysis

Suppose Tidal Goods stores sales amounts in `F7:F18` and regions in `C7:C18`.

1. Select `F7:F18` and name it `Sales`.
2. Select `C7:C18` and name it `Region`.
3. Type a region, such as `North`, in `J3`.
4. In the report cell, enter `=SUMIFS(Sales,Region,J3)`.
5. Change `J3` to another region and confirm that the result updates.

This formula reads like the business question: add **Sales** where **Region** matches the value in `J3`.

2. THE CORE SUMMARY FUNCTIONS

Many business questions begin with summary functions.

Function	Syntax	Business Question
<code>SUM</code>	<code>=SUM(range)</code>	What is the total?
<code>AVERAGE</code>	<code>=AVERAGE(range)</code>	What is typical?
<code>MAX</code>	<code>=MAX(range)</code>	What is the highest value?
<code>MIN</code>	<code>=MIN(range)</code>	What is the lowest value?
<code>COUNT</code>	<code>=COUNT(range)</code>	How many cells contain numbers?
<code>COUNTA</code>	<code>=COUNTA(range)</code>	How many cells contain anything?

Anchor & Oak Events might use these functions to summarize monthly event revenue. `SUM` answers total annual revenue. `AVERAGE` gives a typical month. `MAX` and `MIN` identify the strongest and weakest months. `COUNT` and `COUNTA` help check whether the data is complete.

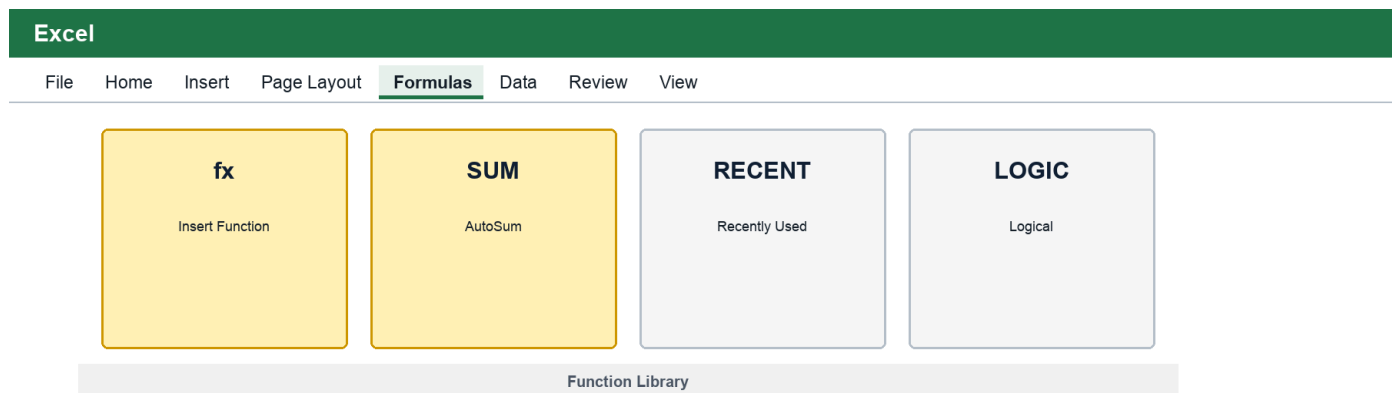
Common Mistake

Do not start a numeric range on a header row. `=COUNT(F6:F18)` may look close, but if row 6 is a header, the formula is asking Excel to count the wrong range. Start on the first data row: `=COUNT(F7:F18)`.

Enter a Summary Function in Windows Excel

You can type a function directly, use **AutoSum**, or use **Insert Function**. Typing is usually fastest once you recognize the pattern; the ribbon tools are useful while learning.

1. Select the empty cell where the answer should appear.
2. Select the **Formulas** tab.
3. In the **Function Library** group, select **AutoSum** for **SUM**, **AVERAGE**, **COUNT**, **MAX**, or **MIN**. Select the AutoSum drop-down arrow to see choices other than Sum.
4. Check the highlighted range. Drag across the correct cells if Excel selected a header, total, or incomplete range.
5. Press **Enter** to accept the formula.
6. Select the result cell and read the formula bar to verify the function and range.



For a function that is not on the AutoSum menu, select **Formulas** → **Insert Function**, search for the function name, select it, and complete its argument boxes. Do not accept the result until the formula answers the intended business question.

3. THE GREAT SHIFT: DYNAMIC ARRAYS AND SPILLING

Older Excel often followed a "one formula, one cell" mindset. If you needed a formula to fill ten rows, you wrote it once and dragged it down.

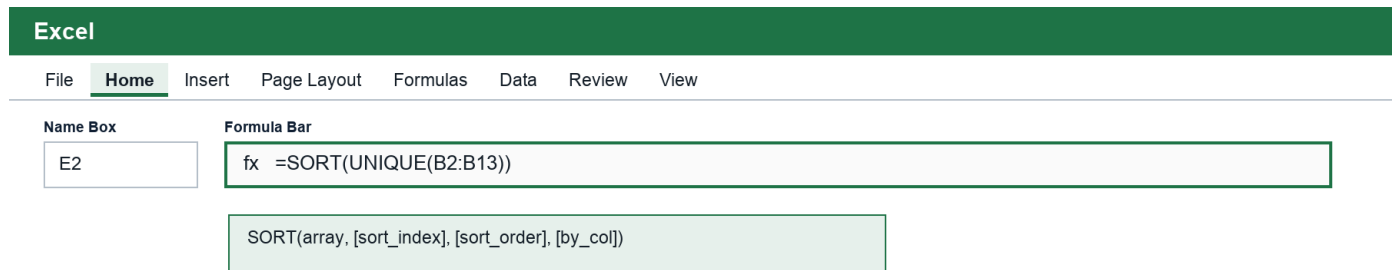
Modern Excel can work differently. A dynamic array formula can return many values from one formula. The results automatically fill nearby empty cells. This is called **spilling**.

Feature	Traditional Calculation	Modern Dynamic Arrays
Output	One formula returns one result.	One formula can return many results.
Manual work	Often requires dragging or copying.	Results spill automatically.
Range behavior	Static ranges can miss new data.	Spilled results can grow or shrink.
Error risk	Copying errors are common.	One formula controls the full output.

Spilling matters because it makes workbooks faster, cleaner, and easier to maintain. If a source table changes, a dynamic array result can update automatically instead of forcing the user to re-copy formulas.

Enter a Function Using the Formula Bar

1. Select the first cell where the result should begin.
2. Type `=` followed by the function name and an opening parenthesis, such as `=UNIQUE(`.
3. Select the source range with the mouse or type the range reference.
4. Read Excel's argument hint below the formula bar. The bold argument is the one Excel expects next.
5. Type commas between arguments and place text criteria inside quotation marks.
6. Close every parenthesis and press **Enter**. Do not press **Ctrl+Shift+Enter** for modern dynamic arrays.



Type the formula, use Excel's argument hint to check the order, then press Enter.

4. THE SPILL RANGE AND THE HASH SIGN

When a dynamic array formula spills, the first cell is the formula cell. The cells around it are spill results.

If the formula starts in `E2`, the whole spilled range can be referenced as:

`=E2#`

The hash sign tells Excel to use the entire spilled result, even if it grows or shrinks later.

Reference Type	Meaning
<code>\$A\$1:\$A\$10</code>	A fixed range. It does not automatically follow a spill.
<code>E2</code>	Only the first cell of the spill.
<code>E2#</code>	The full spilled range that starts in E2.

This is one of the most important ideas in modern Excel. A formula can create a live list, and another formula can refer to the full list with `#`.

Dynamic array result

		=SORT(UNIQUE(B2:B13))	Formula cell
		Events	
		Finance	Spill results
		Marketing	
		Operations	
		Sales	

Reference the changing result as E2#

Only the top-left cell contains the formula. Select another cell in the spill range and Excel outlines the full result. To change the formula, return to the top-left formula cell rather than typing over an individual spill result.

Why It Matters

If a `UNIQUE` list grows from five departments to seven departments, `E2#` grows with it. A downstream formula or drop-down list can stay synced without manual editing.

5. UNIQUE, SORT, AND FILTER

Three dynamic array functions do a lot of everyday data organization work.

Function	Core Purpose	Typical Spill Direction
<code>UNIQUE</code>	Extracts distinct values from a list.	Down rows or across columns.
<code>SORT</code>	Reorders data alphabetically or numerically.	Same shape as the source.
<code>FILTER</code>	Extracts records that meet criteria.	Across columns and down rows.

`UNIQUE` is useful for cleaning repeated categories. `SORT` makes the result organized. Together, they can replace a lot of manual sorting and duplicate removal.

```
=SORT(UNIQUE(Department_Column))
```

`FILTER` extracts rows that meet a condition. Harborside Medical Center could use it to pull patient billing records above a threshold:

```
=FILTER(A2:F200,F2:F200>500,"No matches")
```

The key is the include argument. Excel evaluates each row as `TRUE` or `FALSE`. Rows that evaluate to `TRUE` spill into the result.

Build a Dynamic List Safely

1. Choose a blank area outside the source table with enough empty cells below and to the right.
2. Select the top-left destination cell.

3. Type the formula, such as `=SORT(UNIQUE(B2:B13))`.
4. Press **Enter** once and allow Excel to spill the result.
5. Add or change a source value and confirm that the list updates.
6. Reference the complete result with the top-left cell followed by `#`, such as `=E2#`.

6. LOGIC: IF, AND, AND OR

Functions can also make decisions. The `IF` function checks a condition and returns one result if the condition is true and another if the condition is false.

```
=IF(logical_test,value_if_true,value_if_false)
```

Meridian Advisory Group might flag payroll records like this:

```
=IF([@[Hours Worked]]>40,"Overtime","Regular")
```

Sometimes one condition is not enough.

Function	Use When	Example Pattern
AND	Every condition must be true.	<code>=AND([@Department]="Nursing",[@Hours]>40)</code>
OR	At least one condition must be true.	<code>=OR([@Priority]="High",[@Priority]="Critical")</code>

A common beginner mistake is writing:

```
=OR([@Priority]="Critical","High")
```

Excel cannot guess that "High" should also be compared to `[@Priority]`. You must repeat the full condition:

```
=OR([@Priority]="Critical",[@Priority]="High")
```

Build an IF Formula One Argument at a Time

1. Select the result cell and type `=IF(`.
2. Enter the logical test, such as `D2>40`.
3. Type a comma, then enter the true result in quotation marks: `"Overtime"`.
4. Type another comma, then enter the false result: `"Regular"`.
5. Close the parenthesis and press **Enter**.
6. Test at least one record that should return each outcome.

The completed formula is `=IF(D2>40,"Overtime","Regular")`. If you nest `AND` or `OR` inside `IF`, finish the inner function's closing parenthesis before entering the true and false results.

7. CONDITIONAL REPORTING WITH SUMIFS AND COUNTIFS

Business reporting often asks questions with conditions:

- What is total revenue for one region?
- How many records are late?
- What is the total for one department and one status?

The plural `IFS` functions are usually the best default because they can handle one condition or many.

Function	Use It To	Example
<code>SUMIFS</code>	Add values that meet one or more criteria.	<code>=SUMIFS(Sales,Region,"North",Category,"Kitchen")</code>
<code>COUNTIFS</code>	Count rows that meet one or more criteria.	<code>=COUNTIFS(Status,"Late",Department,"Nursing")</code>
<code>AVERAGEIFS</code>	Average values that meet one or more criteria.	<code>=AVERAGEIFS(Sales,Region,"North")</code>

Tidal Goods Co. might use `SUMIFS` to total sales for a category and region. Harborside Medical Center might use `COUNTIFS` to count late timesheets by department.

Better Practice

Link criteria to input cells when possible. A report that uses `=SUMIFS(Sales,Region,C9)` is easier to update than a formula that hard-codes "North" in every cell.

Build SUMIFS in the Correct Order

1. Select the answer cell and type `=SUMIFS(`.
2. Select the numeric **sum range** first.
3. Type a comma and select the first **criteria range**.
4. Type a comma and select or type its criterion.
5. Repeat the criteria-range and criterion pair for each additional condition.
6. Close the parenthesis and press **Enter**.
7. Confirm that every range has the same number of rows and begins on the same data row.

For example, `=SUMIFS(Sales,Region,C9,Category,D9)` adds Sales only where both criteria match the input cells. Selecting criteria cells instead of typing words into the formula makes the report easier to reuse.

8. XLOOKUP: THE MODERN CONNECTOR

`XLOOKUP` connects tables. It finds a lookup value in one range and returns a matching value from another range.

```
=XLOOKUP(lookup_value,lookup_array,return_array,if_not_found)
```

Meridian Advisory Group might use `XLOOKUP` to find an hourly rate based on an employee ID:

```
=XLOOKUP(A2,EmployeeTable[Employee ID],EmployeeTable[Hourly Rate],"Check ID")
```

Compared with older lookup functions, `XLOOKUP` is more flexible. It can look left or right, can return more than one column, and includes a built-in not-found message.

Build XLOOKUP from the Business Question

1. Identify the value you already have, such as an employee ID. This is the **lookup value**.
2. Select the column where Excel should search. This is the **lookup array**.
3. Select the column containing the answer you want. This is the **return array**.
4. Add a useful not-found message such as "Check ID".

5. Press **Enter**, then test one valid ID and one missing ID.

The lookup and return arrays must cover corresponding rows. XLOOKUP uses an exact match by default, so beginners normally should not add a match-mode argument.

9. DATE AND SCHEDULE HELPERS

Excel dates are numbers behind the scenes, which means functions can calculate schedules.

Function	Purpose	Example
EOMONTH	Returns the last day of a month.	=EOMONTH (A2 , 0)
WORKDAY . INTL	Returns a workday while skipping custom weekends or holidays.	=WORKDAY . INTL (A2 , 10 , "0000011")
SEQUENCE	Generates a list of numbers.	=SEQUENCE (12)

SEQUENCE becomes especially useful when paired with date functions. A monthly report can generate month numbers or month-end dates without typing each row manually.

```
=EOMONTH(Start_Date,SEQUENCE(12)-1)
```

10. TROUBLESHOOTING FUNCTION ERRORS

Most Excel errors are clues. Read the clue before changing the formula.

Error	Likely Meaning
#####	The column is too narrow to display the value.
#NAME?	Excel does not recognize text in the formula, often because of a typo.
#VALUE!	One or more arguments are the wrong type.
#DIV/0!	The formula is dividing by zero or a blank denominator.
#REF!	The formula refers to a deleted or invalid cell.
#SPILL!	The spill range is blocked or cannot expand.

When a formula does not work, ask these questions:

1. Does the formula start with `=`?
2. Are all parentheses closed?
3. Are text criteria inside quotation marks?
4. Does the range start on the first data row?
5. Does the formula need an absolute reference?
6. Is the spill range empty?
7. Does the answer match the business question?

Correct a #SPILL! Error

1. Select the cell displaying `#SPILL!`.

2. Look for Excel's dashed outline showing the intended spill range.
3. Select the warning icon if available and choose **Select Obstructing Cells**.
4. Move or clear the blocking content. Also check for merged cells or a spill formula placed inside an Excel table.
5. Confirm that Excel fills the complete spill range after the obstruction is removed.

Resolve a blocked spill range

	#SPILL!	1. Select the #SPILL! cell.	
		2. Find the dashed intended range.	
	Blocking value	3. Move or clear the blocking value.	
		4. Confirm that the results spill.	

Do not delete the dynamic-array formula simply because the output area is blocked. Fix the obstruction first. If the formula is inside an Excel table, move it to a normal worksheet range because spilled array formulas cannot spill inside tables.

11. THE 80/20 FUNCTION TOOLKIT

Excel has hundreds of functions, but a small toolkit handles most early business analysis work:

Toolkit Area	Functions
Basic summaries	SUM , AVERAGE , MAX , MIN , COUNT , COUNTA
Logic	IF , AND , OR
Conditional reporting	SUMIFS , COUNTIFS , AVERAGEIFS
Lookup	XLOOKUP
Dynamic arrays	FILTER , SORT , UNIQUE , SEQUENCE
Text cleanup	TEXTJOIN
Dates	EOMONTH , WORKDAY .INTL
Multi-row math	SUMPRODUCT

This lesson skips custom function building with LAMBDA . That is a powerful advanced tool, but it is not the first priority for BUS123. The first priority is choosing the right built-in function for the business question.

CHECK YOUR UNDERSTANDING

Answer these before class.

1. What is the main difference between a traditional formula copied down a column and a dynamic array formula?
2. What does the `#` symbol do when placed after the first cell of a spilled result?
3. Which function would return a clean list of unique department names?
4. Which function would extract all rows where a balance is greater than `$500`?
5. Why is `=OR([@Priority]="Critical","High")` incorrect?
6. When would you use `SUMIFS` instead of `SUM`?
7. What does `#SPILL!` usually mean?
8. Why is `XLOOKUP` generally better than older lookup tools for new Excel work?
9. What is one advantage of `=SUM(Monthly_Revenue)` over `=SUM(F7:F18)`?
10. Where would you check which cells a named range currently refers to?

KEY VOCABULARY

Term	Meaning
Function	A named formula that performs a specific task.
Argument	An input a function needs inside its parentheses.
Dynamic Array	A formula result that can return multiple values.
Spill	The behavior where dynamic array results fill neighboring cells.
Spill Range	The full output range created by a dynamic array formula.
Hash Reference	A reference like <code>E2#</code> that points to the full spill range.
Criteria	A rule or condition used by a function.
Structured Reference	A table-based reference that uses column names instead of cell coordinates.
Named Range	A meaningful name assigned to a cell or range, such as <code>Monthly_Revenue</code> .
Name Manager	The Formulas-tab tool used to create, edit, delete, and verify named ranges.

Excel Charts and PivotTables

EXCEL CHARTS AND PIVOTTABLES

Field	Details
Course	BUS123 - Solving Business Problems with Technology
Track	Excel
Module	M04
Lesson	L01
Case	Harborside Medical Center

This lesson is about turning rows of business data into visuals and summaries that help people make decisions. In Module 3, you practiced formulas and functions that calculate accurate numbers. In Module 4, those accurate numbers become charts and PivotTables that a manager can scan, question, and trust.

Harborside Medical Center will be our case. The hospital has monthly data about service lines, payer categories, visits, and revenue. A spreadsheet can store all of those records, but leaders do not want to read every row one at a time. They want to know which service line is strongest, whether visit volume is changing, and which payer mix needs attention.

By the end of this reading, you should be able to choose a chart based on the business question, identify the habits that make charts trustworthy, and explain how a PivotTable summarizes a clean table.

1. START WITH THE BUSINESS QUESTION

A chart is not decoration. A chart is an answer in visual form. Before you click Insert, pause and name the question you are trying to answer.

Business question	Better chart choice	Why it fits
Which service line produced the most revenue?	Column or bar chart	Compares categories by height or length.
Are visits rising or falling across months?	Line chart	Shows change over time.
What share of revenue came from each payer?	Pie or donut chart	Shows parts of a whole when there are only a few pieces.
Does higher visit volume connect with higher revenue?	Scatter chart	Compares two numeric variables.

The mistake beginners make is choosing the chart they like, not the chart that fits the logic of the question. A line chart implies order and movement over time. A pie chart implies parts of one whole. A column chart implies comparison across categories. A scatter chart implies a relationship between two numeric measurements.

Key Idea

The business question chooses the chart type. Design choices come after that.

2. CLEAN TABLES COME BEFORE GOOD CHARTS

Charts and PivotTables are only as trustworthy as the source table behind them. If Excel cannot understand the table boundaries, fields, or record structure, the output may look polished while still being wrong.

A clean Excel table has:

- One header row with clear field names.
- One record per row.
- One kind of data in each column.
- No blank rows inside the data range.
- No manual total row mixed into the source table.
- Consistent categories, spelling, dates, and number formats.

For Harborside, a clean table might have columns such as Month, Service Line, Payer, Visits, Revenue, and Average Revenue per Visit. Each row should represent one service-line record for one month and payer. A row that says "Total" belongs in a summary area, not inside the raw source table.

Convert the Source Range to an Excel Table

1. Select any cell inside the source data.
2. Select **Home** → **Format as Table** and choose a simple, readable style.
3. In the Create Table dialog box, confirm that the range includes every record and field but excludes notes or summary calculations.
4. Select **My table has headers**, then select **OK**.
5. Select the new table and open **Table Design**. Give the table a meaningful name such as `HarborsideData`.
6. Use each field's filter arrow to check for blanks, inconsistent spellings, or accidental totals.

Excel

File **Home** Insert Page Layout Formulas Data Review View

TABLE

Format as Table

Source-data checklist

- ☒ One header row
- ☒ One record per row
- ☒ No blank rows
- ☒ No manual totals

Using an Excel table helps charts and PivotTables recognize the full source range. When new records are added directly below the table, the table can expand to include them.

3. CHART SELECTION IS A LOGIC CHOICE

Different chart types encode information in different ways.

Chart type	Use it when	Watch out for
Column chart	You are comparing a small set of categories.	Long labels may be easier to read in a bar chart.
Bar chart	You are comparing categories with longer text labels.	Sort the bars when ranking matters.
Line chart	The x-axis is ordered time, such as months or quarters.	Do not use it for unrelated categories.
Pie or donut chart	You have two to four pieces of one total.	Too many slices become unreadable fast.
Scatter chart	Both axes are numeric measurements.	Do not use a line chart when numeric x-values are unevenly spaced.

One common trap is the overloaded pie chart. If a chart has eight or nine slices, the reader has to compare tiny angles and colors. A sorted column chart is usually clearer. Another trap is using a line chart for categories such as Imaging, Lab, Urgent Care, and Therapy. Those service lines do not happen in a time sequence, so connecting them with a line creates a false story.

Create a Chart in Windows Excel

1. Start with the business question. For "Which service line produced the most revenue?" you need the **Service Line** and **Revenue** fields.
2. Select the category labels and numeric values, including both headers. Do not include unrelated fields or a grand total unless the total itself belongs in the comparison.
3. Pause and inspect the highlighted range. Each selected row should pair one service line with its revenue value.

Select only the fields that answer the question

Business question: Which service line produced the most revenue?

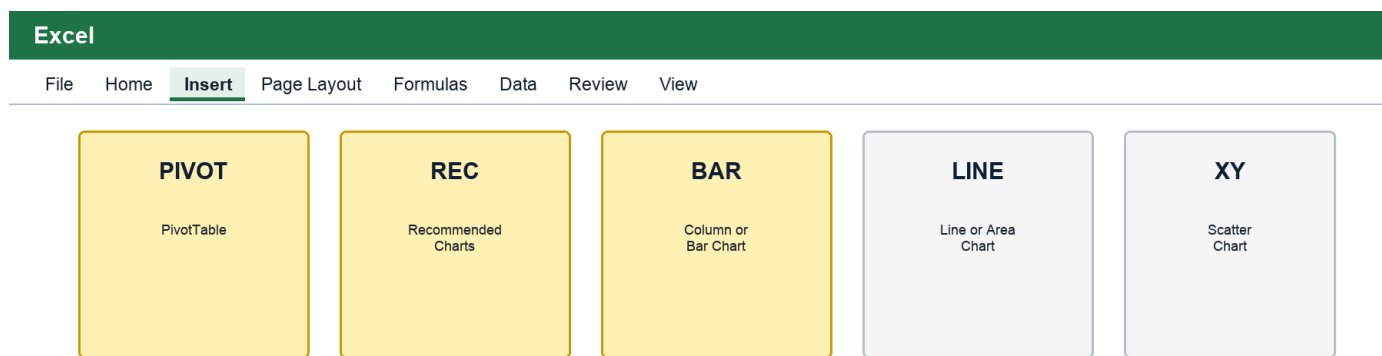
Service Line	Revenue	Visits
Imaging	\$690,000	2,310
Lab	\$420,000	4,920
Therapy	\$610,000	1,740
Urgent Care	\$880,000	3,180

Include

headers +
category labels +
values

Do not include Visits: it does not answer this revenue question.

1. Select the **Insert** tab and find the **Charts** group.
2. Select **Recommended Charts** to preview options or choose a specific chart button such as **Insert Column or Bar Chart**.
3. In the preview, confirm that service lines appear on the category axis and revenue determines the height or length of each bar.
4. Select the appropriate chart and choose **OK**.
5. Move or resize the chart without covering source data needed for review.



Excel's recommendation is a starting point, not the final decision. Reject a suggested chart if it does not match the business question.

4. BUILD THE CHART, THEN REMOVE NOISE

Excel's default chart is a draft. It proves that the data can be charted, but it is rarely ready for a business audience.

A finished chart should have:

- A title that states the takeaway or question.
- Axis labels when the units are not obvious.
- Data labels when exact values matter.
- A limited color palette with emphasis used intentionally.
- No unused legend, heavy gridlines, 3D effects, or decorative clutter.

For example, "Revenue by Service Line" is better than "Chart 1" because it tells the reader what they are looking at. "Urgent Care Leads Q3 Revenue" is even stronger when the chart is meant to support that exact message.

Common Mistake

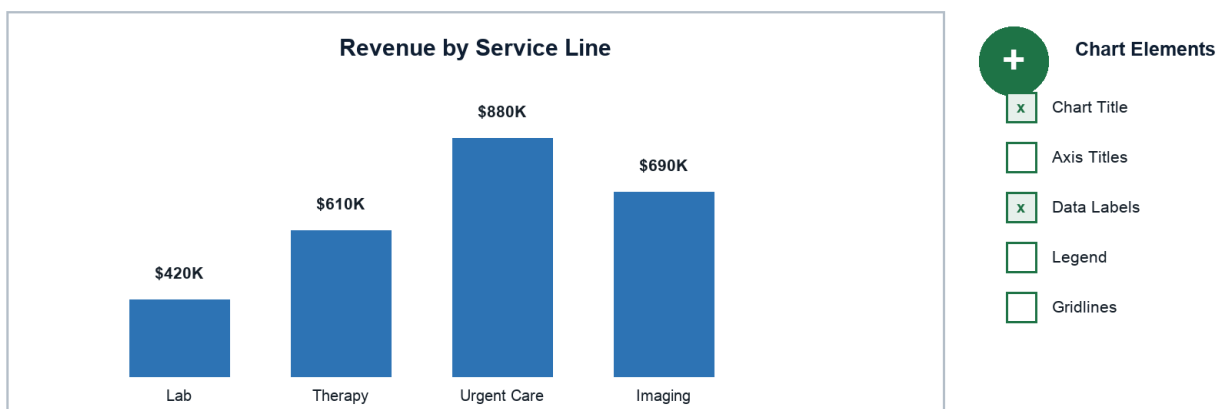
Do not make a chart look more important than the data behind it. A clean source table, correct chart type, and honest title matter more than effects.

Polish the Chart and Remove Noise

1. Select the chart. The **Chart Design** and **Format** tabs appear.

2. Select the green **Chart Elements (+)** button beside the chart, or choose **Chart Design** → **Add Chart Element**.
3. Turn on **Chart Title** and replace the placeholder with a descriptive title.
4. Add **Axis Titles** when the units or categories are not obvious.
5. Add **Data Labels** only when exact values help the decision.
6. Turn off an unnecessary legend or distracting gridlines.
7. Check number formatting. Revenue labels and axes should show currency or clear units such as \$ thousands .
8. Confirm that the visual does not use 3D effects or misleading axis scales.

Polish a chart with Chart Elements



Before finishing, state the conclusion aloud. If the chart does not make that conclusion easier to see, revisit the selected data, chart type, or title.

5. PIVOTTABLES SUMMARIZE MANY ROWS QUICKLY

A PivotTable is a flexible summary engine. Instead of writing a separate formula for each possible view, you drag fields into areas and let Excel summarize the table.

The basic PivotTable areas are:

Area	What it controls	Harborside example
Rows	The categories listed down the left side.	Service Line
Columns	The categories listed across the top.	Payer
Values	The numbers being summarized.	Sum of Revenue or Average of Visits
Filters	The field used to narrow the summary.	Month

A PivotTable can answer "total revenue by service line" by placing Service Line in Rows and Revenue in Values. If the CFO then asks for the same summary by payer, you can move Payer into Rows or Columns instead of rebuilding the report from scratch.

Create a PivotTable from the Clean Table

1. Select any cell in the `HarborsideData` table.
2. Select **Insert** → **PivotTable** → **From Table/Range**.
3. Confirm the table or range shown in the dialog box.
4. Choose **New Worksheet** so the summary has its own workspace, then select **OK**.

Confirm the source and choose a clean workspace

Create PivotTable

Choose the data that you want to analyze

☒ Select a table or range

Table/Range: HarborsideData

Choose where you want the PivotTable report to be placed

☒ New Worksheet

☐ Existing Worksheet

OK

1. In the PivotTable Fields pane, drag **Service Line** to **Rows**.
2. Drag **Revenue** to **Values**.
3. Drag **Month** to **Filters** when the manager needs to switch reporting periods.
4. Compare the PivotTable grand total with a known source total before interpreting the results.

Arrange fields in the PivotTable Fields pane

PivotTable summary

Service Line	Sum of Revenue
Imaging	\$690,000
Lab	\$420,000
Therapy	\$610,000
Urgent Care	\$880,000

PivotTable Fields

☐ Month
☒ Service Line
☐ Payer
☐ Visits
☒ Revenue

Filters
Month

Columns
Drop field here

Rows
Service Line

Values
Sum of Revenue

If the PivotTable Fields pane is hidden, select inside the PivotTable and choose **PivotTable Analyze** → **Field List**.

6. VALUES NEED THE RIGHT SUMMARY SETTING

Dragging a numeric field into Values is not the end of the decision. Excel must know how to summarize that field.

Need	Value Field Setting
Total revenue	Sum
Typical visits per month	Average
Number of records	Count
Largest single value	Max
Smallest single value	Min

Always check the Value Field Settings. If Revenue appears as Count of Revenue instead of Sum of Revenue, the PivotTable is counting records, not adding dollars. The output may still look neat, which makes this error easy to miss.

Correct the Value Field Setting

1. In the PivotTable Fields pane, find the field in the **Values** area.
2. Select its drop-down arrow and choose **Value Field Settings**.
3. On **Summarize Values By**, choose **Sum**, **Average**, **Count**, **Max**, or **Min** based on the business question.
4. Select **Number Format** and apply an appropriate display such as Currency or Number.
5. Select **OK** and verify that the field caption now says **Sum of Revenue**, **Average of Visits**, or the intended summary.
6. Compare the result with a simple source calculation or known total.

Make sure Excel is calculating the intended result

Value Field Settings

Summarize value field by

☒ Sum

☐ Count

☐ Average

☐ Max

☐ Min

Expected field caption

Sum of Revenue

not Count of Revenue

OK

For a quick independent check, select a blank cell and enter `=SUM(HarborsideData[Revenue])`. The formula result should match the PivotTable grand total when no filters are applied. If the totals differ, check the PivotTable source, filters, summary setting, and refresh status before using the result.

7. REFRESH, SLICERS, AND PIVOTCHARTS

Regular formulas usually recalculate when source cells change. PivotTables behave differently. They use a cached snapshot of the source data, so you must refresh them after the source table changes.

In Excel, use Data > Refresh All before presenting a PivotTable or PivotChart. This is especially important when rows have been added, deleted, or corrected.

Slicers are clickable filters for PivotTables. They make it easier for a non-technical audience to switch between months, service lines, or payer categories. A PivotChart turns a PivotTable summary into a chart that responds to the same fields and slicers.

Refresh the PivotTable and Add a Slicer

1. Add or correct a record in the source table.
2. Select inside the PivotTable to display **PivotTable Analyze**.
3. Select **Refresh** to update the selected PivotTable, or open the Refresh menu and choose **Refresh All** to update the workbook's PivotTables and connections.
4. Confirm that the changed record appears in the summary.
5. With the PivotTable selected, choose **PivotTable Analyze** → **Insert Slicer**.
6. Select a field such as **Month**, **Service Line**, or **Payer**, then select **OK**.
7. Select slicer buttons to filter the PivotTable. Hold **Ctrl** to choose multiple items.
8. Select the slicer's **Clear Filter** button and confirm that the full summary returns.

The image shows the Excel ribbon with the following tabs: File, Home, Insert, Page Layout, Formulas, Data, Review, and View. Below the ribbon are three yellow boxes with black borders:

- FIELD**: Field Settings
- REFRESH**: Refresh or Refresh All
- SLICER**: Insert Slicer

To the right of these boxes is a section titled **Before presenting:** with a list of three items:

1. Check Sum vs. Count
2. Refresh the data
3. Test slicer filters

Refresh before presenting, even when the PivotTable appears correct. A polished summary can still be based on an older snapshot.

8. WHAT TO BRING TO CLASS AND THE MANAGEMENT BRIEF

In class, you will work with Harborside's service-line data. Be ready to:

1. Inspect whether a dataset is clean enough for charting.
2. Choose a chart type based on a manager's question.
3. Build and polish a chart from the starter workbook.
4. Plan a PivotTable by selecting Rows, Values, and Filters.
5. Begin the Management Visual Brief for the VP of Ambulatory Services.

The Class Challenge begins in class and finishes as homework. You will convert the supplied tidy records into an Excel Table, calculate four core KPIs, build or link a service-line summary with formulas, a PivotTable, or an equivalent summary, and create one or more charts that help the VP decide which service line needs attention next month.

Your chart type and layout are choices, not a tracing exercise. A sorted bar or column chart may clarify a category comparison, while a donut can be defensible for a small part-to-whole revenue question. Whatever you choose, use an accurate source range, add a specific title, apply readable number formats, and finish with a concise recommendation supported by at least two workbook results.

The goal is not to become an Excel chart designer in one class. The goal is to practice the decision path: audience, question, clean table, summary, visual, verification, recommendation.

CHECK YOUR UNDERSTANDING

Answer these before class.

1. Why should you choose the chart type before clicking Insert?
2. Which chart type would you use to compare revenue across four service lines?
3. Which chart type would you use to show visits by month?
4. Why is a pie chart usually a poor choice for eight payer categories?
5. What is the problem with leaving a manual total row inside the source data?
6. In a PivotTable, where would you place Service Line if you want one row per service line?
7. Why should you check Value Field Settings after dragging a number field into Values?
8. What does Refresh All do for a PivotTable?
9. Which ribbon command creates an Excel table from a clean source range?
10. Where would you drag Revenue to calculate total revenue in a PivotTable?
11. What should you check if the PivotTable displays Count of Revenue instead of Sum of Revenue?
12. Which chart control can add a title, axis titles, or data labels?

KEY VOCABULARY

Term	Meaning
Source table	The clean row-by-row data used to build charts, formulas, and PivotTables.
Chart type	The visual form, such as column, line, pie, donut, bar, or scatter.
Axis	A reference line that organizes chart values or categories.
Data label	A value printed directly on a chart element.
PivotTable	An Excel summary tool that rearranges fields into rows, columns, values, and filters.
Values area	The PivotTable area where numbers are summarized.
Value Field Settings	The menu that controls whether a PivotTable uses Sum, Average, Count, Max, Min, or another summary.
Slicer	A clickable PivotTable filter.
PivotChart	A chart connected to a PivotTable summary.
Refresh All	The command that updates PivotTables and other data connections after source data changes.